Weighted Polynomial Identities, Elliptic Fibrations, and the Easier Waring Problem for Fifth Powers

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Abstract

The easier Waring problem asks for the least positive integer $v(k)$ such that every integer $n \in \mathbb{Z}$ can be expressed as a signed sum of $s = v(k)$ integral k-th powers, $n = \sum_{i=1}^{s} \epsilon_i x_i^k$ with $\epsilon_i \in \{\pm 1\}$ and $x_i \in \mathbb{Z}_{\geq 0}$. For fifth powers ($k = 5$), the bounds $5 \leq v(5) \leq 10$ established by Fuchs and Wright (1939) [2] and recorded in Hardy and Wright [3] have remained at $v(5) \leq 10$ in the classical literature surveyed in [4, 5, 6], where classical polynomial constructions rely on unit-weighted Prouhet–Tarry–Escott (PTE) multigrade identities $\sum_{i=1}^{4}(z+\alpha_i)^5 - \sum_{i=1}^{4}(z+\beta_i)^5 = Cz + D$, for which Bastien's theorem forces $2n \geq 8$ summands and Kleiman–Wright divisibility forces $720 \mid C$ and $360 \mid D$, trapping the modular covering number at $\Delta(5, C) \geq 2$.

In this paper, the upper bound is improved to $v(5) \leq 9$ ($5 \leq v(5) \leq 9$). By introducing non-unit leading weights into paired quinta-power differences and developing a direct integer gap-sum reduction, the classical $16 \mid C$ and $9 \mid C$ divisibility barriers are broken, yielding the explicit 8-term identity (involving 8 fifth-power summands counted with multiplicity across 7 distinct linear forms with coefficients bounded by 39)

$$(2z-38)^5 - (2z-37)^5 + (z+39)^5 - (z+35)^5 - (z-22)^5 - (z+22)^5 + 2(z+6)^5 = 3844230\,z + 27826665,$$

whose slope $C = 3844230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ is square-free and free of prime factors $p \equiv 1 \pmod 5$, so that $y \mapsto y^5 \pmod C$ is a bijection on $\mathbb{Z}/C\mathbb{Z}$ ($\Delta(5, C) = 1$). Consequently, every integer $N \in \mathbb{Z}$ is a sum of at most 9 signed fifth powers. Furthermore, an explicit algebraic-geometric classification of $2k$-term paired linear identities is established via constructive birational correspondences on the open non-degenerate locus: the moduli space is empty ($\dim = -1$) for $k = 2$, birational to a smooth projective hyperelliptic curve of genus $g = 2$ (whose branch-point separation is guaranteed by Fermat's Last Theorem for exponent 5, yielding finitely many rational points by Faltings's theorem for each fixed ordered triple of distinct positive integer weights $a_1 > a_2 > a_3$), birational to a smooth elliptic curve E_a of genus $g = 1$ for $k = 3$ with $a_2 = a_3 = 1$, and contains an explicitly parameterized 2-dimensional open rational subvariety $\mathcal{S} \subset \mathcal{M}(a_1, a_2, a_3, a_4)$ with non-vanishing linear slope $C \neq 0$ for $k = 4$. Using certified points of infinite order on $E_2(\mathbb{Q})$ and $E_6(\mathbb{Q})$ and an exact 4-to-1 paired (at most 8-to-1 unordered) dihedral-torsion fiber bound under $\langle [-1], +P_0 \rangle$, infinitely many primitive, $\mathbb{Q}$-affine inequivalent 6-term linear polynomial identities in $\mathbb{Z}[z]$ are constructed, beating the 8-term PTE barrier over $\mathbb{Q}$. Finally, four structural obstruction and reduction results—the $\mathbb{F}_{11}$-Bastien Obstruction ($m \leq 7 \implies 11 \mid C$, forcing $s \geq m + 5 \geq 10$ for any single linear identity and $s \geq \max(m, 2k+5)$ across families with varying $11 \mid A_i$ constants, so $m = 8$ and $s = 9$ are optimal for the single-identity scheme), the 5-adic Vandermonde Obstruction ($m \leq 6 \implies 25 \mid C$, whereas the modulo-25 congruence obstruction vanishes at $m = 7$ and $v_5(C) = 1$ is achieved in $\mathbb{Z}[z]$ at $m = 8$), the Universal $30 \mid C$ and Fermat–Bastien Congruence Law for $D \pmod{30}$, and the Cauchy–Schwarz Signature Obstruction showing that no non-degenerate $(m-1, 1)$-signature fifth-power zero-sum can have vanishing z^5, z^4, z^3 coefficients over $\mathbb{R}$ (as vanishing of $[z^4]$ and $[z^3]$ forces proportional shifts $b_1/a_1 = \cdots = b_m/a_m$ and identical cancellation $P(z) \equiv 0$) alongside the 25-adic and Fermat–norm reduction within the 7-term odd-symmetric ansatz—are proved, together with a verified catalogue of 20 listed polynomial identities (16 independent up to affine equivalence or specialization) across degrees 1 to 5, a three-tier hybrid sieve for compact 9-term representations (where Tiers 1 and 2 cover $121/144 \approx 84.03\%$ of $\mathbb{Z}$ and Tier 3 covers the remaining $23/144 \approx 15.97\%$), and explicitly separated finite-box computations and congruence-class and heuristic mixed-degree reduction strategies for $s \leq 8$.

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1 Introduction and Literature Review Across Mathematical Branches

In 1934, E. M. Wright [1] introduced a variation of Waring's problem that he termed the "easier" Waring problem: for each integer $k \geq 2$, determine the least positive integer $v(k)$ such that every integer $n \in \mathbb{Z}$ can be expressed in the form

$$n = \epsilon_1 x_1^k + \epsilon_2 x_2^k + \cdots + \epsilon_s x_s^k, \qquad \epsilon_i \in \{+1, -1\}, \quad x_i \in \mathbb{Z}_{\geq 0}, \tag{1}$$

with $s = v(k)$. Throughout this paper, any integer coefficient $c \in \mathbb{Z} \setminus \{0\}$ multiplying a fifth power cX^5 is counted with multiplicity $|c|$ as $|c|$ signed fifth-power summands in (1). When k is odd, $(-x)^k = -x^k$, so (1) is equivalent to representing every integer $n \in \mathbb{Z}$ as a sum of s k-th powers of signed integers $x_i \in \mathbb{Z}$. For any individual integer $N \in \mathbb{Z}$, we write $W(k, N)$ for the least positive integer s such that N admits a representation of the form (1) (so that $v(k) = \sup_{N\in\mathbb{Z}} W(k, N)$). Throughout the literature, the quantity $v(k)$ (sometimes denoted $w(k)$ in expository notes [19, 25]) is contrasted with the classical Waring functions $g(k)$ and $G(k)$, which restrict all summands to be non-negative.

Despite its name—chosen because applying the $(k-1)$-th forward difference operator $\Delta^{k-1}f(z) = \sum_{j=0}^{k-1}(-1)^{k-1-j}\binom{k-1}{j}f(z+j)$ to $f(z) = z^k$ immediately yields the linear polynomial

$$\Delta^{k-1}(z^k) = \sum_{j=0}^{k-1}(-1)^{k-1-j}\binom{k-1}{j}(z+j)^k = k!\,z + \frac{k-1}{2}k!$$

using $\sum_{j=0}^{k-1}\binom{k-1}{j} = 2^{k-1}$ k-th power summands counted with multiplicity, and adjoining at most $\lfloor k!/2 \rfloor$ unit terms $\pm 1^k$ covers all residue classes modulo $k!$ to establish $v(k) \le 2^{k-1} + \lfloor k!/2 \rfloor$ [1] (see also Hardy and Wright [3, Chapter XXI, §21.11, Theorems 400–401])—determining exact values or sharp bounds for $v(k)$ has proved substantially more delicate than the classical problem [5, 6]. Indeed, $v(2) = 3$ is the only exponent for which the exact value of $v(k)$ is known: for cubes, Oltramare (1894) showed $v(3) \le 5$ via $6z = (z+1)^3 + (z-1)^3 - 2z^3$ (using 4 cubic summands plus 1 residue term modulo 6), while the modulo-9 congruence $x^3 \equiv 0, \pm 1 \pmod 9$ gives $v(3) \ge 4$, leaving $v(3) \in \{4, 5\}$ open for over a century.

For fifth powers ($k = 5$), the classical Waring problem was investigated by Wieferich [7] in 1909, who established the first upper bound $g(5) \le 59$, and was finally resolved for all integers in 1964 by Chen Jingrun [8], who proved the exact value $g(5) = 37$. For sufficiently large integers, Vaughan and Wooley [9] established the asymptotic bound $G(5) \le 17$ (Theorem 1.1 of [9]) via smooth-number Weyl differencing. For the easier Waring problem with $k = 5$, Fuchs and Wright [2] proved in 1939 (see also Hardy and Wright [3, Chapter XXI, §21.11] and Borwein and Ingalls [5, p. 16]) that

$$5 \le v(5) \le 10 \qquad (4 < v(5) < 11). \tag{2}$$

To the author's knowledge, within the classical and expository literature consulted—including Wright [4], Hardy and Wright [3], Borwein and Ingalls [5], Vavilov [6], Srikanth and Anuradha Kameswari [17], Weisstein [18], Lugo [19], algebraic identity collections [11, 12, 13, 14, 15], and online discussions [20, 22, 23, 24, 25]—no bound improving upon $v(5) \le 10$ of [2] has been recorded. Regardless of whether unindexed work exists outside the sources consulted, our proof of $v(5) \le 9$ in Section 5 is completely self-contained and rests solely on the explicit 8-term identity (38), the prime factorization of $C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$, and Lemma 5.2, verified independently by exact integer arithmetic in Section 5.2 and Appendix A.

1.1 Survey of Techniques Across Five Branches of Mathematics

To examine the gap $5 \le v(5) \le 10$ systematically, it is instructive to synthesize the mechanisms and inherent limitations (both rigorous algebraic obstructions and analytic heuristics) of five distinct mathematical disciplines that intersect at equation (1), summarized in Table 1. Note that Table 1 serves as a high-level expository survey compressing technical hypotheses from the cited literature; its analytic density statements (Branches 3 and 4) are heuristic expectations rather than rigorous theorems, whereas the algebraic and p-adic obstructions proved in Sections 2–7 are rigorous.

1.2 Summary of Main Contributions

In the author's interpretation, the classical barrier at $v(5) \le 10$ in polynomial constructions stems from restricting polynomial differencing to monic shifts $(z+\alpha_i)^5 - (z+\beta_i)^5$, which corresponds to

Table 1: Cross-disciplinary survey of mathematical methods and heuristic vs. rigorous bottlenecks applicable to $v(5)$ (expository summary; analytic density expectations in Rows 3–4 are heuristic).

Branch of Mathematics	**Key Contributors**	**Core Mechanism for $k = 5$**	**Structural / Heuristic Bottleneck**
1. Multigrade & Prouhet–Tarry–Escott (PTE) Theory	Euler (1750), Prouhet (1851), Wright [1], Fuchs & Wright [2], Hardy & Wright [3], Borwein & Ingalls [5]	Ideal degree-3 PTE solutions $\{\alpha_i\}_{i=1}^4 \overset{3}{=} \{\beta_i\}_{i=1}^4$ annihilate z^5, z^4, z^3, z^2 in $\sum_{i=1}^4((z+\alpha_i)^5 - (z+\beta_i)^5) = Cz + D$, yielding $v(5) \le 8 + \Delta(5, C) = 10$.	*Rigorous obstruction:* Unit leading weights ($A_i = 1$) force $2n \ge 8$ by Bastien's theorem and $720 \mid C$, $360 \mid D$ by Kleiman–Wright divisibility, trapping $\Delta(5, C) \ge 2$.
2. Diophantine Geometry & Rational Waring Problem	Sastry & Chowla (1934), Richmond (1985), Choudhry [11, 12], Wadhawan [13], Piezas [14, 15]	Simultaneous multigrades $\sum_{i=1}^3 x_i^r = \sum_{i=1}^3 y_i^r$ ($r = 1, 5$) reduce to elliptic fibrations $y^2 = f_4(u)$ on quintic surfaces; every rational $N \in \mathbb{Q}$ is an algebraic sum of 6 rational fifth powers [11].	*Rigorous obstruction:* Clearing denominators in rational Waring identities multiplies z by varying 5-th powers unless z^5, z^4, z^3, z^2 vanish identically over $\mathbb{Q}[z]$.
3. Analytic Number Theory & Circle Method	Hardy & Littlewood (1920), Davenport, Vaughan & Wooley [9]	Smooth-number exponential sums and Vinogradov's Mean Value Theorem yield $G(5) \le 17$ [9]; mixed Waring–Kamke methods treat polynomials $Q(z) + \sum_{j=1}^r y_j^5$.	*Analytic / heuristic limitation:* Allowing negative summands destroys bounded lattice-point counting; univariate degree-d polynomials require the heuristic exponent sum $\frac{1}{d} + \frac{r}{5} \ge 1$ for positive expected density.
4. Arithmetic Geometry, *abc* Bounds & Modular Curves	Brownawell & Masser (1986), Liu & Wooley [16], MathOverflow [20], Das, Dey, Koutsianas & Tzanakis [10], Vavilov [6]	Function-field *abc* bounds [16, 20] constrain low-term polynomial identities; Frey–Hellegouarch curves and modular level-lowering rule out arithmetic-progression families of $x_1^5 + x_2^5 + x_3^5 = dw^5$ [10, Theorem 1.1].	*Heuristic vs. rigorous gap:* At the critical exponent $s = k = 5$, heuristic counting density is logarithmic and *abc* heuristics do not rule out $v(5) = 5$; ternary 5-th power equations govern 7-term odd linear identities.
5. Finite Fields & p-adic Valuation Theory	Cauchy–Davenport, Gauss–Jacobi, Weil (1948), Fuchs & Wright [2]	Jacobi sum bounds show $\Delta(5, p) \le 2$ for all primes $p \notin \{11, 31, 41, 61\}$; Hensel's lemma lifts unit 5-th powers mod p ($p \ne 5$) to mod p^e for all $e \ge 1$.	*Rigorous obstruction:* Zero residue $0 \pmod p$ lifts only to $0 \pmod{p^5}$, creating p^2-obstructions whenever $p^2 \mid C$ and $N \equiv D + pk \pmod{p^2}$ with $p \nmid k$.

commuting translation operators $E^{\alpha_i} : z \mapsto z + \alpha_i$. By adjoining dilation operators $S_a : z \mapsto az$ to form weighted linear forms $L_i(z) = A_i z + B_i$ ($A_i \in \mathbb{Z}_{>0}$, $B_i \in \mathbb{Z}$), we obtain the following principal theorem-level results (note that the proof of $v(5) \leq 9$ in Section 5 requires only Identity (38) and Lemma 5.2, while Sections 3, 4, 6, and 7 establish independent geometric classifications, structural obstructions, and multi-tier sieves):

1. **Upper Bound $v(5) \leq 9$ (Section 5):** We establish a root-free rational parameterization and a direct integer gap-sum solver for 8-term weighted linear identities with weights $(2,2,1,1,1,1,1,1)$. Weighting the first pair by $a_1 = 2$ introduces $2^5 = 32$ in the denominator of the rational gap $d_1 = X_1/2^5$, absorbing the $2^4 \cdot 3^2$ factor that forces $720 \mid C$ in classical PTE identities. This yields an explicit 8-summand identity (38) (involving 7 distinct linear forms with $\max|B_i| = 39$) whose slope $C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ (as well as a second parametric identity (40) with $C_{\mathrm{II}} = 7,327,885,530 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17 \cdot 2052629$) satisfies $\Delta(5, C) = 1$, proving constructively that $5 \leq v(5) \leq 9$.

2. **Six-Term Linear Polynomial Identities via Elliptic Fibrations (Section 3):** We prove an explicit algebraic-geometric classification (Theorem 3.2) for $2k$-term paired linear identities with explicit forward and inverse birational maps on the open non-degenerate locus. For $k = 3$ (6 fifth powers), each fixed ordered triple of distinct positive integer weights $a_1 > a_2 > a_3 > 0$ defines an affine sextic curve whose projective completion in weighted projective space $\mathbb{P}(1,1,3)$ has non-vanishing fifth-power Heron-type discriminant $\Delta_t(F) \neq 0$ by Fermat's Last Theorem for exponent 5, and is therefore a smooth hyperelliptic curve $\mathcal{C}_{a_1,a_2,a_3}$ of genus $g = 2$, having only finitely many rational points by Faltings's theorem. When $a_2 = a_3 = 1$ and $a_1 = a > 1$, a nodal singularity at $p = 0$ drops the geometric genus to $g = 1$, yielding the family of smooth elliptic curves $E_a : Y^2 = 3(p+1)\left[(a^{10}-1)p^2 + 3a^{10}p + 3a^{10}\right]$. Using a denominator-clearing and dihedral-torsion fiber classification lemma (Lemma 3.3, which proves that the map from $E_a(\mathbb{Q}) \setminus \mathcal{E}_a$ is 4-to-1 onto affine-equivalence classes of paired 6-term identities via $\langle[-1], +2P_0\rangle$ and at most 8-to-1 onto unordered 6-term identities via $\langle[-1], +P_0\rangle$ after accounting for both perfect matchings of the four unit-weight terms) together with points on $E_2(\mathbb{Q})$ and $E_6(\mathbb{Q})$ certified to have infinite order via explicit integral translation to short Weierstrass models and the Nagell–Lutz theorem, we construct infinitely many primitive, projectively inequivalent 6-term linear polynomial identities in $\mathbb{Z}[z]$, breaking Bastien's 8-term barrier over $\mathbb{Q}$. For $k = 4$, we construct an explicit 2-dimensional open rational subvariety $\mathcal{S} \subset \mathcal{M}(a_1, a_2, a_3, a_4)$ over $\mathbb{Q}$ and verify directly that the linear slope C has leading coefficient $10X_2(X_1^2 - X_2^2)/X_1^2 \neq 0$ in u_2^3 and hence does not vanish identically on $\mathcal{S}$.

3. **Structural Obstruction and Reduction Theorems (Section 4):** For non-degenerate linear identities $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ in $\mathbb{Z}[z]$ with $A_i \in \mathbb{Z}_{>0}$, we establish:

   - *The $\mathbb{F}_{11}$-Bastien Obstruction and 11-adic Rank-Defect Theorem (Theorem 4.1):* Every linear identity of length $m \leq 7$ in $\mathbb{Z}[z]$ satisfies $11 \mid C$, and if $2k = |\{i : 11 \nmid A_i\}|$ ($0 \leq k \leq 3$), the remaining $r = m - 2k$ terms with $11 \mid A_i$ contribute a fixed constant $D \equiv \sum_{i \in S_0} s_i B_i^5 \in \{-r, \ldots, r\} \pmod{11}$ for a single identity (and vary across $\{-r, \ldots, r\} \pmod{11}$ over a family of identities), forcing $s \geq m + 5 \geq 10$ total fifth-power summands to cover $\mathbb{Z}/11\mathbb{Z}$ via a single identity $P_m(z) + \sum_{j=1}^{s-m} y_j^5$ (explaining why $m = 8$ and $s = 8 + 1 = 9$ are optimal for the single-identity scheme) and $s \geq \max(m, 2k+5)$ across a family with varying $11 \mid A_i$ constants (giving $s \geq 9$ whenever $k \geq 2$, and $s \geq 11$ when $k = 3$).
   - *The 5-adic Vandermonde Obstruction for $m \leq 6$ (Proposition 4.2):* By the invertibility of the 5×5 Vandermonde matrix over $\mathbb{F}_5$ and 5-adic lifting to $(\mathbb{Z}/25\mathbb{Z})^\times$, every linear identity with $m \leq 6$ satisfies $25 \mid C$, whereas this specific modulo-25 congruence obstruction vanishes at $m = 7$ (via 3 singletons $+1$ and 2 doubletons $\{-2, -2\}$) and exact valuation $v_5(C) = 1$ ($5 \parallel C$, i.e., $5 \mid C$ and $25 \nmid C$) is achieved globally in $\mathbb{Z}[z]$ across 7 distinct linear forms at $m = 8$.

- *The Universal* $30 \mid C$ *and Fermat–Bastien Congruence Law for* $D \pmod{30}$ *(Theorem 4.3):* Every linear identity in $\mathbb{Z}[z]$ satisfies $30 \mid C$ and $D \equiv \sum_{p \mid A_i} s_i B_i \pmod{p}$ for each prime $p \in \{2,3,5\}$. Whenever $\gcd(A_i, 15) = 1$ for all i (including all unit-weight and 2-power-weighted identities), $15 \mid D$ holds universally, whereas $D \not\equiv 0 \pmod{p}$ is achievable at $m = 8$ whenever $p \mid A_1 = A_2$.
- *The Cauchy–Schwarz Signature Obstruction (Theorem 4.4) and 7-Term Odd-Symmetric Linear Reduction (Theorem 4.6):* No non-degenerate $(m-1,1)$-signature fifth-power zero-sum (such as the Lander–Parkin–Selfridge solution $27^5 + 84^5 + 110^5 + 133^5 - 144^5 = 0$) can have vanishing z^5, z^4, z^3 coefficients over $\mathbb{R}$ (since vanishing of $[z^4]$ and $[z^3]$ forces $b_1/a_1 = \cdots = b_m/a_m$, causing the polynomial to vanish identically), and any 7-term linear identity within the odd-symmetric ansatz reduces to $s_1c_1^5 + s_2c_2^5 + s_3c_3^5 = 16d_0^5$, which in the split-conic subfamily $c_3 = c_1 + c_2$ ($s_1 = s_2 = +1, s_3 = -1$) satisfies $(-d_1)^5 = 2\big[(c_1+c_2)^5 - c_1^5 - c_2^5\big] = 10c_1c_2(c_1+c_2)(c_1^2 + c_1c_2 + c_2^2)$, is ruled out modulo 25 when 2 and 5 divide distinct factors of $\{c_1, c_2, c_1 + c_2\}$, and reduces to the simultaneous system $x^5 + y^5 + 10000z^5 = 0$, $x^{10} + x^5y^5 + y^{10} = w^5$ when 10 divides a single factor.

4. **Catalogue of Polynomial Identities and Structural Minimality Results (Section 6 & Appendix A):** We establish the 3-variable Cauchy–Lamé master quadratic identity, prove that $(z+2)^5 - (z-2)^5 - 2(z+1)^5 + 2(z-1)^5 = 120z^2 + 60$ achieves the minimum positive quadratic coefficient $E = 120$ within the class of origin-symmetric unit-weight fifth-power identities, derive the Pythagorean identity $20a^2b^2z$ and the half-integer shifted $u^2 + v^2 = 2w^2$ identity achieving the minimum positive slope $C = 720$ within the class of unit-weight degree-3, size-4 PTE linear identities, and compile 20 listed verified polynomial identities (16 independent up to affine equivalence or specialization) across degrees 1 to 5 in Table 2 (explicitly marking which identities are used in the proof of $v(5) \le 9$ and the Three-Tier Sieve) with complete term-by-term coefficient certificates and a self-contained Python verification script in Appendix A.

5. **Local-to-Global Analysis, 11^5-Rigidity, and Multi-Tier Sieves (Section 7):** We determine $\Delta(5, p^e)$ for all prime powers p^e via explicit 5-adic lifting (with a complete 25-row modulo-25 unit-first-summand table), finite-field coset evaluation, Gauss–Jacobi character-sum bounds (with full character-indicator expansion), and nested closed cylinder preimages in the compact Hausdorff space $\mathbb{Z}_p^5$, prove the absence of local obstructions at $s = 5$ (noting that local solvability in all $\mathbb{Z}_p$ does not imply a global Hasse principle over $\mathbb{Z}$), formulate the 11^5-rigidity congruence for $N \equiv \pm 4 \pmod{11}$, and construct a three-tier multi-identity sieve in which Tier 1 ($C = 720$) covers $\frac{7}{16} = 43.75\%$ of all integers (reducing $|z| \approx |N|/C$ by a factor of 5,339.2), Tier 2 ($C = 159,930$, standalone coverage $\frac{7}{9} \approx 77.78\%$) covers an additional $\frac{29}{72} \approx 40.28\%$ of integers missed by Tier 1 (reducing the scale by 24.04, for a combined Tier 1 $\cup$ Tier 2 coverage of $\frac{121}{144} \approx 84.03\%$), and Tier 3 ($C_{\mathrm{I}} = 3,844,230$) covers the exact remaining $\frac{23}{144} \approx 15.97\%$ of all integers, alongside positive-density congruence reductions for $s = 8, 7, 6$ and an explicitly labeled heuristic mixed-degree Waring–Kamke reduction strategy.

2 Diagnosis of the Classical $v(5) \le 10$ Bound and Its Barriers

Let $\{\alpha_1, \ldots, \alpha_n\}$ and $\{\beta_1, \ldots, \beta_n\}$ be two disjoint multisets of integers. In the Prouhet–Tarry–Escott (PTE) problem [5, 17], we write $\{\alpha_1, \ldots, \alpha_n\} \overset{d}{=} \{\beta_1, \ldots, \beta_n\}$ if

$$\sum_{i=1}^{n} \alpha_i^r = \sum_{i=1}^{n} \beta_i^r \qquad \text{for } r = 1, 2, \ldots, d. \tag{3}$$

Lemma 2.1 (Bastien's Theorem [5, Proposition 2])**.** *Let K be any field of characteristic 0 or characteristic $p > d$. If $\{\alpha_1, \ldots, \alpha_n\}, \{\beta_1, \ldots, \beta_n\} \subset K$ satisfy $\sum_{i=1}^n \alpha_i^r = \sum_{i=1}^n \beta_i^r$ for $r = 1, \ldots, d$ and $\{\alpha_i\} \neq \{\beta_i\}$ as multisets, then $n \ge d + 1$.*

Proof. Suppose $n \le d$. Since $n! \neq 0$ in K, Newton's symmetric power identities uniquely express the elementary symmetric polynomials $e_1, \dots, e_n$ in terms of the power sums $p_1, \dots, p_n$. Consequently, $\prod_{i=1}^{n}(X - \alpha_i) = \prod_{i=1}^{n}(X - \beta_i)$ in $K[X]$, so $\{\alpha_1, \dots, \alpha_n\} = \{\beta_1, \dots, \beta_n\}$ as multisets. □

For fifth powers ($k = 5$), the binomial expansion of $\sum_{i=1}^{n}(z + \alpha_i)^5 - \sum_{i=1}^{n}(z + \beta_i)^5$ has vanishing z^5, z^4, z^3, z^2 coefficients if and only if $\{\alpha_i\} \overset{3}{=} \{\beta_i\}$ ($d = 3$). By Lemma 2.1, an ideal solution requires $n = 3 + 1 = 4$, consuming $2n = 8$ fifth powers:

$$P_{\mathrm{PTE}}(z) = \sum_{i=1}^{4}(z + \alpha_i)^5 - \sum_{i=1}^{4}(z + \beta_i)^5 = Cz + D, \tag{4}$$

where $C = 5\sum_{i=1}^{4}(\alpha_i^4 - \beta_i^4)$ and $D = \sum_{i=1}^{4}(\alpha_i^5 - \beta_i^5)$. For any integer $C \ge 1$, define the modular fifth-power covering number

$$\Delta(5, C) = \min\left\{ r \ge 1 : \forall a \in \mathbb{Z}/C\mathbb{Z},\ \exists y_1, \dots, y_r \in \mathbb{Z} \text{ such that } \sum_{j=1}^{r} y_j^5 \equiv a \pmod{C} \right\}. \tag{5}$$

If $P_{\mathrm{PTE}}(z) = Cz + D$ with $C > 0$, then for every $N \in \mathbb{Z}$ there exist $y_1, \dots, y_{\Delta(5,C)} \in \mathbb{Z}$ such that $N - D \equiv \sum_{j=1}^{\Delta(5,C)} y_j^5 \pmod{C}$, whence $z = (N - D - \sum y_j^5)/C \in \mathbb{Z}$ expresses N as a sum of $8 + \Delta(5, C)$ fifth powers [5, Lemma 5].

Proposition 2.2 (Universal Divisibility Obstruction for Unit-Weight PTE Identities)**.** *For every non-trivial degree-3, size-4 integer PTE solution $\{\alpha_1, \dots, \alpha_4\} \overset{3}{=} \{\beta_1, \dots, \beta_4\}$ in (4), the integers C and D satisfy*

$$720 \mid C, \qquad 360 \mid D, \qquad \text{and} \qquad \frac{C}{720} \equiv \frac{D}{360} \pmod{2}. \tag{6}$$

Consequently, $\Delta(5, C) \ge \max(\Delta(5, 16), \Delta(5, 9)) = 2$, and no finite collection of unit-weight PTE identities $\{(C_j, D_j)\}_{j=1}^{K}$ can cover $\mathbb{Z}$ with a single additional fifth power y_1^5.

Proof. Let $\Delta_r = \sum_{i=1}^{4}(\alpha_i^r - \beta_i^r)$, so that $\Delta_1 = \Delta_2 = \Delta_3 = 0$, $C = 5\Delta_4$, and $D = \Delta_5$.

1. *Newton's Identities for Δ_4 and Δ_5:* Let $P(X) = \prod_{i=1}^{4}(X - \alpha_i) = X^4 - e_1(\alpha)X^3 + e_2(\alpha)X^2 - e_3(\alpha)X + e_4(\alpha)$ and $Q(X) = \prod_{i=1}^{4}(X - \beta_i)$. Since $p_r(\alpha) = p_r(\beta)$ in $\mathbb{Z}$ for $r = 1, 2, 3$ (and $2! = 2, 3! = 6$ are non-zero in $\mathbb{Q}$), Newton's identities over $\mathbb{Q}$ imply the exact integer equalities $e_r(\alpha) = e_r(\beta) =: e_r \in \mathbb{Z}$ for $r = 1, 2, 3$. For $r = 4$ and $r = 5$, Newton's relations

   $$p_4 - e_1 p_3 + e_2 p_2 - e_3 p_1 + 4e_4 = 0, \qquad p_5 - e_1 p_4 + e_2 p_3 - e_3 p_2 + e_4 p_1 = 0,$$

   subtracted between α and β, yield $e_4(\alpha) - e_4(\beta) = -\frac{1}{4}\Delta_4$ and

   $$D = \Delta_5 = e_1\Delta_4 - \big(e_4(\alpha) - e_4(\beta)\big)e_1 = \frac{5}{4}e_1\Delta_4 = \frac{1}{4}e_1 C. \tag{7}$$

   Moreover, $P(X) - Q(X) = e_4(\alpha) - e_4(\beta) = -\frac{1}{4}\Delta_4$ is a constant in $\mathbb{Z}[X]$. Evaluating at $X = \alpha_i$ (where $P(\alpha_i) = 0$) gives the exact product formula

   $$\frac{\Delta_4}{4} = Q(\alpha_i) = \prod_{j=1}^{4}(\alpha_i - \beta_j) \qquad \text{for every } i \in \{1, 2, 3, 4\}. \tag{8}$$

2. *Self-Contained Proof of $5 \mid C$ and $9 \mid \Delta_4$:* Immediately $5 \mid C = 5\Delta_4$. For the 3-adic valuation $9 \mid \Delta_4$ (established in Kleiman and Wright [5, Corollary 3 and Proposition 7]), note that although the modulo-3 power sums $p_1, p_2, p_3 \pmod 3$ alone do not determine a 4-element residue multiset in $\mathbb{F}_3$ (since $x^3 \equiv x \pmod 3$, e.g. $\{1, 1, 1, 2\}$ and $\{0, 0, 0, 2\}$ share $p_1 \equiv p_3 \equiv 2$ and $p_2 \equiv 1$

$\pmod 3$), the integer equalities $e_1(\alpha) = e_1(\beta)$, $e_2(\alpha) = e_2(\beta)$, and $e_3(\alpha) = e_3(\beta)$ proved in Step 1 (obtained by dividing $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$ by 6 in $\mathbb{Z}$ before reducing modulo 3) imply that the modulo-3 reductions $\bar{P}(X) = \prod_{i=1}^4 (X - \bar{\alpha}_i)$ and $\bar{Q}(X) = \prod_{j=1}^4 (X - \bar{\beta}_j)$ in $\mathbb{F}_3[X]$ share the same X^4, X^3, X^2, X^1 coefficients and satisfy $\bar{P}(X) - \bar{Q}(X) = c \in \mathbb{F}_3$, where $c \equiv -\Delta_4/4 \pmod 3$. If $c \neq 0$ in $\mathbb{F}_3$, then $\bar{P}(x) - \bar{Q}(x) = c \neq 0$ for all $x \in \mathbb{F}_3$, so the root sets $S_\alpha = \{\bar{\alpha}_i\}$ and $S_\beta = \{\bar{\beta}_j\}$ are disjoint non-empty subsets of $\mathbb{F}_3$. Since $|\mathbb{F}_3| = 3$, at least one root set—say S_α—is a singleton $\{r\}$, so $\bar{P}(X) = (X - r)^4$. Then every root $\bar{\beta}_j \in \mathbb{F}_3 \setminus \{r\}$ of $\bar{Q}$ satisfies $(\bar{\beta}_j - r)^2 = 1$ in $\mathbb{F}_3$ and hence $0 = \bar{Q}(\bar{\beta}_j) = (\bar{\beta}_j - r)^4 - c = 1 - c$, forcing $c = 1$ and $\bar{Q}(X) = (X - r)^4 - 1 = ((X - r)^2 - 1)((X - r)^2 + 1)$; since $(X - r)^2 + 1$ has no roots in $\mathbb{F}_3$, $\bar{Q}(X)$ cannot split into 4 linear factors over $\mathbb{F}_3$, a contradiction (equivalently, across all $\binom{3+4-1}{4} = 15$ four-element multisets in $\mathbb{F}_3$, the triple $(e_1, e_2, e_3) \in \mathbb{F}_3^3$ takes 15 distinct values and uniquely determines the residue multiset modulo 3). Thus $c = 0$ and $\bar{P}(X) = \bar{Q}(X)$ in $\mathbb{F}_3[X]$, so α and β have the exact same 4-element residue multiset modulo 3. Since there are 4 elements in each multiset and only 3 residue classes modulo 3, the pigeonhole principle ensures that some residue class modulo 3 occurs at least twice in both α and β. Choosing α_i in that repeated residue class gives at least two factors $\alpha_i - \beta_j$ in (8) that are divisible by 3, whence $3^2 = 9 \mid (\Delta_4/4)$ and $9 \mid \Delta_4$.

3. *Explicit 2-adic Parity Analysis for* $16 \mid \Delta_4$, $720 \mid C$, $360 \mid D$, *and* $(C/720) \equiv (D/360) \pmod 2$*:* Let $r_\alpha, r_\beta \in \{0, 1, 2, 3, 4\}$ denote the number of odd integers in α and β, respectively. Since $x^2 \equiv 1 \pmod 4$ for odd x and $x^2 \equiv 0 \pmod 4$ for even x, the relation $\Delta_2 = 0 \implies \sum \alpha_i^2 \equiv \sum \beta_i^2 \pmod 4$ gives $r_\alpha \equiv r_\beta \pmod 4$, so either $r_\alpha = r_\beta$ or $\{r_\alpha, r_\beta\} = \{4, 0\}$. To exclude $(r_\alpha, r_\beta) = (4, 0)$ (and symmetrically $(0, 4)$), suppose $\alpha_1, \ldots, \alpha_4$ are all odd and $\beta_j = 2b_j$ ($b_j \in \mathbb{Z}$) are all even. Reducing $\Delta_3 = 0$ modulo 8, since $a^3 \equiv a \pmod 8$ for every odd integer a and $(2b_j)^3 = 8b_j^3 \equiv 0 \pmod 8$, we obtain $\sum_{i=1}^4 \alpha_i \equiv \sum_{i=1}^4 \alpha_i^3 = \sum_{j=1}^4 (2b_j)^3 \equiv 0 \pmod 8$. Combining this with $\Delta_1 = 0$ ($\sum \alpha_i = 2 \sum b_j$) gives $\sum_{j=1}^4 b_j = \frac{1}{2} \sum_{i=1}^4 \alpha_i \equiv 0 \pmod 4$. On the other hand, reducing $\Delta_2 = 0$ modulo 8, since $\alpha_i^2 \equiv 1 \pmod 8$ for every odd α_i, we have $4 \equiv \sum_{i=1}^4 \alpha_i^2 = \sum_{j=1}^4 (2b_j)^2 = 4 \sum_{j=1}^4 b_j^2 \pmod 8$, so $\sum_{j=1}^4 b_j^2$ is odd. Because $b_j^2 \equiv b_j \pmod 2$, $\sum_{j=1}^4 b_j \equiv \sum_{j=1}^4 b_j^2 \equiv 1 \pmod 2$ is odd, contradicting $\sum_{j=1}^4 b_j \equiv 0 \pmod 4$. Thus α and β contain the exact same number $r = r_\alpha = r_\beta \in \{0, 1, 2, 3, 4\}$ of odd integers:

   - *Case A ($r \in \{1, 3\}$, so e_1 is odd):* Shifting all elements by $+1$ if $r = 3$ (which preserves all pairwise differences $\alpha_i - \beta_j$ and shifts e_1 by 4) reduces to $r = 1$, where $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3$ are even and α_4, β_4 are odd. Since $(2m)^3 = 8m^3 \equiv 0 \pmod 8$, the cubic relation $\Delta_3 = \sum_{i=1}^4 \alpha_i^3 - \sum_{j=1}^4 \beta_j^3 = 0$ reduced modulo 8 yields

     $$\alpha_4^3 \equiv \beta_4^3 \pmod 8.$$

     Because the cube map $x \mapsto x^3 \pmod 8$ is the identity on odd residues ($\{1^3, 3^3, 5^3, 7^3\} \equiv \{1, 3, 5, 7\} \pmod 8$), it follows that $\alpha_4 \equiv \beta_4 \pmod 8$, so $8 \mid (\alpha_4 - \beta_4)$. Subtracting $\alpha_4 - \beta_4 \equiv 0 \pmod 8$ from $\Delta_1 = \sum_{i=1}^3 \alpha_i - \sum_{j=1}^3 \beta_j + (\alpha_4 - \beta_4) = 0$ gives

     $$\sum_{i=1}^3 \alpha_i \equiv \sum_{j=1}^3 \beta_j \pmod 8, \qquad \text{so} \qquad \sum_{i=1}^3 \alpha_i - \sum_{j=1}^3 \beta_j \equiv 0 \pmod 8.$$

     Taking $i \in \{1, 2, 3\}$, if no pair (α_i, β_j) with $i, j \in \{1, 2, 3\}$ satisfied $\alpha_i \equiv \beta_j \pmod 4$, then all three even integers $\alpha_1, \alpha_2, \alpha_3$ would lie in one even residue class $c \in \{0, 2\} \pmod 4$ and all three even integers $\beta_1, \beta_2, \beta_3$ would lie in the opposite even class $c + 2 \pmod 4$; this would yield $\sum_{i=1}^3 \alpha_i - \sum_{j=1}^3 \beta_j \equiv 3 \cdot 2 \equiv 2 \not\equiv 0 \pmod 4$, contradicting $\sum_{i=1}^3 \alpha_i \equiv \sum_{j=1}^3 \beta_j \pmod 8$ (and hence modulo 4). Hence there exist $i \in \{1, 2, 3\}$ and $j_0 \in \{1, 2, 3\}$ with $4 \mid (\alpha_i - \beta_{j_0})$, while the other two even differences $\alpha_i - \beta_j$ ($j \in \{1, 2, 3\} \setminus \{j_0\}$) are divisible by 2. Thus $4 \cdot 2 \cdot 2 = 16 \mid \prod_{j=1}^4 (\alpha_i - \beta_j) = \Delta_4/4$, giving $64 \mid \Delta_4$.

- *Case B ($r \in \{0,4\}$, so e_1 is even):* Shifting by $+1$ if $r = 4$ reduces to $r = 0$ (all α_i, β_j even), where all four factors $\alpha_1 - \beta_j$ are even, giving $2^4 = 16 \mid (\Delta_4/4)$ and hence $64 \mid \Delta_4$.
- *Case C ($r = 2$, so e_1 is even):* Here $\alpha_1, \alpha_2, \beta_1, \beta_2$ are even and $\alpha_3, \alpha_4, \beta_3, \beta_4$ are odd. Taking $i = 1$ in (8), both $\alpha_1 - \beta_1$ and $\alpha_1 - \beta_2$ are even, so $2 \cdot 2 = 4 \mid \prod_{j=1}^{4}(\alpha_1 - \beta_j) = \Delta_4/4$, which proves $16 \mid \Delta_4$ in all cases (re-proving the Kleiman–Wright divisibility $144 = 16 \cdot 9 \mid \Delta_4$ and $720 = 5 \cdot 144 \mid C$ [5, Corollary 3 and Proposition 7]). Writing $k = \Delta_4/144 \in \mathbb{Z}$, we have
$$\frac{C}{720} = k \qquad \text{and, by (7),} \qquad D = 180 e_1 k \iff \frac{D}{360} = \frac{e_1 k}{2}.$$
In Cases A and B, $64 \mid \Delta_4 \implies 4 \mid k$, so $C/720 = k$ and $D/360 = e_1(k/2)$ are both even integers, giving $(C/720) \equiv (D/360) \equiv 0 \pmod 2$. In Case C, $2 \mid e_1$ implies $D/360 = (e_1/2)k \in \mathbb{Z}$, and if k is even then both $C/720$ and $D/360$ are even. If k is odd ($16 \parallel \Delta_4$, so $4 \parallel \prod_{j=1}^{4}(\alpha_i - \beta_j)$ for every i), then for $i \in \{1,2\}$ neither even factor $\alpha_i - \beta_1, \alpha_i - \beta_2$ can be divisible by 4, and for $i \in \{3,4\}$ neither even factor $\alpha_i - \beta_3, \alpha_i - \beta_4$ can be divisible by 4. This forces $\alpha_1 \equiv \alpha_2 \not\equiv \beta_1 \equiv \beta_2 \pmod 4$ and $\alpha_3 \equiv \alpha_4 \not\equiv \beta_3 \equiv \beta_4 \pmod 4$. Since any two equal even residues mod 4 sum to $0 \pmod 4$ and any two equal odd residues mod 4 sum to $2 \pmod 4$, we have $e_1 = (\alpha_1 + \alpha_2) + (\alpha_3 + \alpha_4) \equiv 0 + 2 = 2 \pmod 4$. Thus $e_1/2$ is odd, so $D/360 = (e_1/2)k \equiv k = C/720 \equiv 1 \pmod 2$.

4. *Covering Failure for $r = 1$:* Modulo $720 = 16 \cdot 9 \cdot 5$, every unit-weight PTE solution has $D_j \equiv 0 \pmod 9$ and $D_j \in \{0,8\} \pmod{16}$. Since $\{y^5 \bmod 9\} = \{0, \pm 1, \pm 2, \pm 4\}$ (7 residues, omitting $\pm 3 \pmod 9$) and $\{d + y^5 \bmod 16 : d \in \{0,8\}\} = \{\text{odd residues mod } 16\} \cup \{0,8\}$ (10 residues, omitting $\{2,4,6,10,12,14\} \pmod{16}$), the union over all unit-weight PTE solutions of $\{D_j + y^5 \bmod 720\}$ covers at most $5 \times 7 \times 10 = 350$ out of 720 residue classes. □

3 Weighted Linear Identities and the Genus Classification

To overcome Proposition 2.2, we replace unit shifts $z + \alpha_i$ by general integer linear forms $L_i(z) = A_i z + B_i$ ($A_i \in \mathbb{Z}_{>0}, B_i \in \mathbb{Z}$) and study polynomial identities of the form
$$P_m(z) = \sum_{i=1}^{m} s_i (A_i z + B_i)^5 = Cz + D, \qquad s_i \in \{+1, -1\}, \quad A_i \in \mathbb{Z}_{>0}, \quad C \neq 0. \tag{9}$$

3.1 Vandermonde Nullspace and Paired Weighted Formulations

Writing $A_i z + B_i = A_i(z + t_i)$ with $t_i = B_i/A_i \in \mathbb{Q}$ and $w_i = s_i A_i^5 \in \mathbb{Z} \setminus \{0\}$, the vanishing of the coefficients of z^5, z^4, z^3, z^2 in (9) is equivalent to the weighted moment system
$$\sum_{i=1}^{m} w_i t_i^r = 0 \quad (r = 0,1,2,3), \qquad C = 5\sum_{i=1}^{m} w_i t_i^4 \neq 0, \qquad D = \sum_{i=1}^{m} w_i t_i^5. \tag{10}$$

Lemma 3.1 (Vandermonde Lower Bound)**.** *Any solution to (10) over a field of characteristic 0 requires at least $m \geq 5$ distinct shifts t_i. Moreover, if the terms are grouped into k opposite-sign pairs with equal leading weights,*
$$P_{2k}(z) = \sum_{i=1}^{k} a_i^5 \Big[(z + u_i + d_i)^5 - (z + u_i - d_i)^5 \Big], \qquad a_i \in \mathbb{Z}_{>0}, \quad u_i, d_i \in \mathbb{Q}, \tag{11}$$
then z^5 vanishes automatically, and setting $X_i = a_i^5 d_i \neq 0$, the vanishing of z^4, z^3, z^2 is equivalent to
$$\sum_{i=1}^{k} X_i = 0, \qquad \sum_{i=1}^{k} X_i u_i = 0, \qquad 3\sum_{i=1}^{k} X_i u_i^2 + \sum_{i=1}^{k} \frac{X_i^3}{a_i^{10}} = 0. \tag{12}$$

Proof. Suppose a solution to (10) involves $s \leq 4$ distinct shifts $\tilde{t}_1, \ldots, \tilde{t}_s$. Grouping terms with identical shifts by setting $\tilde{w}_j = \sum_{t_i = \tilde{t}_j} w_i$ yields $\sum_{j=1}^{s} \tilde{w}_j \tilde{t}_j^r = 0$ for $0 \leq r \leq 3$ and $C = 5\sum_{j=1}^{s} \tilde{w}_j \tilde{t}_j^4$. Since $\tilde{t}_1, \ldots, \tilde{t}_s$ are distinct and $s \leq 4$, the $4 \times s$ Vandermonde matrix $(\tilde{t}_j^r)_{0 \leq r \leq 3, 1 \leq j \leq s}$ has full column rank s, forcing $\tilde{w}_1 = \cdots = \tilde{w}_s = 0$ and hence $C = 0$, a contradiction. Thus at least $s \geq 5$ distinct shifts (and hence $m \geq s \geq 5$ terms) are required.

For the paired ansatz (11), expanding $(z + u_i + d_i)^5 - (z + u_i - d_i)^5 = 2d_i\big(5(z + u_i)^4 + 10d_i^2(z + u_i)^2 + d_i^4\big)$ and collecting powers of z yields:

$$
\begin{aligned}
[z^4]\, P_{2k}(z) &= 10 \sum_{i=1}^{k} a_i^5 d_i = 10 \sum_{i=1}^{k} X_i, \\
[z^3]\, P_{2k}(z) &= 40 \sum_{i=1}^{k} a_i^5 d_i u_i = 40 \sum_{i=1}^{k} X_i u_i, \\
[z^2]\, P_{2k}(z) &= 20 \sum_{i=1}^{k} a_i^5 d_i (3u_i^2 + d_i^2) = 20 \left(3 \sum_{i=1}^{k} X_i u_i^2 + \sum_{i=1}^{k} \frac{X_i^3}{a_i^{10}} \right),
\end{aligned}
$$

using $d_i^2 = X_i^2 / a_i^{10}$. □

3.2 The Genus Classification Theorem for $2k$-Term Paired Identities

Theorem 3.2 (Algebraic Geometry Classification of Paired Fifth-Power Identities)**.** *Let $\mathcal{M}(a_1, \ldots, a_k)$ denote the open projective moduli locus of non-degenerate solutions $(u_1, \ldots, u_k, d_1, \ldots, d_k)$ (modulo translation $u_i \mapsto u_i + \tau$ and simultaneous scaling $(u_i, d_i) \mapsto (\lambda u_i, \lambda d_i)$ for $\lambda \in \mathbb{Q}^\times$, restricted to the Zariski-open locus where $d_i \neq 0$, the centers u_i are not all equal, and $C \neq 0$) to the paired system (12):*

1. Case $k = 2$ (4 fifth powers): $\mathcal{M}(a_1, a_2) = \emptyset$ (dim $= -1$)*. Removing the z^2-vanishing condition yields a 0-dimensional family of pure quadratic identities $P_4(z) = Ez^2 + D$ parameterized by $a_1^4 B_1 = a_2^4 B_2$.*

2. Case $k = 3$ (6 fifth powers) with fixed distinct integer weights $a_1 > a_2 > a_3 > 0$: *$\mathcal{M}(a_1, a_2, a_3)$ is birationally equivalent (via explicit rational maps defined outside the branch and degenerate locus $pq(p + q)F(p, q) = 0$) to the smooth projective hyperelliptic curve $\mathcal{C}_{a_1,a_2,a_3}$ of genus $g = 2$ in weighted projective space $\mathbb{P}(1, 1, 3)$ with coordinates $[p : q : y]$ of weights $(1, 1, 3)$, defined by the weighted-homogeneous sextic equation:*

$$
\mathcal{C}_{a_1,a_2,a_3} : \quad y^2 = -3pq(p + q)\Big[a_2^{10} a_3^{10} p^3 + a_1^{10} a_3^{10} q^3 - a_1^{10} a_2^{10} (p + q)^3\Big]. \tag{13}
$$

   By Fermat's Last Theorem for exponent 5, the cubic factor $F(p, q)$ has non-zero discriminant and three distinct roots in $\mathbb{P}^1(\mathbb{C})$ coprime to $pq(p + q)$, so $\mathcal{C}_{a_1,a_2,a_3}$ is smooth of genus $g = 2$. By Faltings's Theorem (Mordell's Conjecture) applied to $\mathcal{C}_{a_1,a_2,a_3}$, the rational point set $\mathcal{C}_{a_1,a_2,a_3}(\mathbb{Q})$ (and hence $\mathcal{M}(a_1, a_2, a_3)(\mathbb{Q})$) is finite for every fixed ordered triple of distinct positive integers $a_1 > a_2 > a_3$.

3. Case $k = 3$ (6 fifth powers) with two equal weights $a_1 = a > 1$ and $a_2 = a_3 = 1$: *The curve $\mathcal{C}_{a,1,1}$ acquires a nodal singularity at $p = 0$, factoring out p^2 and normalizing in the affine chart $q = 1$ to the smooth projective curve of genus $g = 1$ (an elliptic curve over $\mathbb{Q}$):*

$$
E_a : \quad Y^2 = 3(p + 1)\Big[(a^{10} - 1)p^2 + 3a^{10} p + 3a^{10}\Big], \tag{14}
$$

   with universal rational 4-torsion points $P_0 = (0, 3a^5)$, $3P_0 = -P_0 = (0, -3a^5)$, and $2P_0 = (-1, 0)$ (which lie in the degenerate locus $p(p + 1) = 0$). Whenever rank$(E_a(\mathbb{Q})) \geq 1$ *(in particular for $a = 2$ and $a = 6$, certified in Theorem 3.4), there exist infinitely many primitive, translation- and scaling-inequivalent 6-term linear polynomial identities in $\mathbb{Z}[z]$.*

4. Case $k = 4$ (8 fifth powers, open rational subvariety): *For any weight vector $(a_1, a_2, a_3, a_4) \in \mathbb{Z}^4_{>0}$, the moduli locus $\mathcal{M}(a_1, a_2, a_3, a_4)$ contains an explicit 2-dimensional open rational subvariety $\mathcal{S} \subset \mathcal{M}(a_1, a_2, a_3, a_4)$ (defined on the open slice $X_1X_2u_2 \neq 0$, $X_1 \neq \pm X_2$, with $X_3 = -X_1, X_4 = -X_2$) on which $C \neq 0$ and which admits an explicit root-free rational parameterization over $\mathbb{Q}$.*

Proof. (1) For $k = 2$, $\sum_{i=1}^2 X_i = 0 \implies X_2 = -X_1 \neq 0$, and $\sum_{i=1}^2 X_i u_i = X_1(u_1 - u_2) = 0 \implies u_1 = u_2$. Then the third equation of (12) gives $X_1^3(a_1^{-10} - a_2^{-10}) = 0 \implies a_1 = a_2 \implies d_2 = -d_1$, so the 4 terms cancel identically ($C = 0$).

(2) For $k = 3$, let $p = u_2 - u_3$, $q = u_3 - u_1$, and $r = u_1 - u_2 = -(p + q)$. Note that $p + q + r = 0$ and $pu_1 + qu_2 + ru_3 = 0$. On the degenerate locus $u_1 = u_2 = u_3$, we have $p = q = r = 0$; translating $u_i \mapsto u_i - u_1 = 0$ reduces $P_6(z)$ to an odd polynomial where $[z^2] = 0 \iff X_1^3/a_1^{10} + X_2^3/a_2^{10} - (X_1 + X_2)^3/a_3^{10} = 0$. On the open non-degenerate locus where (u_1, u_2, u_3) are not all equal, the vectors $(1, 1, 1)$ and (u_1, u_2, u_3) are linearly independent in $\mathbb{Q}^3$, and their 1-dimensional orthogonal complement is spanned by $(p, q, r) = (p, q, -(p + q))$. Since $X_i \neq 0$ for all $i = 1, 2, 3$, we have $p \neq 0, q \neq 0, p + q \neq 0$, and $(X_1, X_2, X_3) = \kappa(p, q, -(p + q))$ for a unique non-zero scalar $\kappa \in \mathbb{Q}^\times$. Direct algebraic expansion yields the exact identity

$$\begin{aligned} pu_1^2 + qu_2^2 + ru_3^2 &= (u_2 - u_3)u_1^2 + (u_3 - u_1)u_2^2 + (u_1 - u_2)u_3^2 \\ &= (u_1 - u_2)(u_1 - u_3)(u_2 - u_3) = r(-q)p = -pqr = pq(p + q). \end{aligned}$$

Substituting $(X_1, X_2, X_3) = \kappa(p, q, -(p + q))$ into the third equation of (12) and dividing by $\kappa \neq 0$ gives

$$3pq(p + q) + \kappa^2 \left(\frac{p^3}{a_1^{10}} + \frac{q^3}{a_2^{10}} - \frac{(p + q)^3}{a_3^{10}} \right) = 0 \iff \kappa^2 = \frac{-3pq(p + q)}{p^3/a_1^{10} + q^3/a_2^{10} - (p + q)^3/a_3^{10}}. \quad (15)$$

Define the homogeneous cubic polynomial

$$F(p, q) := a_2^{10}a_3^{10}p^3 + a_1^{10}a_3^{10}q^3 - a_1^{10}a_2^{10}(p + q)^3 = a_1^{10}a_2^{10}a_3^{10} \left(\frac{p^3}{a_1^{10}} + \frac{q^3}{a_2^{10}} - \frac{(p + q)^3}{a_3^{10}} \right),$$

so that (15) is defined on the open subset $F(p, q) \neq 0$ and becomes $\kappa^2 = -3a_1^{10}a_2^{10}a_3^{10}pq(p + q)/F(p, q)$. Clearing the denominator $F(p, q)$ by introducing the weighted projective coordinate

$$y := \frac{\kappa F(p, q)}{a_1^5 a_2^5 a_3^5}$$

gives the weighted-homogeneous sextic equation in $\mathbb{P}(1, 1, 3)$:

$$y^2 = \frac{\kappa^2 F(p, q)^2}{a_1^{10}a_2^{10}a_3^{10}} = -3pq(p + q)F(p, q),$$

which is precisely (13). To exhibit the explicit birational equivalence between $\mathcal{M}(a_1, a_2, a_3)$ and the Zariski-open subset $\mathcal{U} = \{[p : q : y] \in \mathcal{C}_{a_1,a_2,a_3} : pq(p + q)F(p, q) \neq 0 \text{ and } C(p, q) \neq 0\}$, fix the translation gauge $u_i \mapsto u_i + \tau$ uniquely by setting $u_3 = 0$, so that $(u_1, u_2, u_3) = (-q, p, 0)$. Under simultaneous scaling $(u_i, d_i) \mapsto (\lambda u_i, \lambda d_i)$ ($\lambda \in \mathbb{Q}^\times$), $(p, q) \mapsto (\lambda p, \lambda q)$, $\kappa = X_1/p$ is invariant of weight 0, and $y \mapsto \lambda^3 y$ has weight 3 in $\mathbb{P}(1, 1, 3)$. Conversely, given any rational point $[p : q : y] \in \mathcal{U}(\mathbb{Q})$ outside the exceptional branch locus $pq(p + q)F(p, q) = 0$ (which corresponds to $y = 0$, where $\kappa = 0$ or $F(p, q) = 0$), the inverse rational map recovers the unique moduli class in $\mathcal{M}(a_1, a_2, a_3)(\mathbb{Q})$ via

$$\kappa = \frac{a_1^5 a_2^5 a_3^5 y}{F(p, q)}, \qquad (u_1, u_2, u_3) = (-q, p, 0), \qquad (d_1, d_2, d_3) = \kappa \left(\frac{p}{a_1^5}, \frac{q}{a_2^5}, -\frac{p + q}{a_3^5} \right).$$

Furthermore, expanding the linear coefficient $C = [z^1]\,P_6(z) = 40\sum_{i=1}^{3} X_i u_i (u_i^2 + d_i^2)$ at $(u_1, u_2, u_3) = (-q, p, 0)$ yields

$$C = 40\kappa pq\left[(p^2 - q^2) + \kappa^2\left(\frac{q^2}{a_2^{10}} - \frac{p^2}{a_1^{10}}\right)\right].$$

To verify that C does not vanish identically on $\mathcal{C}_{a_1,a_2,a_3}$ inside the open non-degenerate locus $pq(p+q)F(p,q) \neq 0$, evaluate at $p = q \neq 0$: here $pq(p+q) = 2q^3 \neq 0$ and

$$F(q,q) = q^3\Big(a_2^{10}a_3^{10} + a_1^{10}a_3^{10} - 8a_1^{10}a_2^{10}\Big),$$

where the bracketed factor satisfies $a_2^{10}a_3^{10} + a_1^{10}a_3^{10} - 8a_1^{10}a_2^{10} < 2a_1^{10}a_2^{10} - 8a_1^{10}a_2^{10} = -6a_1^{10}a_2^{10} < 0$ since $a_1 > a_2 > a_3 > 0$. Thus $F(q,q) \neq 0$ for every $q \neq 0$ and $\kappa^2 = -6a_1^{10}a_2^{10}a_3^{10}q^3/F(q,q) > 0$ ($\kappa \neq 0$), so $p = q \neq 0$ lies in the open locus $pq(p+q)F(p,q) \neq 0$ and gives

$$C(q,q) = 40\kappa^3 q^4\left(\frac{1}{a_2^{10}} - \frac{1}{a_1^{10}}\right) \neq 0$$

because $a_1 > a_2 > 0$. Since C is non-zero at $p = q \neq 0$ on the irreducible curve $\mathcal{C}_{a_1,a_2,a_3}$, $C = 0$ defines at most finitely many points on $\mathcal{C}_{a_1,a_2,a_3}$, and $C \neq 0$ holds on a dense Zariski-open subset of $\mathcal{C}_{a_1,a_2,a_3}$.

When $a_1 > a_2 > a_3 > 0$, $F(p,q)$ has non-zero leading coefficient $a_2^{10}(a_3^{10} - a_1^{10}) \neq 0$ on p^3, and satisfies $F(0,q) = a_1^{10}(a_3^{10} - a_2^{10})q^3 \neq 0$, $F(p,0) = a_2^{10}(a_3^{10} - a_1^{10})p^3 \neq 0$, and $F(p,-p) = a_3^{10}(a_2^{10} - a_1^{10})p^3 \neq 0$, so $F(p,q)$ shares no roots with the linear factors $p, q, p+q$. In the affine coordinate $t = p/q$, the cubic $F(t,1) = a_2^{10}(a_3^{10} - a_1^{10})t^3 - 3a_1^{10}a_2^{10}t^2 - 3a_1^{10}a_2^{10}t + a_1^{10}(a_3^{10} - a_2^{10})$ has discriminant

$$\Delta_t(F) = -27a_1^{20}a_2^{20}a_3^{20}\big(a_1^5 + a_2^5 + a_3^5\big)\big(a_1^5 + a_2^5 - a_3^5\big)\big(a_1^5 - a_2^5 + a_3^5\big)\big(a_1^5 - a_2^5 - a_3^5\big).$$

For any ordered triple of distinct positive integers $a_1 > a_2 > a_3 \geq 1$, the first three Heron-type fifth-power factors are strictly positive since $a_1^5 + a_2^5 + a_3^5 > 0$, $a_1^5 + a_2^5 - a_3^5 > a_2^5 > 0$, and $a_1^5 - a_2^5 + a_3^5 > a_3^5 > 0$. (Note that the fourth factor $a_1^5 - a_2^5 - a_3^5$ need not be positive when a_2 is large; for example, at $(a_1, a_2, a_3) = (101, 100, 99)$, $101^5 - 100^5 - 99^5 = -8,999,799,998 < 0$.) However, smoothness of $\mathcal{C}_{a_1,a_2,a_3}$ requires only that $a_1^5 - a_2^5 - a_3^5 \neq 0$. If $a_1^5 - a_2^5 - a_3^5 = 0$ held for positive integers $a_1, a_2, a_3 \in \mathbb{Z}_{>0}$, then $(x, y, z) = (a_2, a_3, a_1)$ would form a positive integer solution to Fermat's quintic equation $x^5 + y^5 = z^5$, which is impossible by Fermat's Last Theorem for exponent 5 (proved by Dirichlet and Legendre in 1825, and in general by Wiles [21]). Therefore $a_1^5 - a_2^5 - a_3^5 \neq 0$, and hence $\Delta_t(F) \neq 0$ for every ordered triple of positive integers $a_1 > a_2 > a_3 \geq 1$. Consequently, $F(t,1)$ has three distinct roots in $\mathbb{C}$ (none equal to 0 or -1), so the homogeneous sextic $-3pq(p+q)F(p,q)$ has six distinct roots in $\mathbb{P}^1(\mathbb{C})$ and defines a smooth projective hyperelliptic curve $\mathcal{C}_{a_1,a_2,a_3} \subset \mathbb{P}(1,1,3)$ of genus $g = \frac{6-2}{2} = 2$. Faltings's theorem implies $|\mathcal{C}_{a_1,a_2,a_3}(\mathbb{Q})| < \infty$ for each fixed triple $a_1 > a_2 > a_3 \geq 1$.

(3) When $a_1 = a > 1$ and $a_2 = a_3 = 1$, normalize the scaling gauge by setting $q = 1$. Then $q^3 - (p+q)^3 = 1 - (p+1)^3 = -p(p^2 + 3p + 3)$, so the denominator $H(p)$ of (15) factors as

$$H(p) := \frac{p^3}{a^{10}} + 1 - (p+1)^3 = -\frac{p}{a^{10}}\Big[(a^{10} - 1)p^2 + 3a^{10}p + 3a^{10}\Big] =: -\frac{p\,Q_a(p)}{a^{10}},$$

where $Q_a(p) = (a^{10} - 1)p^2 + 3a^{10}p + 3a^{10}$. Cancelling $-p$ on the open locus $p \neq 0$ between the numerator $-3p(p+1)$ and the denominator $-pQ_a(p)/a^{10}$ in (15) gives

$$\kappa^2 = \frac{-3p(p+1)}{-p\,Q_a(p)/a^{10}} = \frac{3a^{10}(p+1)}{(a^{10} - 1)p^2 + 3a^{10}p + 3a^{10}},$$

which is birationally equivalent to (14) via the substitution $Y = \kappa Q_a(p)/a^5 = \kappa\big((a^{10} - 1)p^2 + 3a^{10}p + 3a^{10}\big)/a^5$ and its inverse $\kappa = a^5 Y/Q_a(p)$ (noting that $Q_a(p) > 0$ for all $p \in \mathbb{R}$ since its

discriminant is $9a^{20} - 12a^{10}(a^{10} - 1) = 3a^{10}(4 - a^{10}) < 0$ for all $a \geq 2$). Thus the cubic in (14) has three distinct roots in $\mathbb{C}$ and its projective completion in $\mathbb{P}^2$ is a smooth elliptic curve E_a over $\mathbb{Q}$. At $p = 0$, $Y = \pm 3a^5$, and the tangent line $Y = 3a^5(p + 1)$ intersects E_a with multiplicity 2 at $p = 0$ and multiplicity 1 at $T = (-1, 0)$, so $2P_0 = T$ and $P_0 = (0, 3a^5)$ has exact order 4. The passage from rational points on $E_a(\mathbb{Q})$ to infinitely many primitive integer identities in $\mathbb{Z}[z]$ is established in Lemma 3.3 immediately below.

(4) For $k = 4$ with any weight vector $(a_1, a_2, a_3, a_4) \in \mathbb{Z}^4_{>0}$, restricting to the subvariety where $X_3 = -X_1$, $X_4 = -X_2$, and fixing the translation gauge $u_4 = 0$ gives $u_3 = u_1 + \frac{X_2}{X_1}u_2$ on the open locus $X_1X_2u_2 \neq 0$. Expanding $3\sum_{i=1}^4 X_iu_i^2 = 3X_1u_1^2 + 3X_2u_2^2 - 3X_1\big(u_1 + \frac{X_2}{X_1}u_2\big)^2 = -\frac{3X_2u_2}{X_1}\big(2X_1u_1 + (X_2 - X_1)u_2\big)$ cancels the quadratic term $3X_1u_1^2$, leaving a linear equation in u_1 in (12) for every weight vector (a_1, a_2, a_3, a_4):

$$u_1 = \frac{X_1^4(a_1^{-10} - a_3^{-10}) + X_1X_2^3(a_2^{-10} - a_4^{-10}) - 3X_2(X_2 - X_1)u_2^2}{6X_1X_2u_2}.$$

In particular, for weights $(a, 1, 1, 1)$ ($a > 1$), this gives the explicit rational parameterization

$$u_1 = \frac{-X_1^4(1 - a^{-10}) - 3X_2(X_2 - X_1)u_2^2}{6X_1X_2u_2}. \tag{16}$$

To verify explicitly that the linear slope $C = 40\sum_{i=1}^4 X_iu_i(u_i^2 + d_i^2)$ does not vanish identically on this slice (as required by the definition of $\mathcal{M}(a_1, a_2, a_3, a_4)$), substitute $X_3 = -X_1$, $X_4 = -X_2$, $u_4 = 0$, $u_3 = u_1 + \frac{X_2}{X_1}u_2$, and the formula for u_1 into $C = 40\big[X_1u_1(u_1^2 + X_1^2a_1^{-10}) + X_2u_2(u_2^2 + X_2^2a_2^{-10}) - X_1u_3(u_3^2 + X_1^2a_3^{-10})\big]$. Direct algebraic expansion expresses C as a Laurent polynomial in u_2:

$$C(X_1, X_2, u_2) = \frac{10X_2(X_1^2 - X_2^2)}{X_1^2}u_2^3 + 20X_2\Big[X_2^2(a_2^{-10} + a_4^{-10}) - X_1^2(a_1^{-10} + a_3^{-10})\Big]u_2 + \frac{10\Big[X_1^6(a_1^{-10} - a_3^{-10})^2 - X_2^6(a_2^{-10} - a_4^{-10})^2\Big]}{3X_2}u_2^{-1}.$$

Whenever $X_1X_2 \neq 0$ and $X_1^2 \neq X_2^2$, the leading coefficient $\frac{10X_2(X_1^2 - X_2^2)}{X_1^2}$ of u_2^3 is non-zero, so $C(X_1, X_2, u_2)$ is not the zero polynomial on the slice. Thus the open condition $C(X_1, X_2, u_2) \neq 0$ (together with $X_1X_2u_2 \neq 0$ and distinctness of the centers u_i) defines a non-empty 2-dimensional Zariski-open rational subvariety $\mathcal{S} \subset \mathcal{M}(a_1, a_2, a_3, a_4)$ over $\mathbb{Q}$ as (X_1, X_2, u_2) varies in $(\mathbb{Q}^\times)^3$ modulo simultaneous scaling. □

To make the passage from rational points on $E_a(\mathbb{Q})$ to primitive integer identities in Theorem 3.2(3) completely explicit and self-contained—and to account for both perfect matchings of the four unit-weight terms—we record the exact denominator-clearing and fiber-bound properties in the following lemma:

Lemma 3.3 (Denominator Clearing, Exact Exceptional Locus, and Dihedral-Torsion Fiber Bound on $E_a(\mathbb{Q})$)**.** *Let $a \geq 2$ be an integer, and let $(p, Y) \in E_a(\mathbb{Q})$ be a rational point on (14).*

1. Exact Factorization of $C(p)$ on E_a: *Substituting $a_1 = a, a_2 = a_3 = 1, q = 1$, and $\kappa^2 = 3a^{10}(p + 1)/Q_a(p)$ into the linear coefficient $C(p) = 40\kappa p\big[(p^2 - 1) + \kappa^2(1 - p^2/a^{10})\big]$ gives the exact factored rational function*

$$C(p) = \frac{40\kappa(a^{10} - 1)\,p^3(p + 1)(p + 2)}{Q_a(p)} = \frac{40a^5(a^{10} - 1)\,Yp^3(p + 1)(p + 2)}{Q_a(p)^2}. \tag{17}$$

Since $a \geq 2$ and $Q_a(p) > 0$ for all $p \in \mathbb{R}$, we have $C(p) = 0$ on $E_a(\mathbb{Q})$ if and only if $p \in \{0, -1, -2\}$. Moreover, the three pairs have non-zero gaps $d_i \neq 0$ whenever $p \notin \{0, -1\}$, and distinct centers

$(u_1, u_2, u_3) = (-1, p, 0)$ *whenever* $p \notin \{0, -1\}$. *At* $p = -2$, *the right-hand side of* (14) *is* $-3(4a^{10} - 4 - 6a^{10} + 3a^{10}) = -3(a^{10} - 4) < 0$ *for all* $a \geq 2$, *so* $p = -2$ *yields no real or rational points on* $E_a(\mathbb{Q})$. *Thus the entire degenerate and* $C = 0$ *exceptional set on* $E_a(\mathbb{Q})$ *consists solely of the* 3 *finite torsion points*

$$\mathcal{E}_a = \Big\{(0, 3a^5),\ (0, -3a^5),\ (-1, 0)\Big\} \subset E_a(\mathbb{Q})_{\text{tors}}.$$

2. Weight-Preserving Denominator Clearing: *For any point* $(p, Y) \in E_a(\mathbb{Q}) \setminus \mathcal{E}_a$, *write the six rational shifts as* $\beta_{2i-1} = a_i(u_i + d_i) \in \mathbb{Q}$ *and* $\beta_{2i} = a_i(u_i - d_i) \in \mathbb{Q}$ *for* $i = 1, 2, 3$, *where* $(a_1, a_2, a_3) = (a, 1, 1)$, *so that*

$$P_6(z) = \sum_{i=1}^{3} \Big[(a_i z + \beta_{2i-1})^5 - (a_i z + \beta_{2i})^5\Big] = C(p)\, z + D(p) \qquad \text{in } \mathbb{Q}[z].$$

Let $L \in \mathbb{Z}_{>0}$ *be the least common multiple of the denominators of* $\beta_1, \ldots, \beta_6$ *in lowest terms, so* $L\beta_j \in \mathbb{Z}$ *for* $1 \leq j \leq 6$ *and* $\gcd(L, L\beta_1, \ldots, L\beta_6) = 1$. *Let* $c = \gcd(L\beta_1, \ldots, L\beta_6) \in \mathbb{Z}_{>0}$ *(or any positive common divisor* $c \mid \gcd(L\beta_1, \ldots, L\beta_6)$, *such as* $c = \gcd(a, L\beta_1, \ldots, L\beta_6)$ *or* $c = 1$*), so that* $c \mid L\beta_j$ *holds for every* $1 \leq j \leq 6$. *Substituting* $z = \frac{c}{L}\tilde{z}$ *and multiplying the identity by* $(L/c)^5 \in \mathbb{Q}^\times$ *yields an integral identity*

$$\sum_{i=1}^{3} \Big[(A_i \tilde{z} + B_{2i-1})^5 - (A_i \tilde{z} + B_{2i})^5\Big] = \tilde{C}\tilde{z} + \tilde{D} \qquad \text{in } \mathbb{Z}[\tilde{z}],$$

with $B_j = L\beta_j/c \in \mathbb{Z}$, *preserving the exact integer leading weights* $(A_1, A_2, A_3) = (a, 1, 1)$ *(so* $A_1 : A_2 : A_3 = a : 1 : 1$*), non-zero slope* $\tilde{C} = (L/c)^4 C(p) = L^4 c^{-4} C(p) \neq 0$, *and constant term* $\tilde{D} = (L/c)^5 D(p) = L^5 c^{-5} D(p)$.

3. Exact 4-to-1 Paired and At Most 8-to-1 Unordered Bounds on Affine Equivalence Classes:

   (a) Paired 6-term identities: *Two points* $P = (p, Y)$ *and* $P' = (p', Y')$ *in* $E_a(\mathbb{Q}) \setminus \mathcal{E}_a$ *yield the same paired 6-term identity in* $\mathbb{Z}[z]$ *up to affine change of variable* $z \mapsto \alpha z + \beta$ $(\alpha \in \mathbb{Q}^\times, \beta \in \mathbb{Q})$, *permutation of the two weight-1 pairs, and simultaneous pair reversal if and only if* $P' \in \{\pm P, \pm(P + 2P_0)\}$, *i.e.,* $p' = p$ *with* $Y' = \pm Y$, *or* $p' = \frac{-p}{p+1}$ *with* $Y' = -\frac{Y}{(p+1)^2}$ *(for* $P + 2P_0$*) or* $+\frac{Y}{(p+1)^2}$ *(for* $-(P + 2P_0)$*). Thus the map from* $E_a(\mathbb{Q}) \setminus \mathcal{E}_a$ *to affine-equivalence classes of paired 6-term identities is strictly 4-to-1.*

   (b) Unordered 6-term identities (accounting for both unit-weight matchings): *In an unordered 6-term identity* $(az + B_1)^5 - (az + B_2)^5 + (z + B_3)^5 - (z + B_4)^5 + (z + B_5)^5 - (z + B_6)^5 = Cz + D$, *the two positive weight-1 shifts* $\{B_3, B_5\}$ *and the two negative weight-1 shifts* $\{B_4, B_6\}$ *admit exactly two perfect matchings into opposite-sign pairs:* $\mathcal{M}_1 = \{(B_3, B_4), (B_5, B_6)\}$ *and* $\mathcal{M}_2 = \{(B_3, B_6), (B_5, B_4)\}$. *With* $P_0 = (0, 3a^5)$ *and* $3P_0 = -P_0 = (0, -3a^5)$, *switching from* $\mathcal{M}_1$ *to* $\mathcal{M}_2$ *corresponds on* $E_a(\mathbb{Q})$ *to translating* $P = (p, Y)$ *by the 4-torsion points* $3P_0 = -P_0$ *and* $+P_0$, *which sends* p *to*

$$p_{\mathcal{M}_2}(P + 3P_0) = \frac{-2\kappa}{\kappa - 1}, \qquad \tilde{p}_{\mathcal{M}_2}(P + P_0) = \frac{-2\kappa}{\kappa + 1} \qquad \left(\kappa = \frac{a^5 Y}{Q_a(p)} = \frac{3a^5(p+1)}{Y}\right). \tag{18}$$

   Consequently, two points $P, P' \in E_a(\mathbb{Q}) \setminus \mathcal{E}_a$ *yield the same unordered 6-term identity up to affine change of variable* $z \mapsto \alpha z + \beta$, *arbitrary permutation of summands, and overall sign if and only if* P' *lies in the dihedral-torsion coset*

$$\mathcal{O}(P) = \Big\{\pm P,\ \pm(P + P_0),\ \pm(P + 2P_0),\ \pm(P + 3P_0)\Big\} \subset E_a(\mathbb{Q})$$

   of size $|\mathcal{O}(P)| \leq 8$ *under the order-8 group* $\langle [-1], +P_0\rangle \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}/4\mathbb{Z}$. *Hence the map from* $E_a(\mathbb{Q}) \setminus \mathcal{E}_a$ *to* $\mathbb{Q}$*-affine equivalence classes of unordered primitive 6-term linear identities is at most 8-to-1.*

Proof. (1) Substituting $q = 1, a_1 = a, a_2 = a_3 = 1$, and $\kappa^2 = 3a^{10}(p+1)/Q_a(p)$ into $C(p)$ gives

$$\begin{aligned}\frac{C(p)}{40\kappa p} &= (p^2-1) + \frac{3a^{10}(p+1)}{Q_a(p)}\left(1 - \frac{p^2}{a^{10}}\right)\\ &= \frac{p+1}{Q_a(p)}\Big[(p-1)\big((a^{10}-1)p^2 + 3a^{10}p + 3a^{10}\big) + 3(a^{10}-p^2)\Big]\\ &= \frac{p+1}{Q_a(p)}\Big[(a^{10}-1)p^3 + (2a^{10}-2)p^2\Big] = \frac{(a^{10}-1)p^2(p+1)(p+2)}{Q_a(p)},\end{aligned}$$

which yields (17). Since $a \geq 2$, $a^{10} - 1 > 0$ and $Q_a(p) > 0$ for all $p \in \mathbb{R}$, so $C(p) = 0 \iff \kappa p(p+1)(p+2) = 0 \iff p \in \{0, -1, -2\}$. As shown above, $p = -2$ gives $Y^2 = -3(a^{10} - 4) < 0$, so $(p, Y) \notin E_a(\mathbb{Q})$. Thus $\mathcal{E}_a = \{(0, \pm 3a^5), (-1, 0)\}$.

(2) Setting $z = \frac{c}{L}\tilde{z}$ in $P_6(z) = C(p)z + D(p)$ and multiplying by $(L/c)^5 \in \mathbb{Q}^\times$ multiplies each linear form $a_i z + \beta_j = a_i \frac{c}{L}\tilde{z} + \beta_j$ by $L/c \neq 0$ to give

$$\frac{L}{c}\left(a_i \frac{c}{L}\tilde{z} + \beta_j\right) = a_i\tilde{z} + \frac{L\beta_j}{c} = A_i\tilde{z} + B_j,$$

where $A_i = a_i \in \mathbb{Z}_{>0}$ and $B_j = L\beta_j/c \in \mathbb{Z}$ because $c \mid L\beta_j$ for every $1 \leq j \leq 6$ by definition of c. On the right-hand side, $(L/c)^5\big(C(p)\frac{c}{L}\tilde{z} + D(p)\big) = \tilde{C}\tilde{z} + \tilde{D}$ scales $C(p) \neq 0$ by $(L/c)^5 \cdot (c/L) = (L/c)^4 = L^4c^{-4} \neq 0$, so $\tilde{C} = L^4c^{-4}C(p) \neq 0$.

(3) Under any affine substitution $z \mapsto \alpha z + \beta$ ($\alpha \in \mathbb{Q}^\times, \beta \in \mathbb{Q}$) and overall sign ± 1, the 6 fifth-power terms split uniquely by the absolute value of their z^5-coefficients into one opposite-sign pair of weight $a|\alpha|$ (centered at u_1) and four unit-weight terms of weight $|\alpha|$ (two with sign $+1$, say β_3, β_5, and two with sign -1, say β_4, β_6, after normalizing the weight-a pair). Under $z \mapsto \alpha z + \beta$, any pair center u transforms affinely as $u \mapsto \alpha^{-1}(u - \beta)$.

- *Fixed pairing $\mathcal{M}_1 = \{(\beta_3, \beta_4), (\beta_5, \beta_6)\}$:* The three centers $u_1, u_2 = \frac{\beta_3+\beta_4}{2}, u_3 = \frac{\beta_5+\beta_6}{2}$ have affine-invariant difference ratio

$$\frac{u_2 - u_3}{u_3 - u_1} = \frac{p - 0}{0 - (-1)} = p.$$

  Swapping the two weight-1 pairs permutes $u_2 \leftrightarrow u_3$, which replaces p by $\frac{u_3-u_2}{u_2-u_1} = \frac{-p}{p+1}$ and κ by $-\kappa(p+1)$ (corresponding on E_a to translation by the 2-torsion point $2P_0 = (-1, 0)$, which sends $(p, Y) \mapsto \big(\frac{-p}{p+1}, -\frac{Y}{(p+1)^2}\big)$), while overall negation flips the sign of κ (and hence $Y \mapsto -Y$). This gives the exact 4-element orbit $\{\pm P, \pm(P + 2P_0)\}$.

- *Alternative pairing $\mathcal{M}_2 = \{(\beta_3, \beta_6), (\beta_5, \beta_4)\}$:* In the normalized gauge $(u_1, u_2, u_3) = (-1, p, 0)$ where $(\beta_3, \beta_4, \beta_5, \beta_6) = (p + \kappa, p - \kappa, -\kappa(p+1), \kappa(p+1))$, matching β_3 with β_6 and β_5 with β_4 yields the alternative weight-1 pair centers and half-gaps:

$$\begin{aligned} u_2' &= \frac{\beta_3 + \beta_6}{2} = \frac{p + \kappa(p+2)}{2}, & u_3' &= \frac{\beta_5 + \beta_4}{2} = \frac{p - \kappa(p+2)}{2},\\ d_2' &= \frac{\beta_3 - \beta_6}{2} = \frac{p(1-\kappa)}{2}, & d_3' &= \frac{\beta_5 - \beta_4}{2} = -\frac{p(1+\kappa)}{2}.\end{aligned}$$

  Then $u_2' - u_3' = \kappa(p+2)$, $u_3' - u_1 = \frac{p-\kappa(p+2)+2}{2} = \frac{(p+2)(1-\kappa)}{2}$, and $u_2' - u_1 = \frac{(p+2)(1+\kappa)}{2}$. Since $p \neq -2$ on $E_a(\mathbb{Q})$, the factor $p + 2 \neq 0$ cancels identically, giving the two ordered center-difference ratios of $\mathcal{M}_2$:

$$p_{\mathcal{M}_2}(P + 3P_0) = \frac{u_2' - u_3'}{u_3' - u_1} = \frac{2\kappa}{1-\kappa} = \frac{-2\kappa}{\kappa - 1}, \qquad \tilde{p}_{\mathcal{M}_2}(P + P_0) = \frac{u_3' - u_2'}{u_2' - u_1} = \frac{-2\kappa}{\kappa + 1}.$$

Substituting $\kappa = \frac{3a^5(p+1)}{Y}$ and using the curve equation $Y^2 = 3(p+1)Q_a(p) = 3(p+1)\big[(a^{10}-1)p^2 + 3a^{10}p + 3a^{10}\big]$ shows that

$$\tilde{p}_{\mathcal{M}_2} = \frac{-6a^5(p+1)}{Y + 3a^5(p+1)} = \frac{-6a^5(p+1)\big(Y - 3a^5(p+1)\big)}{Y^2 - 9a^{10}(p+1)^2} = \frac{2a^5\big(3a^5(p+1) - Y\big)}{(a^{10}-1)p^2},$$

which is precisely the p-coordinate of the 4-torsion translate $P + P_0$ on $E_a(\mathbb{Q})$ for $P_0 = (0, 3a^5)$ (since the secant slope from $P_0 = (0, 3a^5)$ to (p, Y) is $\lambda = (Y - 3a^5)/p$), and similarly $p_{\mathcal{M}_2} = \frac{-2\kappa}{\kappa - 1} = \frac{2a^5(3a^5(p+1)+Y)}{(a^{10}-1)p^2}$ is the p-coordinate of $P + 3P_0 = P - P_0$ (obtained by $P_0 \mapsto -P_0$). Thus the two matchings $\mathcal{M}_1$ and $\mathcal{M}_2$, together with pair permutation and overall sign, generate the coset $\mathcal{O}(P) = \{\pm(P + kP_0) : k \in \{0, 1, 2, 3\}\}$ of size at most 8, so the map to unordered affine-equivalence classes is at most 8-to-1. □

3.3 Explicit Six-Term Linear Polynomial Identities on E_2 and E_6

To apply the Nagell–Lutz theorem unambiguously, we record the exact algebraic substitutions transforming E_a in (14) into reduced integral models and into a short integral Weierstrass equation $Y_S^2 = X_S^3 + \mathcal{A}_a X_S + \mathcal{B}_a$:

- When $3 \nmid a$, $a^2 \equiv 1 \pmod 3$ implies $a^{10} \equiv 1 \pmod 3$, so setting $M_a = (a^{10} - 1)/3 \in \mathbb{Z}$ gives $(a^{10}-1)p^2 + 3a^{10}p + 3a^{10} = 3(M_a p^2 + a^{10}p + a^{10})$. Substituting $Y = 3y$ into (14) and dividing both sides by 9 yields the simplified cubic model

$$E_a^{\text{int}}: \quad y^2 = (p+1)\big(M_a p^2 + a^{10}p + a^{10}\big), \qquad \kappa = \frac{a^5 y}{M_a p^2 + a^{10}p + a^{10}}. \tag{19}$$

- For general $a \geq 2$ (including $3 \mid a$), let $N_a = 3(a^{10} - 1) \in \mathbb{Z}_{>0}$. Multiplying both sides of (14) by N_a^2 gives

$$(N_a Y)^2 = (N_a p + N_a)\big((N_a p)^2 + 9a^{10}(N_a p) + 9a^{10}N_a\big).$$

  Setting $x_W = N_a p$ and $y_W = N_a Y$ and expanding the right-hand side $(x_W + N_a)(x_W^2 + 9a^{10}x_W + 9a^{10}N_a) = x_W^3 + (N_a + 9a^{10})x_W^2 + 18a^{10}N_a x_W + 9a^{10}N_a^2$ yields the monic integral Weierstrass model

$$E_a^W: \quad y_W^2 = x_W^3 + (N_a + 9a^{10})x_W^2 + 18a^{10}N_a x_W + 9a^{10}N_a^2, \qquad \kappa = \frac{3a^5 y_W}{x_W^2 + 9a^{10}x_W + 9a^{10}N_a}. \tag{20}$$

- *Explicit Integral Translation to Short Weierstrass Form:* Observe that for every integer $a \geq 2$, the x_W^2-coefficient of (20) is

$$N_a + 9a^{10} = 3(a^{10} - 1) + 9a^{10} = 12a^{10} - 3 = 3(4a^{10} - 1),$$

  which is always divisible by 3 in $\mathbb{Z}$. Setting $\beta_a := 4a^{10} - 1 \in \mathbb{Z}$ and applying the integral translation

$$X_S = x_W + \beta_a = N_a p + (4a^{10} - 1), \qquad Y_S = y_W = N_a Y \tag{21}$$

  eliminates the quadratic term and transforms (20) into the short integral Weierstrass model

$$E_a^{\text{short}}: \quad Y_S^2 = X_S^3 + \mathcal{A}_a X_S + \mathcal{B}_a, \tag{22}$$

  with exact integer coefficients

$$\mathcal{A}_a = 3\big(2a^{20} - 10a^{10} - 1\big) \in \mathbb{Z}, \qquad \mathcal{B}_a = -(a^{10} + 2)\big(7a^{20} - 26a^{10} + 1\big) \in \mathbb{Z}.$$

Because $\beta_a = 4a^{10} - 1 \in \mathbb{Z}$ is an integer, a point $P \in E_a(\mathbb{Q})$ has $X_S(P) \in \mathbb{Z}$ on the short Weierstrass model (22) if and only if $x_W(P) = X_S(P) - \beta_a \in \mathbb{Z}$ on (20). By the standard Nagell–Lutz theorem applied directly to the short integral Weierstrass equation (22), every non-zero rational torsion point $P \in E_a(\mathbb{Q})_{\text{tors}}$ must satisfy $X_S(P), Y_S(P) \in \mathbb{Z}$ (and hence $x_W(P), y_W(P) \in \mathbb{Z}$).

Theorem 3.4 (Explicit Six-Term Linear Identities in $\mathbb{Z}[z]$)**.** *The elliptic curves E_2 ($2^{10} = 1024, M_2 = 341, N_2 = 3069, \beta_2 = 4095, \mathcal{A}_2 = 6,260,733, \mathcal{B}_2 = -7,503,557,634$) and E_6 ($6^{10} = 60466176, N_6 = 181398525, \beta_6 = 241864703$) have positive Mordell–Weil rank over $\mathbb{Q}$, containing the certified infinite-order points*

$$P_1 = (p, y) = \left(\frac{144}{25}, \frac{43888}{125}\right) \in E_2^{\text{int}}(\mathbb{Q}), \quad P_1 + 3P_0 = P_1 - P_0 = (p, y) = \left(\frac{260}{81}, -\frac{132308}{729}\right) \in E_2^{\text{int}}(\mathbb{Q})$$

on (19) (with $2P_1 = \left(\frac{71784764}{188101225}, \frac{116043399920052}{2579808300875}\right) \in E_2^{\text{int}}(\mathbb{Q})$), and

$$Q_1 = (p, Y) = \left(\frac{16}{5}, \frac{659568}{5}\right) \in E_6(\mathbb{Q})$$

on (14). Clearing denominators via Lemma 3.3(2) with $c = \gcd(L\beta_1, \dots, L\beta_6)$ yields three explicit 6-term linear polynomial identities in $\mathbb{Z}[z]$:

$$(2z - 3636)^5 - (2z - 4221)^5 + (z + 7763)^5 - (z + 4847)^5 + (z - 6138)^5 - (z + 6138)^5 = C_2 z + D_2, \tag{23}$$

$$(2z - 4690)^5 - (2z - 5860)^5 + (z + 16817)^5 - (z + 13567)^5 + (z - 10985)^5 - (z + 10985)^5 = 16C_2 z + D_2', \tag{24}$$

$$(6z - 29436)^5 - (6z - 29454)^5 + (z + 19349)^5 - (z + 12059)^5 + (z - 15309)^5 - (z + 15309)^5 = C_6 z + D_6, \tag{25}$$

where the slopes and constants are given by

$$\begin{aligned}
C_2 &= 13,972,565,906,763,750 = 2 \cdot 3^6 \cdot 5^4 \cdot 11 \cdot 13^3 \cdot 31 \cdot 97 \cdot 211, \\
D_2 &= 8,797,920,299,940,110,625, \qquad D_2' = 570,150,770,968,195,560,000, \\
C_6 &= 539,939,269,033,164,000 = 2^5 \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 13^2 \cdot 101 \cdot 151 \cdot 311, \\
D_6 &= 842,901,695,917,396,315,200.
\end{aligned}$$

Here $P_1 + 3P_0$ and P_1 represent the two distinct paired matchings $\mathcal{M}_1$ and $\mathcal{M}_2$ in the same dihedral-torsion coset $\mathcal{O}(P_1)$ of Lemma 3.3(3) (so (24) and (23) are paired-inequivalent over $\mathcal{M}(2,1,1)$ while related as unordered sums over $\mathbb{Q}[z]$ via $z_{in\,(24)} = 2z_{in\,(23)} - 1291$, which reduces the lattice slope from $16C_2$ to C_2 with $v_2(C_2) = 1$ and odd D_2), whereas the multiples mP_1 ($m = 1, 2, 3, \dots$) lie in pairwise distinct $\langle[-1], +P_0\rangle$-cosets $\mathcal{O}(mP_1)$ and produce infinitely many unordered $\mathbb{Q}$-affine inequivalent primitive 6-term linear identities in $\mathbb{Z}[z]$.

Proof. 1. *Infinite-Order Certificates via Short Weierstrass Models and Nagell–Lutz:* By the integral translation (21), every non-zero rational torsion point on the short integral Weierstrass model E_a^{short} in (22) must have integral coordinates $X_S = x_W + \beta_a \in \mathbb{Z}$ and $Y_S = y_W \in \mathbb{Z}$, which is equivalent to $x_W, y_W \in \mathbb{Z}$ since $\beta_a = 4a^{10} - 1 \in \mathbb{Z}$.

- For $a = 2$ ($2^{10} = 1024, M_2 = 341, N_2 = 3069, \beta_2 = 4095$), substituting $p = \frac{144}{25}$ into (19) gives $341(144)^2 + 1024(144)(25) + 1024(25)^2 = 7,070,976 + 3,686,400 + 640,000 = 11,397,376 = 3376^2$, so

$$(p + 1)\left(341p^2 + 1024p + 1024\right) = \frac{169}{25} \cdot \frac{3376^2}{625} = \left(\frac{13 \cdot 3376}{125}\right)^2 = \left(\frac{43888}{125}\right)^2,$$

confirming $P_1 = \left(\frac{144}{25}, \frac{43888}{125}\right) \in E_2^{\text{int}}(\mathbb{Q})$. On E_2^W and E_2^{short}, P_1 maps to

$$x_W(P_1) = N_2 p = 3069 \cdot \frac{144}{25} = \frac{441936}{25} \notin \mathbb{Z}, \qquad X_S(P_1) = x_W(P_1) + 4095 = \frac{544311}{25} \notin \mathbb{Z}$$

(with 5^2 in the denominator), and $P_1 + 3P_0$ at $p = 260/81$ maps to $x_W(P_1 + 3P_0) = 3069 \cdot (260/81) = 88660/9 \notin \mathbb{Z}$ and $X_S(P_1 + 3P_0) = 125515/9 \notin \mathbb{Z}$ (with 3^2 in the denominator). Hence P_1 is not a torsion point on $E_2^{\text{short}}(\mathbb{Q})$ and has infinite order, proving $\operatorname{rank}(E_2(\mathbb{Q})) \geq 1$.

- For $a = 6$ ($a^{10} = 60466176$, $N_6 = 181398525$, $\beta_6 = 241864703$), evaluating the right-hand side of (14) at $p = \frac{16}{5}$ gives

$$3\left(\frac{16}{5}+1\right)\left[60466175\left(\frac{16}{5}\right)^2 + 3(60466176)\left(\frac{16}{5}\right) + 3(60466176)\right] = \frac{63}{5} \cdot \frac{6905166080}{5} = \left(\frac{659568}{5}\right)^2,$$

so $Q_1 = \left(\frac{16}{5}, \frac{659568}{5}\right) \in E_6(\mathbb{Q})$. On E_6^W, Q_1 has integral coordinates $(x_W, y_W) = (N_6 p, N_6 Y) = (580475280, 23928932467440) \in \mathbb{Z}^2$, and doubling Q_1 by the chord-and-tangent formula yields

$$\begin{aligned} x_W(2Q_1) &= \left(\frac{3x_W^2 + 2(N_6 + 9 \cdot 6^{10})x_W + 18 \cdot 6^{10} N_6}{2y_W}\right)^2 - (N_6 + 9 \cdot 6^{10}) - 2x_W \\ &= -\frac{1151448732500}{22801} = -\frac{1151448732500}{151^2} \notin \mathbb{Z}, \end{aligned}$$

so $X_S(2Q_1) = x_W(2Q_1) + \beta_6 = -1151448732500/22801 + 241864703 = 4363308360603/22801 = 4363308360603/151^2 \notin \mathbb{Z}$. Since $x_W(2Q_1) \notin \mathbb{Z}$ (and equivalently $X_S(2Q_1) \notin \mathbb{Z}$) on the short integral Weierstrass model E_6^{short}, $2Q_1$ (and hence Q_1) is not a torsion point by the Nagell–Lutz theorem, proving that $Q_1 \in E_6(\mathbb{Q})$ has infinite order and $\operatorname{rank}(E_6(\mathbb{Q})) \geq 1$.

2. *Complete Coordinate Transformations, Dual Matchings, and Coefficient Verification of* (23)–(25)*:*

- *Identity* (23) *from* $P_1 + 3P_0$ *on* E_2*:* At $p = \frac{260}{81}$ on (19), we have $p + 1 = \frac{341}{81}$ and $341\left(\frac{260}{81}\right)^2 + 1024\left(\frac{260}{81}\right) + 1024 = \frac{51332544}{6561} = \frac{341 \cdot 388^2}{81^2}$, so $y = -\frac{341 \cdot 388}{729} = -\frac{132308}{729}$ and $|\kappa| = \frac{32 \cdot 132308/729}{51332544/6561} = \frac{72}{97}$ ($\kappa = -\frac{72}{97}$). Setting $(u_1, u_2, u_3) = \left(-1, \frac{260}{81}, 0\right)$ and

$$(d_1, d_2, d_3) = |\kappa|(p/32, 1, -(p+1)) = \left(\frac{65}{873}, \frac{72}{97}, -\frac{2728}{873}\right) = \left(\frac{585}{7857}, \frac{5832}{7857}, -\frac{24552}{7857}\right),$$

the six rational shifts $\beta_{2i-1} = a_i(u_i + d_i)$ and $\beta_{2i} = a_i(u_i - d_i)$ in Lemma 3.3(2) are

$$(\beta_1, \ldots, \beta_6) = \left(-\frac{1616}{873}, -\frac{1876}{873}, \frac{31052}{7857}, \frac{19388}{7857}, -\frac{2728}{873}, \frac{2728}{873}\right).$$

Here $L = \operatorname{lcm}(873, 7857) = 7857$, giving $(L\beta_1, \ldots, L\beta_6) = (-14544, -16884, 31052, 19388, -24552, 24552)$ with greatest common divisor $c = \gcd(L\beta_1, \ldots, L\beta_6) = 4$. Multiplying the linear forms by $L/c = \frac{7857}{4}$ via $z = \frac{4}{7857}\tilde{z}$ yields $B_j = L\beta_j/4 = (-3636, -4221, 7763, 4847, -6138, 6138)$ and $\tilde{C} = (7857/4)^4 C(p) = C_2$ in (23). Direct binomial expansion in $\mathbb{Z}[z]$ confirms that the coefficients of z^5, z^4, z^3, z^2 vanish identically:

$$\begin{aligned} [z^4] &= 5\,[16(-3636 + 4221) + (7763 - 4847) + (-6138 - 6138)] = 5\,[9360 + 2916 - 12276] = 0, \\ [z^3] &= 10\,\left[8\left((-3636)^2 - (-4221)^2\right) + (7763^2 - 4847^2) + 0\right] = 10\,[-36770760 + 36770760] = 0, \\ [z^2] &= 10\,\left[4\left((-3636)^3 - (-4221)^3\right) + (7763^3 - 4847^3) - 2 \cdot 6138^3\right] \\ &= 10\,[108540649620 + 353958190524 - 462498840144] = 0. \end{aligned}$$

In accordance with Lemma 3.3(3)(b), the four unit-weight terms of (23) have positive shifts $\{7763, -6138\}$ and negative shifts $\{4847, 6138\}$. While Matching $\mathcal{M}_1 = \{(7763, 4847), (-6138, 6138)\}$ has centers $(u_1, u_2, u_3) = (-1964.25, 6305, 0)$ and ratio $p =$

$\frac{6305-0}{0-(-1964.25)} = \frac{260}{81}$, the alternative Matching $\mathcal{M}_2 = \{(7763, 6138), (-6138, 4847)\}$ has centers $u_2' = \frac{7763+6138}{2} = 6950.5$ (with half-gap $d_2' = \frac{7763-6138}{2} = 812.5$) and $u_3' = \frac{-6138+4847}{2} = -645.5$ (with half-gap $d_3' = -5492.5$), yielding the normalized center-difference ratio

$$\tilde{p}_{\mathcal{M}_2}((P_1 + 3P_0) + P_0) = \frac{u_2' - u_3'}{u_3' - u_1} = \frac{6950.5 - (-645.5)}{-645.5 - (-1964.25)} = \frac{7596}{1318.75} = \frac{144}{25} = \frac{-2\kappa}{\kappa + 1} \quad \left(\text{for } \kappa = -\frac{72}{97}\right),$$

which is the exact p-coordinate of $P_1 = (P_1 + 3P_0) + P_0$ below.

- *Identity* (24) *from* P_1 *on* E_2*:* At $p = \frac{144}{25}$ and $y = \frac{43888}{125}$ on (19), we have $\kappa = \frac{32 \cdot 43888/125}{11397376/625} = \frac{130}{211}$. Setting $(u_1, u_2, u_3) = \big(-1, \frac{144}{25}, 0\big)$ and

$$(d_1, d_2, d_3) = \kappa\big(p/32, 1, -(p+1)\big) = \left(\frac{117}{1055}, \frac{130}{211}, -\frac{4394}{1055}\right),$$

the six rational shifts are

$$(\beta_1, \ldots, \beta_6) = \left(-\frac{1876}{1055}, -\frac{2344}{1055}, \frac{33634}{5275}, \frac{27134}{5275}, -\frac{4394}{1055}, \frac{4394}{1055}\right),$$

with $L = \operatorname{lcm}(1055, 5275) = 5275$, $(L\beta_1, \ldots, L\beta_6) = (-9380, -11720, 33634, 27134, -21970, 21970)$, and $c = \gcd(L\beta_1, \ldots, L\beta_6) = 2$. Multiplying the linear forms by $L/c = \frac{5275}{2}$ via $z = \frac{2}{5275}\tilde{z}$ yields $B_j = L\beta_j/2 = (-4690, -5860, 16817, 13567, -10985, 10985)$ and $\tilde{C} = (5275/2)^4 C(p) = 16C_2$ in (24). Direct binomial expansion in $\mathbb{Z}[z]$ confirms each degree explicitly:

$$\begin{aligned}
[z^4] &= 5\,[16(-4690 + 5860) + (16817 - 13567) + (-10985 - 10985)] = 5\,[18720 + 3250 - 21970] = 0,\\
[z^3] &= 10\,\big[8\big((-4690)^2 - (-5860)^2\big) + (16817^2 - 13567^2) + 0\big] = 10\,[-98748000 + 98748000] = 0,\\
[z^2] &= 10\,\big[4\big((-4690)^3 - (-5860)^3\big) + (16817^3 - 13567^3) - 2 \cdot 10985^3\big]\\
&= 10\,[392273388000 + 2258851455250 - 2651124843250] = 0,\\
[z^1] &= 5\,\big[2(4690^4 - 5860^4) + (16817^4 - 13567^4) + 0\big] = 223,561,054,508,220,000 = 16C_2,\\
[z^0] &= (-4690)^5 - (-5860)^5 + 16817^5 - 13567^5 = 570,150,770,968,195,560,000 = D_2'.
\end{aligned}$$

- *Identity* (25) *from* Q_1 *on* E_6*:* At $a = 6$, $p = \frac{16}{5}$, $Y = \frac{659568}{5}$ on (14), we have $\kappa = \frac{3a^5}{Y}(p+1) = \frac{1458}{1963}$, $(u_1, u_2, u_3) = \big(-1, \frac{16}{5}, 0\big)$, and

$$(d_1, d_2, d_3) = \kappa\big(p/6^5, 1, -(p+1)\big) = \left(\frac{3}{9815}, \frac{1458}{1963}, -\frac{30618}{9815}\right).$$

The six rational shifts are

$$(\beta_1, \ldots, \beta_6) = \left(-\frac{58872}{9815}, -\frac{58908}{9815}, \frac{38698}{9815}, \frac{24118}{9815}, -\frac{30618}{9815}, \frac{30618}{9815}\right),$$

with $L = 9815$ and $c = \gcd(L\beta_1, \ldots, L\beta_6) = 2$. Clearing denominators by $L/c = \frac{9815}{2}$ via $z = \frac{2}{9815}\tilde{z}$ gives $B_j = L\beta_j/2 = (-29436, -29454, 19349, 12059, -15309, 15309)$ and $\tilde{C} = (9815/2)^4 C(p) = C_6$ in (25), verified degree-by-degree by:

$$\begin{aligned}
[z^4] &= 5\,[1296(-29436 + 29454) + (19349 - 12059) - 2(15309)] = 5\,[23328 + 7290 - 30618] = 0,\\
[z^3] &= 10\,\big[216\big((-29436)^2 - (-29454)^2\big) + (19349^2 - 12059^2)\big] = 10\,[-228964320 + 228964320] = 0,\\
[z^2] &= 10\,\big[36\big((-29436)^3 - (-29454)^3\big) + (19349^3 - 12059^3) - 2 \cdot 15309^3\big]\\
&= 10\,[1685463653088 + 5490338644170 - 7175802297258] = 0. \qquad \square
\end{aligned}$$

4 Structural Obstruction Theorems for Fifth-Power Polynomial Identities

We now establish four structural obstruction and reduction results explaining the arithmetic constraints on m-term fifth-power identities $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ in $\mathbb{Z}[z]$ with non-zero leading coefficients $A_i \in \mathbb{Z} \setminus \{0\}$.

4.1 The $\mathbb{F}_{11}$-Bastien Obstruction and 11-adic Rank-Defect Theorem

Theorem 4.1 (The $\mathbb{F}_{11}$-Bastien Obstruction and 11-adic Rank-Defect Theorem)**.** *Let $m \leq 7$, and suppose $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ in $\mathbb{Z}[z]$ with $s_i \in \{\pm 1\}$, $A_i \in \mathbb{Z} \setminus \{0\}$, and $B_i \in \mathbb{Z}$.*

1. Universal $11 \mid C$ Divisibility: *We necessarily have $11 \mid C$.*
2. 11-adic Rank-Defect Law: *Let $S_1 = \{i : 11 \nmid A_i\}$ and $S_0 = \{i : 11 \mid A_i\}$. Then $|S_1| = 2k$ is even ($0 \leq k \leq 3$), the terms in S_1 cancel in pairs modulo 11 and contribute $0 \pmod{11}$ to D, and the $r = m - 2k$ terms in S_0 contribute $D \equiv \sum_{i \in S_0} s_i B_i^5 \in \{-r, \ldots, r\} \pmod{11}$. Consequently, if s denotes the total number of fifth-power summands in an attempted representation of the form $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 + \sum_{j=1}^{s-m} y_j^5$ (comprising the m linear forms plus $s - m \geq 0$ additional fifth powers y_j^5):*
   (a) Single-identity bound: *For any single linear identity $P_m(z) = Cz + D$ ($m \leq 7$), $D \pmod{11}$ is fixed, so the reachable residues modulo 11 lie in the window $D + [-(s-m), s-m] \pmod{11}$ of size at most $2(s-m)+1$; covering all 11 residue classes in $\mathbb{Z}/11\mathbb{Z}$ therefore requires $s - m \geq 5$, so $s \geq m + 5 \geq 10$ (since $m \geq 5$ by Lemma 3.1; in particular, the 6-term identity (23) requires $s = 6 + 5 = 11$ summands, and $m = 8$ with $s = 8 + 1 = 9$ is optimal for the single-identity scheme).*
   (b) Family bound: *Across a family of linear identities of length $m \leq 7$ whose $r = m - 2k$ terms in S_0 have varying constants $B_i \pmod{11}$, covering all 11 residue classes in $\mathbb{Z}/11\mathbb{Z}$ requires $r + (s - m) = s - 2k \geq 5$, so $s \geq \max(m, 2k + 5)$ total fifth powers (for $m = 6$, $s \geq \max(6, 2k+5) \in \{6, 7, 9, 11\}$ for $k = 0, 1, 2, 3$, giving $s \geq 9$ whenever $k \geq 2$, and $s \geq 11$ when $k = 3$, i.e., when all six A_i are coprime to 11).*

Proof. Reduce the polynomial identity $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ modulo 11 in $\mathbb{F}_{11}[z]$. For each $i \in S_0$, $11 \mid A_i \implies (A_i z + B_i)^5 \equiv B_i^5 \in \{0, \pm 1\} \pmod{11}$ is a constant polynomial in $\mathbb{F}_{11}[z]$. For each $i \in S_1$, $11 \nmid A_i$, so Fermat's Little Theorem implies $A_i^{10} \equiv 1 \pmod{11}$ and hence $A_i^5 \equiv \pm 1 \pmod{11}$. Setting $\tilde{s}_i = s_i A_i^5 \in \{\pm 1\}$ and $\gamma_i \equiv B_i A_i^{-1} \pmod{11}$, we have

$$\sum_{i \in S_1} \tilde{s}_i (z + \gamma_i)^5 \equiv Cz + \left(D - \sum_{i \in S_0} s_i B_i^5 \right) \pmod{11} \quad \text{in } \mathbb{F}_{11}[z].$$

Partition $S_1 = S_+ \sqcup S_-$ according to $\tilde{s}_i = +1$ or -1, with $m_+ = |S_+|$ and $m_- = |S_-|$. Vanishing of the z^5 coefficient in $\mathbb{F}_{11}[z]$ gives $m_+ - m_- \equiv 0 \pmod{11}$. Since $m_+ + m_- = |S_1| \leq m \leq 7$, we have $|m_+ - m_-| \leq 7 < 11$, which forces $m_+ = m_- = k \leq 3$ and $|S_1| = 2k$.

Now compare the coefficients of z^4, z^3, z^2 in $\mathbb{F}_{11}[z]$:

$$5\left(\sum_{i \in S_+} \gamma_i - \sum_{j \in S_-} \gamma_j \right) \equiv 0, \quad 10\left(\sum_{i \in S_+} \gamma_i^2 - \sum_{j \in S_-} \gamma_j^2 \right) \equiv 0, \quad 10\left(\sum_{i \in S_+} \gamma_i^3 - \sum_{j \in S_-} \gamma_j^3 \right) \equiv 0 \pmod{11}.$$

Because $5, 10 \in \mathbb{F}_{11}^{\times}$ and $\mathrm{char}(\mathbb{F}_{11}) = 11 > 3 \geq k$, Lemma 2.1 (Bastien's theorem over $\mathbb{F}_{11}$) forces the multisets $\{\gamma_i : i \in S_+\}$ and $\{\gamma_j : j \in S_-\}$ to be identical in $\mathbb{F}_{11}$. Therefore,

$$C \equiv 5\left(\sum_{i \in S_+} \gamma_i^4 - \sum_{j \in S_-} \gamma_j^4 \right) \equiv 0 \pmod{11}, \qquad D - \sum_{i \in S_0} s_i B_i^5 \equiv \sum_{i \in S_+} \gamma_i^5 - \sum_{j \in S_-} \gamma_j^5 \equiv 0 \pmod{11}.$$

Finally, since $C \equiv 0 \pmod{11}$ and fifth powers modulo 11 lie in $\{-1, 0, 1\}$, for any single identity $P_m(z) = Cz + D$ the constant $D \pmod{11}$ is fixed and $Cz + D + \sum_{j=1}^{s-m} y_j^5 \equiv D + \sum_{j=1}^{s-m} y_j^5 \in D + [-(s-m), s-m] \pmod{11}$, which requires $2(s-m) + 1 \geq 11 \iff s - m \geq 5$, so $s \geq m + 5 \geq 10$ (as $m \geq 5$ by Lemma 3.1). Across a family of identities where the $r = m - 2k$ terms in S_0 vary modulo 11, $Cz + D + \sum_{j=1}^{s-m} y_j^5 \equiv \sum_{i \in S_0} s_i B_i^5 + \sum_{j=1}^{s-m} y_j^5 \pmod{11}$ is a sum of $r + (s-m) = s - 2k$ fifth powers mod 11; representing $\pm 5 \pmod{11}$ requires $s - 2k \geq 5 \implies s \geq \max(m, 2k + 5)$. □

Proposition 4.2 (The 5-adic Vandermonde Obstruction for $m \leq 6$)**.** *For every linear polynomial identity $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ in $\mathbb{Z}[z]$ with $m \leq 6$, we have $25 \mid C$ ($v_5(C) \geq 2$). At $m = 7$, this particular modulo-25 congruence obstruction vanishes (as a local congruence test modulo 25, without implying the existence of a global 7-term identity in $\mathbb{Z}[z]$), and at $m = 8$ (including 7 distinct linear forms with multiplicity 8), exact valuation $v_5(C) = 1$ (i.e., $5 \parallel C$, so $5 \mid C$ and $25 \nmid C$) is achieved globally in $\mathbb{Z}[z]$.*

Proof. Since $[z^4] = 5\sum_{i=1}^{m} s_i A_i^4 B_i = 0$, $[z^3] = 10\sum_{i=1}^{m} s_i A_i^3 B_i^2 = 0$, $[z^2] = 10\sum_{i=1}^{m} s_i A_i^2 B_i^3 = 0$, and $C = 5\sum_{i=1}^{m} s_i A_i B_i^4$, dividing the exact integer equations by $5, 10, 10, 5$ and reducing modulo 5 shows that any term with $5 \mid A_i$ contributes $0 \pmod 5$ to $\frac{1}{5}[z^4]$, $\frac{1}{10}[z^3]$, $\frac{1}{10}[z^2]$, $\frac{1}{5}C$ (and $0 \pmod{25}$ to $[z^5] = \sum_{i=1}^{m} s_i A_i^5$). For each term i with $5 \nmid A_i$, Fermat's Little Theorem gives $A_i^4 \equiv 1 \pmod 5$; setting $u_i = s_i A_i^5 \in \mathbb{Z}$ and $t_i \equiv B_i A_i^{-1} \in \mathbb{F}_5$, we have $s_i A_i^{5-r} B_i^r \equiv u_i t_i^r \pmod 5$ for $r = 0, 1, 2, 3, 4$. Grouping terms with $5 \nmid A_i$ by their shift residue $t \in \mathbb{F}_5 = \{0, 1, 2, 3, 4\}$ via $W_t = \sum_{5 \nmid A_i, t_i = t} u_i$, vanishing of $[z^5]$, $\frac{1}{5}[z^4]$, $\frac{1}{10}[z^3]$, $\frac{1}{10}[z^2]$ modulo 5 yields the linear system over $\mathbb{F}_5$:

$$\sum_{t \in \mathbb{F}_5} W_t t^r \equiv 0 \pmod 5 \quad (r = 0, 1, 2, 3), \qquad \frac{C}{5} \equiv \sum_{t \in \mathbb{F}_5} W_t t^4 \pmod 5.$$

Because the 5×5 Vandermonde matrix $(t^r)_{0 \leq r \leq 4, t \in \mathbb{F}_5}$ over $\mathbb{F}_5$ is invertible and $\sum_{t \in \mathbb{F}_5} t^r \equiv 0 \pmod 5$ for $0 \leq r \leq 3$ while $\sum_{t \in \mathbb{F}_5} t^4 \equiv 4 \equiv -1 \pmod 5$, the 1-dimensional nullspace of the first four rows in $\mathbb{F}_5^5$ is spanned by $(1, 1, 1, 1, 1)$. Thus $v_5(C) = 1$ ($\frac{1}{5}C \not\equiv 0 \pmod 5$) holds if and only if

$$W_0 \equiv W_1 \equiv W_2 \equiv W_3 \equiv W_4 \equiv c \in \mathbb{F}_5^\times = \{+1, -1, +2, -2\} \pmod 5.$$

In particular, all five residue classes $t \in \mathbb{F}_5$ must be non-empty. Now examine $u_i = s_i A_i^5$ modulo 25: as A_i mod 5 runs over $\{+1, -1, +2, -2\}$, A_i^5 mod 25 runs over $\{+1, -1, +7, -7\} \pmod{25}$, so reduction modulo 5 is a bijection between the four 5-adic unit fifth powers $\{+1, -1, +7, -7\} \pmod{25}$ and $\mathbb{F}_5^\times = \{+1, -1, +2, -2\} \pmod 5$. Scaling every linear form by an integer λ coprime to 5 multiplies each u_i by $\lambda^5 \in \{\pm 1, \pm 7\} \pmod{25}$ (and by $\lambda \in \{\pm 1, \pm 2\} \pmod 5$), so we may normalize $c \equiv +1 \pmod 5$:

- Any singleton class $t \in \mathbb{F}_5$ (containing 1 term u_i) has $u_i \equiv +1 \pmod 5$, which lifts uniquely to $u_i \equiv +1 \pmod{25}$.
- Any doubleton class $t \in \mathbb{F}_5$ (containing 2 terms u_a, u_b) has $u_a + u_b \equiv +1 \pmod 5$; the only two unordered pairs in $\{+1, -1, +2, -2\} \pmod 5$ summing to $+1 \pmod 5$ are $\{-1, +2\}$ (since $-1 + 2 = +1$) and $\{-2, -2\}$ (since $-2 - 2 = -4 \equiv +1 \pmod 5$), which lift uniquely modulo 25 to $(-1) + (+7) = +6 \pmod{25}$ and $(-7) + (-7) = -14 \pmod{25}$, respectively.

For $m \leq 6$, the 5 non-empty classes $t \in \mathbb{F}_5$ can consist either of 5 singletons (plus at most one $5 \mid A_i$ term, which contributes $0 \pmod{25}$), giving $[z^5] = \sum_{i=1}^{m} s_i A_i^5 \equiv 5(+1) = 5 \not\equiv 0 \pmod{25}$, or of 4 singletons and 1 doubleton (when $m = 6$), giving either $[z^5] \equiv 4(+1) + 6 = 10 \not\equiv 0 \pmod{25}$ (for a $\{-1, +2\}$ doubleton) or $[z^5] \equiv 4(+1) + (-14) = -10 \not\equiv 0 \pmod{25}$ (for a $\{-2, -2\}$ doubleton). All cases contradict $[z^5] = 0$, proving $25 \mid C$ for all $m \leq 6$. At $m = 7$, partitioning the 7 terms into 3 singleton classes $+1 \pmod 5$ and 2 doubleton classes $\{-2, -2\} \pmod 5$ (comprising $3 \times 1 + 2 \times 2 = 7$ terms in total) gives the modulo-25 fifth-power sum $3(+1) + 2(-14) = -25 \equiv 0 \pmod{25}$, so the modulo-25 Vandermonde lift obstruction vanishes at $m = 7$. Finally, at $m = 8$, Theorem 5.3 exhibits explicit identities in $\mathbb{Z}[z]$ with $v_5(C) = 1$ (in particular, Identity (38) uses 7 distinct linear forms with multiplicity 8, realizing 3 singletons, 1 doubleton, and 1 tripleton modulo 5). □

4.2 The Universal $30 \mid C$ and Fermat–Bastien Congruence Law for $D \pmod{30}$

Theorem 4.3 (Universal $30 \mid C$ and Fermat–Bastien Law for $D \pmod{30}$)**.** *Let $\sum_{i=1}^{m} s_i(A_i z + B_i)^5 = Cz + D$ be any linear polynomial identity in $\mathbb{Z}[z]$ with $A_i \in \mathbb{Z} \setminus \{0\}$ and $B_i \in \mathbb{Z}$. Then:*

1. Universal Slope Divisibility: $30 \mid C$ *holds for every* $m \geq 1$.
2. Constant Term Congruence at $p \in \{2,3,5\}$: *For each prime* $p \in \{2,3,5\}$,
$$D \equiv \sum_{p \mid A_i} s_i B_i \pmod{p}. \tag{26}$$
3. Consequences for $\gcd(C,D)$: *Whenever* $\gcd(A_i,15) = 1$ *for all* $i = 1,\dots,m$ *(in particular, for every unit-weight identity* $A_i = 1$ *and every 2-power-weighted identity* $A_i \in \{1,2\}$*), the index set* $\{i : p \mid A_i\}$ *is empty for* $p \in \{3,5\}$*, so* $15 \mid D$ *and* $15 \mid \gcd(C,D)$ *hold universally for all* $m \geq 1$*. By contrast, whenever* $p \in \{2,3,5\}$ *divides* $A_1 = A_2$, $D \equiv B_1 - B_2 \not\equiv 0 \pmod{p}$ *is achievable at* $m = 8$.

Proof. (1) Since $C = 5\sum_{i=1}^{m} s_i A_i B_i^4$, we have $5 \mid C$ in $\mathbb{Z}$, and the integer $C/5 = \sum_{i=1}^{m} s_i A_i B_i^4$ is well-defined. Likewise, since $[z^3] = 10\sum_{i=1}^{m} s_i A_i^3 B_i^2 = 0$ in $\mathbb{Z}$, dividing by 10 in $\mathbb{Z}$ gives the exact integer equation $\sum_{i=1}^{m} s_i A_i^3 B_i^2 = 0$. For all integers $A, B \in \mathbb{Z}$, $AB^2(B^2 - A^2) = AB^2(B-A)(B+A)$ is a product of three factors that is divisible by both 2 and 3, hence divisible by 6. Summing over $i = 1,\dots,m$ with signs s_i yields

$$\frac{C}{5} = \sum_{i=1}^{m} s_i A_i B_i^4 = \sum_{i=1}^{m} s_i A_i B_i^2 (B_i^2 - A_i^2) + \sum_{i=1}^{m} s_i A_i^3 B_i^2 = \sum_{i=1}^{m} s_i A_i B_i^2 (B_i^2 - A_i^2) \equiv 0 \pmod{6}.$$

Since $6 \mid (C/5)$, we have $30 \mid C$.

(2) The primes p satisfying $(p-1) \mid 4$ are precisely $p \in \{2,3,5\}$. For each $p \in \{2,3,5\}$ and any $A_i, B_i \in \mathbb{Z}$, Fermat's Little Theorem gives $B_i^5 \equiv B_i \pmod{p}$ and $A_i^4 \equiv 1 \pmod{p}$ if $p \nmid A_i$ (and $A_i^4 \equiv 0 \pmod{p}$ if $p \mid A_i$). Since $[z^4] = 5\sum_{i=1}^{m} s_i A_i^4 B_i = 0$ in $\mathbb{Z}$, dividing by 5 in $\mathbb{Z}$ gives $\sum_{i=1}^{m} s_i A_i^4 B_i = 0$. Subtracting this from $D = \sum_{i=1}^{m} s_i B_i^5$ modulo p gives

$$D = \sum_{i=1}^{m} s_i (B_i^5 - A_i^4 B_i) \equiv \sum_{p \nmid A_i} s_i (B_i - B_i) + \sum_{p \mid A_i} s_i (B_i - 0) = \sum_{p \mid A_i} s_i B_i \pmod{p}.$$

(3) If $\gcd(A_i,15) = 1$ for all $i = 1,\dots,m$, the sum on the right-hand side of (26) is empty for $p = 3$ and $p = 5$, yielding $D \equiv 0 \pmod{3}$ and $D \equiv 0 \pmod{5}$, so $15 \mid D$. When $p \mid A_1 = A_2$, $D \not\equiv 0 \pmod{p}$ is achieved at $m = 8$:

- For $p = 2$: by (38) ($D_{\mathrm{I}} = 27826665 \equiv 1 \pmod{2}$).
- For $p = 3$: by the 8-term identity with weights $(3,3,1,1,1,1,1,1)$ and gap-sum parameters $(\delta_1,\delta_2,\delta_3,\delta_4) = (2,-16,14,-160)$, $(\sigma_1,\sigma_2,\sigma_3,\sigma_4) = (-6,-342,-482,-10)$:
$$(3z-2)^5 - (3z-4)^5 + (z-179)^5 - (z-163)^5 + (z-234)^5 - (z-248)^5 + (z-85)^5 - (z+75)^5 = C^{(3)}z + D^{(3)},$$
where $[z^4] = 5(81\cdot 2 - 16 + 14 - 160) = 0$, $[z^3] = 10\big(27(4-16) + (-16)(-342) + 14(-482) + (-160)(-10)\big) = 10(-324 + 5472 - 6748 + 1600) = 0$, $[z^2] = 0$, $C^{(3)} = -2,216,246,400$, and $D^{(3)} = 161,024,169,680 \equiv -2 - (-4) = 2 \pmod{3}$.
- For $p = 5$: by the 8-term identity with weights $(5,5,1,1,1,1,1,1)$ and gap-sum parameters $(\delta_1,\delta_2,\delta_3,\delta_4) = (2,16,20,-1286)$, $(\sigma_1,\sigma_2,\sigma_3,\sigma_4) = (-10,4814,-4112,-6)$:
$$(5z-4)^5 - (5z-6)^5 + (z+2415)^5 - (z+2399)^5 + (z-2046)^5 - (z-2066)^5 + (z-646)^5 - (z+640)^5 = C^{(5)}z + D^{(5)},$$
where $[z^4] = 5(625\cdot 2 + 16 + 20 - 1286) = 0$, $[z^3] = 10\big(125(16-36) + 16(4814) + 20(-4112) + (-1286)(-6)\big) = 10(-2500 + 77024 - 82240 + 7716) = 0$, $[z^2] = 0$, $C^{(5)} = 1,017,973,008,000$, and $D^{(5)} = 4,252,444,454,692,752 \equiv -4 - (-6) = 2 \pmod{5}$. □

4.3 The Cauchy–Schwarz Signature Obstruction and 7-Term Odd Linear Reduction

Theorem 4.4 (Cauchy–Schwarz Signature Obstruction for $(m-1,1)$ Zero-Sums)**.** *Let $a_1,\dots,a_m \in \mathbb{Z}_{>0}$ satisfy a fifth-power zero-sum of signature $(m-1,1)$:*

$$a_1^5 + a_2^5 + \cdots + a_{m-1}^5 - a_m^5 = 0$$

(such as the Lander–Parkin–Selfridge solution $27^5+84^5+110^5+133^5-144^5=0$ for $m=5$). Then for any real shifts $b_1,\dots,b_m \in \mathbb{R}$, the polynomial $P(z)=\sum_{i=1}^{m-1}(a_iz+b_i)^5-(a_mz+b_m)^5$ has vanishing z^4 and z^3 coefficients if and only if $\frac{b_1}{a_1}=\frac{b_2}{a_2}=\cdots=\frac{b_m}{a_m}$, in which case $P(z)\equiv 0$ identically. In particular, no non-degenerate polynomial identity $P(z)\not\equiv 0$ of degree ≤ 2 (or linear identity $Cz+D$ with $C\neq 0$) can be formed from an $(m-1,1)$-signature fifth-power zero-sum over $\mathbb{R}$.

Proof. Let $w_i = a_i^5 > 0$ for $1 \le i \le m-1$, so $a_m^5 = \sum_{i=1}^{m-1} w_i$, and let $t_i = b_i/a_i \in \mathbb{R}$. Vanishing of $[z^4]$ and $[z^3]$ requires

$$\sum_{i=1}^{m-1} w_i t_i = a_m^5 t_m \iff \sum_{i=1}^{m-1} w_i(t_i - t_m) = 0, \qquad \sum_{i=1}^{m-1} w_i t_i^2 = a_m^5 t_m^2.$$

Expanding $\sum_{i=1}^{m-1} w_i(t_i-t_m)^2$ gives

$$\sum_{i=1}^{m-1} w_i(t_i-t_m)^2 = \sum_{i=1}^{m-1} w_i t_i^2 - 2t_m \sum_{i=1}^{m-1} w_i t_i + t_m^2 \sum_{i=1}^{m-1} w_i = a_m^5 t_m^2 - 2a_m^5 t_m^2 + a_m^5 t_m^2 = 0.$$

Since $w_i > 0$ for all $1 \le i \le m-1$ and $t_i \in \mathbb{R}$, $\sum_{i=1}^{m-1} w_i(t_i-t_m)^2 = 0$ forces $t_1 = t_2 = \cdots = t_m$, so $a_iz+b_i = a_i(z+t_m)$ and $P(z) = \big(\sum_{i=1}^{m-1} a_i^5 - a_m^5\big)(z+t_m)^5 \equiv 0$. □

Remark 4.5 (Two-Shift Quartic Decomposition on the 7-Term LPS Surface). Adjoining a single weighted pair $(az+B_6)^5-(az+B_7)^5 = a^5\big[(z+t_6)^5-(z+t_7)^5\big]$ $(a>0)$ to the Lander–Parkin–Selfridge 5-tuple $(27,84,110,133,-144)$ changes the signature from $(4,1)$ to $(5,2)$, breaking the Cauchy–Schwarz obstruction. Here $B_6, B_7 \in \mathbb{R}$ denote the raw additive constants of the added pair and $t_6 = B_6/a, t_7 = B_7/a$ denote the corresponding normalized shifts (matching the convention $t_i = B_i/A_i$ of (10)). Setting $t_5 = 0$, $S_r(t) = \sum_{i=1}^{4} w_i t_i^r$ with $w_i = a_i^5$, and defining the normalized center u and normalized half-gap d explicitly under both equivalent conventions by

$$u = \frac{t_6+t_7}{2} = \frac{B_6+B_7}{2a}, \qquad d = \frac{t_6-t_7}{2} = \frac{B_6-B_7}{2a} \neq 0$$

(so that $t_6 = u+d$ and $t_7 = u-d$, and the added pair contributes $a^5\big[(u+d)^r-(u-d)^r\big]$ to the r-th weighted moment in (10)), the moment equations (10) for $[z^4],[z^3],[z^2]$ become

$$S_1 + 2a^5 d = 0, \qquad S_2 + 4a^5 du = 0, \qquad S_3 + 2a^5 d(3u^2+d^2) = 0.$$

Substituting $2a^5 d = -S_1$, $u = \frac{S_2}{2S_1}$, and $d = -\frac{S_1}{2a^5}$ into the third equation gives $S_3 - S_1\left(\frac{3S_2^2}{4S_1^2} + \frac{S_1^2}{4a^{10}}\right) = 0$; multiplying by $4S_1$ yields the quartic surface equation

$$4S_1(t)S_3(t) - 3S_2(t)^2 = \frac{S_1(t)^4}{a^{10}}.$$

On the two-shift slice $t_3 = t_4 = 0$ (where $S_r = w_1t_1^r + w_2t_2^r$), direct algebraic expansion yields the exact sum-of-squares identity

$$\begin{aligned} 4S_1(t)S_3(t) - 3S_2(t)^2 &= 4(w_1t_1+w_2t_2)(w_1t_1^3+w_2t_2^3) - 3(w_1t_1^2+w_2t_2^2)^2 \\ &= w_1^2t_1^4 + 4w_1w_2t_1^3t_2 - 6w_1w_2t_1^2t_2^2 + 4w_1w_2t_1t_2^3 + w_2^2t_2^4 \\ &= \big(w_1t_1^2+w_2t_2^2\big)^2 + 4w_1w_2t_1t_2(t_1-t_2)^2. \end{aligned}$$

For positive shifts $t_1, t_2 > 0$ on this two-shift slice $t_3 = t_4 = 0$, writing $\lambda = t_1/t_2 > 0$, the homogeneous ratio $R(\lambda) := \big(4S_1(t)S_3(t) - 3S_2(t)^2\big)/S_1(t)^4$ has exact derivative

$$R'(\lambda) = \frac{12\lambda w_1 w_2 (w_1 + w_2)(\lambda - 1)}{(\lambda w_1 + w_2)^5}.$$

Thus on the positive quadrant $t_1, t_2 > 0$ of the slice $t_3 = t_4 = 0$, $R(\lambda)$ is strictly decreasing on $\lambda \in (0, 1]$ from $R(0^+) = w_2^{-2} = 84^{-10}$ to its unique global minimum $R(1) = (w_1 + w_2)^{-2} = (27^5 + 84^5)^{-2}$ at $t_1 = t_2$, and strictly increasing on $\lambda \in [1, \infty)$ from $R(1) = (27^5 + 84^5)^{-2}$ to $R(\infty) = w_1^{-2} = 27^{-10}$.

Theorem 4.6 (Algebraic, 25-adic, and Fermat–Norm Reduction Within the 7-Term Odd-Symmetric Ansatz)**.** *Consider the 7-term odd-symmetric ansatz (3 origin-symmetric pairs plus 1 singleton):*

$$P_7(z) = \sum_{i=1}^{3} s_i \Big[(c_i z + a_i)^5 + (c_i z - a_i)^5\Big] - (d_1 z)^5 = Cz, \qquad c_i, a_i, d_1 \in \mathbb{Z} \setminus \{0\}. \tag{27}$$

Within this specified 7-term odd-symmetric family, the following reductions hold:

1. *By odd symmetry ($P_7(-z) = -P_7(z)$), $[z^4] = [z^2] = [z^0] = 0$ hold identically. Setting $Y_i = c_i a_i$, $[z^3] = 0$ is equivalent to the ternary conic $s_1 c_1 Y_1^2 + s_2 c_2 Y_2^2 + s_3 c_3 Y_3^2 = 0$, which splits rationally whenever $s_1 c_1 + s_2 c_2 + s_3 c_3 = 0$.*

2. *Vanishing of $[z^5]$ requires $2(s_1 c_1^5 + s_2 c_2^5 + s_3 c_3^5) = d_1^5$, forcing $2 \mid d_1$ ($d_1 = 2d_0$) and reducing to the ternary fifth-power equation*

$$s_1 c_1^5 + s_2 c_2^5 + s_3 c_3^5 = 16\, d_0^5 = 2^4 d_0^5. \tag{28}$$

   (For arithmetic-progression specializations of $x_1^5 + x_2^5 + x_3^5 = dw^5$ with restricted common difference, modular level-lowering obstructions are established by Das, Dey, Koutsianas, and Tzanakis [10, Theorem 1.1].)

3. *In the natural split-conic subfamily $c_3 = c_1 + c_2$ ($s_1 = s_2 = +1, s_3 = -1$), the $[z^5] = 0$ condition $d_1^5 = 2\big(c_1^5 + c_2^5 - (c_1 + c_2)^5\big)$ satisfies the exact algebraic factorization*

$$(-d_1)^5 = 2\Big[(c_1 + c_2)^5 - c_1^5 - c_2^5\Big] = 10\, c_1 c_2 (c_1 + c_2)\big(c_1^2 + c_1 c_2 + c_2^2\big). \tag{29}$$

   For coprime $c_1, c_2 \in \mathbb{Z} \setminus \{0\}$, the four factors $c_1, c_2, c_1 + c_2$, and $c_1^2 + c_1 c_2 + c_2^2$ are pairwise coprime and $\gcd(10, c_1^2 + c_1 c_2 + c_2^2) = 1$, forcing $c_1^2 + c_1 c_2 + c_2^2 = w^5$ for some odd integer $w \geq 1$. Moreover:

   (a) *If the primes 2 and 5 divide distinct elements of $\{c_1, c_2, c_1 + c_2\}$, then (29) has no integer solutions (ruled out modulo 25).*

   (b) *If 10 divides a single element of $\{c_1, c_2, c_1 + c_2\}$, then (29) reduces to the simultaneous generalized Fermat and norm-form system*

$$x^5 + y^5 + 10000\, z^5 = 0, \qquad x^{10} + x^5 y^5 + y^{10} = w^5, \qquad \gcd(x, y) = 1, \quad 10 \nmid xy. \tag{30}$$

Proof. Parts (1) and (2) follow immediately by expanding $(c_i z + a_i)^5 + (c_i z - a_i)^5 = 2c_i^5 z^5 + 20 c_i^3 a_i^2 z^3 + 10 c_i a_i^4 z$. For Part (3), substituting $s_1 = s_2 = +1$, $s_3 = -1$, and $c_3 = c_1 + c_2$ into $[z^5]\, P_7(z) = 2(s_1 c_1^5 + s_2 c_2^5 + s_3 c_3^5) - d_1^5 = 0$ gives the intermediate relation

$$d_1^5 = 2\Big[c_1^5 + c_2^5 - (c_1 + c_2)^5\Big] = -2\Big[(c_1 + c_2)^5 - c_1^5 - c_2^5\Big],$$

or equivalently $(-d_1)^5 = -d_1^5 = 2\big[(c_1 + c_2)^5 - c_1^5 - c_2^5\big]$. By the Cauchy–Lamé quinta-power polynomial identity $(c_1 + c_2)^5 - c_1^5 - c_2^5 = 5c_1 c_2 (c_1 + c_2)(c_1^2 + c_1 c_2 + c_2^2)$, multiplying the entire

parenthesized difference by the outer factor 2 yields $2 \cdot 5c_1c_2(c_1+c_2)(c_1^2+c_1c_2+c_2^2) = 10c_1c_2(c_1+c_2)(c_1^2+c_1c_2+c_2^2)$ in (29).

Assume $\gcd(c_1,c_2)=1$ by homogeneity. Then $\gcd(c_1,c_1+c_2)=\gcd(c_2,c_1+c_2)=1$. Since $c_1^2+c_1c_2+c_2^2=(c_1+c_2)^2-c_1c_2$, any prime p dividing both $c_1c_2(c_1+c_2)$ and $c_1^2+c_1c_2+c_2^2$ would divide two of $\{c_1,c_2,c_1+c_2\}$, contradicting $\gcd(c_1,c_2)=1$. Thus $\gcd\left(c_1c_2(c_1+c_2),c_1^2+c_1c_2+c_2^2\right)=1$. Moreover, $c_1^2+c_1c_2+c_2^2\equiv 1 \pmod 2$ is always odd, and $4(c_1^2+c_1c_2+c_2^2)=(2c_1+c_2)^2+3c_2^2\not\equiv 0 \pmod 5$ since $-3\equiv 2 \pmod 5$ is a quadratic non-residue modulo 5. Hence $\gcd(10,c_1^2+c_1c_2+c_2^2)=1$. Since $c_1^2+c_1c_2+c_2^2=\left(c_1+\frac{1}{2}c_2\right)^2+\frac{3}{4}c_2^2>0$, unique factorization in (29) forces

$$c_1^2+c_1c_2+c_2^2=w^5 \qquad \text{and} \qquad 10\,c_1c_2(c_1+c_2)=u^5,$$

where $u=-d_1/w\in\mathbb{Z}\setminus\{0\}$. Since $10\mid u^5$ and $10=2\cdot 5$ is square-free, both 2 and 5 divide u, so $10\mid u$. Writing $u=-10z$ for a non-zero integer $z\in\mathbb{Z}\setminus\{0\}$ gives $u^5=-10^5z^5$, and dividing $10c_1c_2(c_1+c_2)=-10^5z^5$ by -10 yields the exact product equation

$$c_1\,c_2\,\big(-(c_1+c_2)\big)=10^4z^5=2^4\cdot 5^4z^5=10000\,z^5.$$

Because the three non-zero integers $c_1,c_2,-(c_1+c_2)$ are pairwise coprime with sum 0 and product $2^4\cdot 5^4z^5$, unique factorization in $\mathbb{Z}$ implies that each prime $p\in\{2,5\}$ divides a unique element of $\{c_1,c_2,-(c_1+c_2)\}$ to valuation $v_p=4+5k_p\equiv 4 \pmod 5$, while every prime $q\notin\{2,5\}$ has valuation $v_q\equiv 0 \pmod 5$ on each element. Consequently, absorbing fifth powers of 2 and 5 into the respective fifth-power factors:

- In case (a), 2 and 5 divide distinct terms, so up to permutation $\{c_1,c_2,-(c_1+c_2)\}=\{16x^5,625y^5,z^5\}$ with $5\nmid xz$. Summing to 0 gives $16x^5+z^5=-625y^5\equiv 0 \pmod{25}$. However, for $5\nmid x,z$, we have $x^5,z^5\in\{\pm1,\pm7\} \pmod{25}$, so $16x^5\in\{\pm16,\pm112\}\equiv\{\pm9,\pm12\} \pmod{25}$, which is disjoint from $-z^5\in\{\pm1,\pm7\} \pmod{25}$, a contradiction.
- In case (b), 2 and 5 divide the same term, so up to permutation $\{c_1,c_2,-(c_1+c_2)\}=\{x^5,y^5,2^45^4z^5\}=\{x^5,y^5,10000z^5\}$ with $\gcd(x,y)=1$ and $10\nmid xy$. Summing to 0 gives $x^5+y^5+10000z^5=0$, and substituting $\{c_1,c_2\}=\{x^5,y^5\}$ (or $\{x^5,-(x^5+y^5)\}$, which preserves $c_1^2+c_1c_2+c_2^2=x^{10}+x^5y^5+y^{10}$) into $c_1^2+c_1c_2+c_2^2=w^5$ yields (30). □

5 Proof of $v(5)\le 9$ via Weighted Identities

We now combine the rational parameterization (16) and a direct integer gap-sum solver for 4 weighted pairs $(2,2,1,1,1,1,1,1)$ to prove $v(5)\le 9$.

5.1 Direct Integer Gap-Sum Formulation and Root-Free Parameterization

Consider the 8-term weighted polynomial

$$P_8(z)=(2z+B_1)^5-(2z+B_2)^5+\sum_{i=2}^{4}\Big[(z+B_{2i-1})^5-(z+B_{2i})^5\Big], \qquad B_j\in\mathbb{Z}. \tag{31}$$

Proposition 5.1 (Direct Integer Gap-Sum Equations and Rational Parameterization)**.** *Let $\delta_i=B_{2i-1}-B_{2i}\in\mathbb{Z}$ and $\sigma_i=B_{2i-1}+B_{2i}\in\mathbb{Z}$ for $i=1,2,3,4$, with $\sigma_i\equiv\delta_i \pmod 2$.*

1. *The polynomial $P_8(z)$ in (31) is linear ($[z^5]=[z^4]=[z^3]=[z^2]=0$) if and only if*

$$16\delta_1+\delta_2+\delta_3+\delta_4=0, \tag{32}$$

$$8\delta_1\sigma_1+\delta_2\sigma_2+\delta_3\sigma_3+\delta_4\sigma_4=0, \tag{33}$$

$$12\delta_1\sigma_1^2+3\big(\delta_2\sigma_2^2+\delta_3\sigma_3^2+\delta_4\sigma_4^2\big)=-\big(4\delta_1^3+\delta_2^3+\delta_3^3+\delta_4^3\big). \tag{34}$$

2. *Eliminating* $\sigma_3 = -(8\delta_1\sigma_1 + \delta_2\sigma_2 + \delta_4\sigma_4)/\delta_3$ *(for* $\delta_3 \neq 0$*) after dividing* (34) *by* 3 *reduces* (34) *to the quadratic equation in* σ_2*:*

$$\delta_2(\delta_2+\delta_3)\sigma_2^2 + 2\delta_2(8\delta_1\sigma_1+\delta_4\sigma_4)\sigma_2 + 4\delta_1\delta_3\sigma_1^2 + (8\delta_1\sigma_1+\delta_4\sigma_4)^2 + \delta_3\delta_4\sigma_4^2 - \delta_3 R = 0, \quad (35)$$

where $R = -(4\delta_1^3+\delta_2^3+\delta_3^3+\delta_4^3)/3 \in \mathbb{Z}$*.*

3. *In particular, setting* $a = 2, X_1 = 32m, X_2 = 32n, X_3 = -32m, X_4 = -32n$*, and* $u_2 = mw, u_4 = 0$ *in* (16) *yields the root-free rational shifts*

$$u_1 = \frac{-1023m^2 + 3nw^2(m-n)}{6nw}, \quad u_2 = mw, \quad u_3 = \frac{-1023m^2 + 3nw^2(m+n)}{6nw}, \quad u_4 = 0, \quad (36)$$

with $d_1 = m, d_2 = 32n, d_3 = -32m, d_4 = -32n$*. When* $m = 4, w = 4$*, and* n *is any odd integer, all* 8 *scaled shifts* $b_j(n) = a_i(u_i \pm d_i)\frac{n}{2}$ *are integers divisible by* 8*, so* $B_j(n) = b_j(n)/8 \in \mathbb{Z}$ *gives an integer family* $P_8(z;n) = C(n)z + D(n)$ *with*

$$C(n) = \frac{30\,n^3\left(85n^4 - 678n^2 + 116281\right)}{8} \equiv 2 \pmod 4 \qquad (\forall n \equiv 1 \pmod 2). \quad (37)$$

Proof. Substituting $B_{2i-1} = (\sigma_i + \delta_i)/2$ and $B_{2i} = (\sigma_i - \delta_i)/2$ into (31) and expanding via Lemma 3.1 with $u_i = \sigma_i/(2a_i)$ and $d_i = \delta_i/(2a_i)$ directly produces (32)–(34) (noting that $4\delta_1^3 + \delta_2^3 + \delta_3^3 + \delta_4^3 \equiv 16\delta_1 + \delta_2 + \delta_3 + \delta_4 = 0 \pmod 3$ by Fermat's Little Theorem, so $R \in \mathbb{Z}$). For Part (2), dividing both sides of (34) by the outer factor 3 first yields

$$4\delta_1\sigma_1^2 + \delta_2\sigma_2^2 + \delta_3\sigma_3^2 + \delta_4\sigma_4^2 = R.$$

Setting $T = 8\delta_1\sigma_1 + \delta_4\sigma_4$, substituting $\sigma_3 = -(T + \delta_2\sigma_2)/\delta_3$ (for $\delta_3 \neq 0$), and multiplying through by δ_3 gives

$$4\delta_1\delta_3\sigma_1^2 + \delta_2\delta_3\sigma_2^2 + (T+\delta_2\sigma_2)^2 + \delta_3\delta_4\sigma_4^2 - \delta_3 R = 0,$$

which expands immediately to (35). For Part (3), substituting $a = 2$ (so $1 - a^{-10} = \frac{1023}{1024} = \frac{1023}{32^2}$), $X_1 = 32m, X_2 = 32n$, and $u_2 = mw$ into (16) gives

$$u_1 = \frac{-(32m)^4(1023/32^2) - 3(32n)(32n-32m)m^2w^2}{6(32m)(32n)mw} = \frac{-32^2\cdot 1023m^4 + 3\cdot 32^2 n(m-n)m^2w^2}{6\cdot 32^2 m^2 nw},$$

which simplifies by cancelling 32^2m^2 to (36), with $u_3 = u_1 + \frac{X_2}{X_1}u_2 = u_1 + nw$. Substituting $m = 4, w = 4$ into (36) and scaling the shifts by $n/16$ gives the 8 polynomial shifts $B_j(n) = b_j(n)/8$, where

$$b_1(n) = -2n^2 + 12n - 682, \quad b_2(n) = -2n^2 + 4n - 682, \quad b_3(n) = 16n^2 + 8n, \quad b_4(n) = -16n^2 + 8n,$$
$$b_5(n) = n^2 - 60n - 341, \quad b_6(n) = n^2 + 68n - 341, \quad b_7(n) = -16n^2, \quad b_8(n) = 16n^2.$$

To verify reproducibility of $[z^4] = [z^3] = [z^2] = 0$ directly from $B_j(n) = b_j(n)/8$, compute the gap-sum variables $\delta_i = B_{2i-1}(n) - B_{2i}(n)$ and $\sigma_i = B_{2i-1}(n) + B_{2i}(n)$:

$$(\delta_1,\delta_2,\delta_3,\delta_4) = \left(n, 4n^2, -16n, -4n^2\right), \qquad (\sigma_1,\sigma_2,\sigma_3,\sigma_4) = \left(-\frac{n^2-4n+341}{2}, 2n, \frac{n^2+4n-341}{4}, 0\right).$$

Then:

- $[z^4] = 0$: $16\delta_1 + \delta_2 + \delta_3 + \delta_4 = 16n + 4n^2 - 16n - 4n^2 = 0$.
- $[z^3] = 0$: $8\delta_1\sigma_1 + \delta_2\sigma_2 + \delta_3\sigma_3 + \delta_4\sigma_4 = -4n(n^2 - 4n + 341) + 8n^3 - 4n(n^2 + 4n - 341) + 0 = 0$.

- $[z^2] = 0$: Since $12\delta_1\sigma_1^2 = 12n\frac{(n^2-4n+341)^2}{4} = 3n(n^2-4n+341)^2$, $3\delta_2\sigma_2^2 = 3(4n^2)(4n^2) = 48n^4$, $3\delta_3\sigma_3^2 = 3(-16n)\frac{(n^2+4n-341)^2}{16} = -3n(n^2+4n-341)^2$, and $3\delta_4\sigma_4^2 = 0$, the difference of squares $(n^2-4n+341)^2 - (n^2+4n-341)^2 = (2n^2)(-8n+682) = -16n^3+1364n^2$ yields

$$\begin{aligned} 12\delta_1\sigma_1^2 + 3\big(\delta_2\sigma_2^2 + \delta_3\sigma_3^2 + \delta_4\sigma_4^2\big) &= 3n\Big[(n^2-4n+341)^2 - (n^2+4n-341)^2\Big] + 48n^4 \\ &= 3n\big(-16n^3+1364n^2\big) + 48n^4 = 4092n^3, \end{aligned}$$

  which equals $-\big(4\delta_1^3 + \delta_2^3 + \delta_3^3 + \delta_4^3\big) = -\big(4n^3 + 64n^6 - 4096n^3 - 64n^6\big) = 4092n^3$ identically in $\mathbb{Q}[n]$.

For every odd integer n, $n^2 \equiv 1 \pmod 8$, so each $b_j(n) \equiv 0 \pmod 8$ and $B_j(n) = b_j(n)/8 \in \mathbb{Z}$. Direct expansion of $P_8(z;n)$ with these shifts $B_j(n)$ yields (37). Since $n^2 \equiv 1 \pmod 8 \implies 85n^4 - 678n^2 + 116281 \equiv 8 \pmod{16}$, the integer $(85n^4 - 678n^2 + 116281)/8$ is always odd, so $v_2(C(n)) = 1$ ($2^1 \parallel C(n)$) for every odd n. □

5.2 The Main Theorem: Proof that $v(5) \le 9$

Lemma 5.2 (Bijectivity Criterion for Modular Fifth Powers)**.** *Let $C \ge 1$ be a positive integer. Then $\Delta(5, C) = 1$ (that is, the fifth-power map $y \mapsto y^5 \pmod C$ is a bijection from $\mathbb{Z}/C\mathbb{Z}$ onto itself) if and only if:*

1. *C is square-free ($p^2 \nmid C$ for every prime p), and*
2. *Every prime factor $p \mid C$ satisfies $p \not\equiv 1 \pmod 5$.*

Proof. By the Chinese Remainder Theorem, $y \mapsto y^5 \pmod C$ is a bijection on $\mathbb{Z}/C\mathbb{Z}$ if and only if $y \mapsto y^5 \pmod{p^e}$ is a bijection on $\mathbb{Z}/p^e\mathbb{Z}$ for every prime power $p^e \parallel C$. If $e \ge 2$, then $p \mid y^5 \implies p \mid y \implies p^5 \mid y^5 \implies p^2 \mid y^5$, so the residue class $p \bmod p^e$ has no fifth root and $y \mapsto y^5$ is not surjective. Thus $e = 1$ (C square-free) is necessary. When $e = 1$, $\mathbb{Z}/p\mathbb{Z} = \{0\} \cup (\mathbb{Z}/p\mathbb{Z})^\times$, where $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$. The map $y \mapsto y^5$ is an automorphism of $(\mathbb{Z}/p\mathbb{Z})^\times$ if and only if $\gcd(5, p-1) = 1$, i.e., $p \not\equiv 1 \pmod 5$ (noting that for $p = 5$, $p - 1 = 4$ is coprime to 5 and $y^5 \equiv y \pmod 5$ by Fermat's Little Theorem). □

Theorem 5.3 (Upper Bound $v(5) \le 9$)**.** *Every integer $N \in \mathbb{Z}$ is a sum of at most 9 signed fifth powers of integers ($v(5) \le 9$, so $5 \le v(5) \le 9$). Specifically:*

1. Compact Identity ($C_{\mathrm{I}} = 3,844,230$, $\max|B_i| = 39$, 8 summands across 7 distinct linear forms): *For all $z \in \mathbb{Z}$,*

$$(2z-38)^5 - (2z-37)^5 + (z+39)^5 - (z+35)^5 - (z-22)^5 - (z+22)^5 + 2(z+6)^5 = C_{\mathrm{I}}z + D_{\mathrm{I}}, \tag{38}$$

   where

$$C_{\mathrm{I}} = 3,844,230 = 2 \times 3 \times 5 \times 13 \times 9857, \qquad D_{\mathrm{I}} = 27,826,665.$$

   Equivalently, shifting $z \mapsto z - 6$ converts the repeated term $2(z+6)^5$ into $+2z^5$:

$$(2z-50)^5 - (2z-49)^5 + (z+33)^5 - (z+29)^5 - (z-28)^5 - (z+16)^5 + 2z^5 = 3,844,230\,z + 4,761,285. \tag{39}$$

2. Parametric Identity ($C_{\mathrm{II}} = 7,327,885,530$, from $(a, m, n, w) = (2, 28, 1, 4)$ in (36) after dividing the resulting integer shifts $a_i(u_i \pm d_i) = (-66672, -66784, 144, 80, -34256, -32464, -32, 32)$ by their common factor 16): *For all $z \in \mathbb{Z}$,*

$$(2z-4167)^5 - (2z-4174)^5 + (z+9)^5 - (z+5)^5 + (z-2141)^5 - (z-2029)^5 + (z-2)^5 - (z+2)^5 = C_{\mathrm{II}}z + D_{\mathrm{II}}, \tag{40}$$

   where

$$C_{\mathrm{II}} = 7,327,885,530 = 2 \times 3 \times 5 \times 7 \times 17 \times 2052629, \qquad D_{\mathrm{II}} = -10,178,299,260,675.$$

3. *For both* $C \in \{C_{\mathrm{I}}, C_{\mathrm{II}}\}$, *we have* $\Delta(5, C) = 1$. *Hence for every integer* $N \in \mathbb{Z}$, *there exists a unique integer representative* $y \in (-C_{\mathrm{I}}/2, C_{\mathrm{I}}/2] = [-1922114, 1922115]$ *such that* $y^5 \equiv N - D_{\mathrm{I}} \pmod{C_{\mathrm{I}}}$, *and setting* $z = (N - D_{\mathrm{I}} - y^5)/C_{\mathrm{I}} \in \mathbb{Z}$ *gives the explicit 9-term fifth-power representation*

$$N = (2z-38)^5 - (2z-37)^5 + (z+39)^5 - (z+35)^5 - (z-22)^5 - (z+22)^5 + (z+6)^5 + (z+6)^5 + y^5. \tag{41}$$

Proof. To verify (38), note that grouping the 8 fifth powers into the four pairs $[(2z-38)^5 - (2z-37)^5]$, $[(z+39)^5 - (z+35)^5]$, $[(z+6)^5 - (z-22)^5]$, and $[(z+6)^5 - (z+22)^5]$ gives the integer gap-sum solution $(\delta_1, \delta_2, \delta_3, \delta_4) = (-1, 4, 28, -16)$ and $(\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (-75, 74, -16, 28)$ of Proposition 5.1(1), since $16(-1) + 4 + 28 - 16 = 0$ and $8(-1)(-75) + 4(74) + 28(-16) + (-16)(28) = 600 + 296 - 448 - 448 = 0$. Expanding the left-hand side of (38) directly by binomial coefficients $\binom{5}{r}$ confirms each degree:

- $[z^5]$: $(2^5 - 2^5) + (1 - 1 - 1 - 1 + 2) = 0 + 0 = 0$.
- $[z^4]$: $5\,[16(-38) - 16(-37) + 39 - 35 - (-22) - 22 + 2(6)] = 5\,[-16 + 4 + 22 - 22 + 12] = 0$.
- $[z^3]$: $10\,\big[8((-38)^2 - (-37)^2) + (39^2 - 35^2) - ((-22)^2 + 22^2) + 2(6^2)\big] = 10(600 + 296 - 968 + 72) = 0$.
- $[z^2]$: $10\,\big[4((-38)^3 - (-37)^3) + (39^3 - 35^3) - ((-22)^3 + 22^3) + 2(6^3)\big] = 10(-16876 + 16444 + 432) = 0$.
- $[z^1]$: $5\,\big[2((-38)^4 - (-37)^4) + (39^4 - 35^4) - ((-22)^4 + 22^4) + 2(6^4)\big] = 5(421950 + 812816 - 468512 + 2592) = 5 \times 768846 = 3,844,230 = C_{\mathrm{I}}$.
- $[z^0]$: $(-38)^5 - (-37)^5 + 39^5 - 35^5 - 0 + 2(6^5) = -79235168 + 69343957 + 90224199 - 52521875 + 15552 = 27,826,665 = D_{\mathrm{I}}$.

Similarly, direct expansion confirms (40) (see Appendix A, Entry 3), where evaluating (36) at $(a, m, n, w) = (2, 28, 1, 4)$ produces $(u_1, u_2, u_3, u_4) = (-33364, 112, -33360, 0)$ and $(d_1, d_2, d_3, d_4) = (28, 32, -896, -32)$, whose raw shifts $a_i(u_i \pm d_i) = (-66672, -66784, 144, 80, -34256, -32464, -32, 32)$ have greatest common divisor 16; dividing by 16 yields $B_j = (-4167, -4174, 9, 5, -2141, -2029, -2, 2)$.

Now factor $C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ and $C_{\mathrm{II}} = 7,327,885,530 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17 \cdot 2052629$ (where $p_1 = 9857$ and $p_2 = 2052629$ are verified prime by trial division up to $\lfloor\sqrt{9857}\rfloor = 99$ and $\lfloor\sqrt{2052629}\rfloor = 1432$, as well as by primitive-root orders $\mathrm{ord}_{9857}(5) = 9856 = 2^7 \cdot 7 \cdot 11$ and $\mathrm{ord}_{2052629}(2) = 2052628 = 2^2 \cdot 513157$). Modulo 5, the prime factors $\{2, 3, 5, 13, 9857\}$ of C_{I} are congruent to $\{2, 3, 0, 3, 2\} \pmod 5$, and the prime factors $\{2, 3, 5, 7, 17, 2052629\}$ of C_{II} are congruent to $\{2, 3, 0, 2, 2, 4\} \pmod 5$; none is congruent to $1 \pmod 5$. By Lemma 5.2, $\Delta(5, C_{\mathrm{I}}) = \Delta(5, C_{\mathrm{II}}) = 1$ (confirmed also by direct enumeration of all 3,844,230 fifth powers modulo C_{I}). Thus (41) represents every $N \in \mathbb{Z}$ as a sum of 9 signed fifth powers. □

6 Catalogue of Polynomial Identities and Structural Minimality Results

Beyond proving $v(5) \leq 9$, the operator and gap-sum methods produce a collection of 20 listed low-complexity polynomial identities (16 independent up to affine equivalence or specialization) across degrees 1 to 5, including two identities that achieve provable coefficient minimality within their specified structural classes ($E = 120$ in Theorem 6.1 and $C = 720$ in Theorem 6.2).

6.1 The Three-Variable Cauchy–Lamé Master Quadratic Identity

Theorem 6.1 (Three-Variable Cauchy–Lamé Master Quadratic and Minimal $120z^2+60$ Identity). *1. For all $x,y,z\in\mathbb{Z}$, the 6 signed fifth powers satisfy the exact factored identity*

$$(z+x+y)^5-(z-x-y)^5-(z+x)^5+(z-x)^5-(z+y)^5+(z-y)^5=10\,xy(x+y)\big(6z^2+x^2+xy+y^2\big). \tag{42}$$

2. Setting $x=y=1$ yields the 6-summand (4-distinct-form) pure quadratic identity

$$(z+2)^5-(z-2)^5-2(z+1)^5+2(z-1)^5=120\,z^2+60=60(2z^2+1). \tag{43}$$

3. Universal Divisibility and Minimality of $E=120$ Within the Class of Origin-Symmetric Unit-Weight Identities: *In any general unit-weight quadratic identity $\sum_{i=1}^{m}s_i(z+B_i)^5=Ez^2+Cz+D$ in $\mathbb{Z}[z]$, $60\mid E$ holds universally, and in any origin-symmetric unit-weight quadratic identity $\sum_{i=1}^{k}c_i\big[(z+b_i)^5-(z-b_i)^5\big]=Ez^2+D$, $120\mid E$ holds universally. Thus (43) achieves the minimum possible positive quadratic coefficient $E=120=5!$ within the class of origin-symmetric unit-weight fifth-power identities.*

Proof. (1)–(2) For any b, $(z+b)^5-(z-b)^5=10bz^4+20b^3z^2+2b^5$. Applying this to $b\in\{x+y,-x,-y\}$, the z^4 coefficient is $10[(x+y)-x-y]=0$, the z^2 coefficient is $20[(x+y)^3-x^3-y^3]=60xy(x+y)$, and the constant term is $2[(x+y)^5-x^5-y^5]=10xy(x+y)(x^2+xy+y^2)$ by the classical Cauchy–Lamé factorization. Factoring out $10xy(x+y)$ yields (42), and setting $x=y=1$ gives (43).

(3) In any general unit-weight quadratic identity, vanishing of $[z^4]$ gives $\sum s_iB_i=0$, while $E=10\sum s_iB_i^3$. Since $B^3-B=B(B-1)(B+1)$ is a product of three consecutive integers, $6\mid(B_i^3-B_i)$ for every $B_i\in\mathbb{Z}$. Summing over i yields $\sum s_iB_i^3\equiv\sum s_iB_i=0\pmod 6$, so $60=10\times6\mid E$. In any origin-symmetric unit-weight quadratic identity $\sum_{i=1}^{k}c_i\big[(z+b_i)^5-(z-b_i)^5\big]=Ez^2+D$, vanishing of $[z^4]$ gives $\sum_{i=1}^{k}c_ib_i=0$ and the quadratic coefficient is $E=20\sum_{i=1}^{k}c_ib_i^3$. Since $\sum_{i=1}^{k}c_ib_i^3\equiv\sum_{i=1}^{k}c_ib_i=0\pmod 6$, it follows that $20\times6=120\mid E$. □

6.2 Pythagorean and Half-Integer Shifted $u^2+v^2=2w^2$ Linear Identities

Theorem 6.2 (Pythagorean and Minimal Unit-Weight PTE Slope $C=720$ Linear Identities). *1.* Pythagorean Fifth-Power Identity: *For every Pythagorean triple $a^2+b^2=c^2$ $(a,b,c\in\mathbb{Z})$,*

$$(z+c)^5+(z-c)^5+2z^5-(z+a)^5-(z-a)^5-(z+b)^5-(z-b)^5=20a^2b^2z. \tag{44}$$

In particular, for $(a,b,c)=(3,4,5)$,

$$(z+5)^5+(z-5)^5+2z^5-(z+4)^5-(z-4)^5-(z+3)^5-(z-3)^5=2880\,z. \tag{45}$$

2. Half-Integer Shifted $u^2+v^2=2w^2$ Identity: *For any odd integers u,v,w satisfying $u^2+v^2=2w^2$,*

$$\begin{aligned}&\left(z+\frac{u-1}{2}\right)^5+\left(z-\frac{u+1}{2}\right)^5+\left(z+\frac{v-1}{2}\right)^5+\left(z-\frac{v+1}{2}\right)^5\\&\quad-2\left(z+\frac{w-1}{2}\right)^5-2\left(z-\frac{w+1}{2}\right)^5=\frac{5}{16}\big(u^4+v^4-2w^4\big)(2z-1).\end{aligned} \tag{46}$$

Setting $(u,v,w)=(7,1,5)$ ($7^2+1^2=2\cdot5^2=50$) yields the minimal-slope unit-weight degree-3, size-4 PTE linear identity in $\mathbb{Z}[z]$ ($C=720,\max|B_i|=4$, 8 summands across 6 distinct linear forms; cf. Fuchs and Wright [2] and Borwein and Ingalls [5, p. 16]):

$$(z-4)^5+(z-1)^5+z^5+(z+3)^5-2(z-3)^5-2(z+2)^5=720\,z-360. \tag{47}$$

Proof. (1) Since $(z+x)^5+(z-x)^5 = 2z^5+20x^2z^3+10x^4z$, the left-hand side of (44) expands to

$$2(1+1-1-1)z^5+20(c^2+0^2-a^2-b^2)z^3+10(c^4-a^4-b^4)z = 10\big[(a^2+b^2)^2-a^4-b^4\big]z = 20a^2b^2z.$$

(2) When $u^2+v^2=2w^2$ with u,v,w odd, the symmetric tuples $\{\pm u,\pm v\}$ and $\{\pm w,\pm w\}$ satisfy $\sum\alpha_i^r=\sum\beta_i^r$ for $r=1,2,3$, so $\sum_{\alpha\in\{\pm u,\pm v\}}(X+\alpha)^5-2\big[(X+w)^5+(X-w)^5\big]=10(u^4+v^4-2w^4)X$. Substituting $X=2z-1$, every term $2z-1\pm\alpha$ is even; dividing by $2^5=32$ gives the coefficient $\frac{10}{32}=\frac{5}{16}$ in (46). Setting $(u,v,w)=(7,1,5)$ gives $\frac{5}{16}(2401+1-1250)(2z-1)=\frac{5}{16}(1152)(2z-1)=360(2z-1)=720z-360$. By Proposition 2.2, $720\mid C$ for all unit-weight degree-3, size-4 PTE identities, so $C=720$ is the minimum positive slope within that structural class. □

6.3 Master Table of Verified Polynomial Identities Across Degrees 1 to 5

Table 2 compiles the complete set of 20 listed verified polynomial identities (16 independent up to affine equivalence or specialization: Entry 2 is Entry 1 under $z\mapsto z-6$, Entry 9b is Entry 9 under $z\mapsto 2z-1291$, Entry 12 is Entry 11 at $x=y=1$, and Entry 13 is Entry 14 at $(a,b,k)=(1,2,1)$) established in this work across degrees 1 to 5, separating the total number of fifth-power summands counted with multiplicity from the number of distinct linear forms, and marking which identities are used directly in the proof of Theorem 5.3 and the Three-Tier Sieve (Proposition 7.4). Complete machine-verified expansions and the self-contained verification script are provided in Appendix A.

7 Local-to-Global Analysis, 11^5-Rigidity, and Multi-Tier Sieves

7.1 Exact Determination of $\Delta(5,p^e)$ and Absence of Local Obstructions at $s=5$

Proposition 7.1 (Complete Classification of $\Delta(5,p^e)$)**.** *For every prime p and integer $e\geq 1$, the modular fifth-power covering number $\Delta(5,p^e)$ is given by*

$$\Delta(5,p^e)=\begin{cases}5 & \text{if } p=11 \text{ and } e\geq 1,\\ 3 & \text{if } p\in\{31,41,61\} \text{ and } e\geq 1, \text{ or if } p=5 \text{ and } e\geq 2,\\ 2 & \text{if } p\equiv 1 \pmod 5 \text{ with } p\geq 71 \text{ and } e\geq 1, \text{ or if } p\not\equiv 1 \pmod 5 \text{ with } p\neq 5 \text{ and } e\geq 2,\\ 1 & \text{if } e=1 \text{ and } p\not\equiv 1 \pmod 5.\end{cases} \tag{48}$$

In particular, $\max_{m\geq 1}\Delta(5,m)=\max_{p,e}\Delta(5,p^e)=5$. Every integer $N\in\mathbb{Z}$ is locally representable as a sum of 5 fifth powers in $\mathbb{Z}_p$ for every prime p (and in $\mathbb{R}$), establishing the absence of local obstructions at $s=5$ (note that local representability across all completions $\mathbb{Z}_p$ and $\mathbb{R}$ does not imply a global local-to-global principle in $\mathbb{Z}$), whereas $N\equiv\pm 5 \pmod{11}$ is not representable as a sum of 4 fifth powers in $\mathbb{Z}_{11}$, giving the lower bound $v(5)\geq 5$.

Proof. The case $e=1, p\not\equiv 1 \pmod 5$ follows from Lemma 5.2. When $p\not\equiv 1 \pmod 5$ ($p\neq 5$) and $e\geq 2$, $\gcd(5,\phi(p^e))=\gcd(5,p^{e-1}(p-1))=1$, so every p-adic unit is a single fifth power mod p^e, while any multiple of p is the sum of two units ($pk\equiv 1+(pk-1) \pmod{p^e}$), giving $\Delta(5,p^e)=2$.

For $p=5$ and $e\geq 2$, we prove by induction on $e\geq 2$ that the image of $(\mathbb{Z}/5^e\mathbb{Z})^\times$ under $y\mapsto y^5 \pmod{5^e}$ is the exact preimage $U_e=\{a\in\mathbb{Z}/5^e\mathbb{Z} : a\equiv\pm 1,\pm 7 \pmod{25}\}$. For $e=2$, $(5k+r)^5\equiv r^5 \pmod{25}$ takes values $\{\pm 1,\pm 7\} \pmod{25}$ on $r\in\{\pm 1,\pm 2\}$. For the induction step from $e\geq 2$ to $e+1$, note that $2(e-1)+1=2e-1\geq e+1$, so for any $y\in\mathbb{Z}$ with $5\nmid y$ and any $t\in\mathbb{Z}$,

$$\big(y+5^{e-1}t\big)^5 = y^5+5y^4\big(5^{e-1}t\big)+10y^3\big(5^{2e-2}t^2\big)+\cdots\equiv y^5+5^ey^4t \pmod{5^{e+1}}.$$

Since $y^4\equiv 1 \pmod 5$, as t varies over $\{0,1,2,3,4\} \pmod 5$, $y^5+5^ey^4t$ hits all 5 lifts of y^5 mod 5^e to $\mathbb{Z}/5^{e+1}\mathbb{Z}$, completing the induction. Modulo 25, the 5-element set of fifth powers $S_{25}=$

Table 2: Catalogue of 20 listed verified polynomial identities (16 independent up to affine equivalence or specialization) for fifth powers in $\mathbb{Z}[z]$ and $\mathbb{Z}[x, y, z]$ across degrees 1 to 5, distinguishing summand count (with multiplicity) from distinct linear forms.

Category & Degree	Summands	Forms	Exact Polynomial Identity	Role & Parameters
1. Linear (deg 1), $\Delta(5, C) = 1$ (Compact)	8	7	$(2z-38)^5 - (2z-37)^5 + (z+39)^5 - (z+35)^5 - (z-22)^5 - (z+22)^5 + 2(z+6)^5 = 3844230z + 27826665$	**Thm 5.3 & Tier 3**; $\max\|B_i\| = 39$, $C = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$
2. Linear (deg 1), $\Delta(5, C) = 1$ ($+2z^5$ form)	8	7	$(2z-50)^5 - (2z-49)^5 + (z+33)^5 - (z+29)^5 - (z-28)^5 - (z+16)^5 + 2z^5 = 3844230z + 4761285$	Shifted Thm 5.3; at $z = 0$, $2z^5 = 0$ vanishes (6 summands)
3. Linear (deg 1), $\Delta(5, C) = 1$ (Parametric)	8	8	$(2z-4167)^5 - (2z-4174)^5 + (z+9)^5 - (z+5)^5 + (z-2141)^5 - (z-2029)^5 + (z-2)^5 - (z+2)^5 = 7327885530z - 10178299260675$	Thm 5.3(2); $C = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17 \cdot 2052629$
4. Linear (deg 1), Single-Digit Weighted	8	8	$(2z-7)^5 - (2z-8)^5 + (z-9)^5 - (z+5)^5 + (z-6)^5 - (z-2)^5 + (z+9)^5 - (z+7)^5 = 39930z - 11715$	Small-coeff. example; $\max\|B_i\| = 9$, $C = 2 \cdot 3 \cdot 5 \cdot 11^3$
5. Linear (deg 1), $(2, 2, 1, 1)$-Weighted	8	7	$2(2z-13)^5 - (2z-15)^5 - (2z-12)^5 + (z-15)^5 - (z+2)^5 + (z+14)^5 - (z+13)^5 = 159930z - 327255$	**Tier 2 Sieve**; $\max\|B_i\| = 15$, $C = 2 \cdot 3^2 \cdot 5 \cdot 1777$
6. Linear (deg 1), Symmetric Pair	8	8	$(2z-14)^5 - (2z-15)^5 + (z-15)^5 - (z-2)^5 + (z+2)^5 - (z+7)^5 + (z+15)^5 - (z+13)^5 = 229350z - 166485$	Small-coeff. example; $\max\|B_i\| = 15$, $C = 2 \cdot 3 \cdot 5^2 \cdot 11 \cdot 139$
7. Linear (deg 1), Minimal Unit PTE	8	6	$(z-4)^5 + (z-1)^5 + z^5 + (z+3)^5 - 2(z-3)^5 - 2(z+2)^5 = 720z - 360$	**Tier 1 Sieve**; $\max\|B_i\| = 4$, proved minimal unit PTE $C = 720$
8. Linear (deg 1), Pythagorean $(3, 4, 5)$	8	7	$(z+5)^5 + (z-5)^5 + 2z^5 - (z+4)^5 - (z-4)^5 - (z+3)^5 - (z-3)^5 = 2880z$	Thm 6.2(1); $\max\|B_i\| = 5$, $D = 0$
9. Linear (deg 1), 6-Term Elliptic $(E_2, P_1 + 3P_0)$	6	6	$(2z-3636)^5 - (2z-4221)^5 + (z+7763)^5 - (z+4847)^5 + (z-6138)^5 - (z+6138)^5 = C_2 z + D_2$	Thm 3.4; 6 summands; $v_2(C_2) = 1$, D_2 odd
9b. Linear (deg 1), 6-Term Elliptic (E_2, P_1)	6	6	$(2z-4690)^5 - (2z-5860)^5 + (z+16817)^5 - (z+13567)^5 + (z-10985)^5 - (z+10985)^5 = 16C_2 z + D_2'$	Thm 3.4; 6 summands; $p = 144/25$ on E_2
10. Linear (deg 1), 6-Term Elliptic (E_6, Q_1)	6	6	$(6z-29436)^5 - (6z-29454)^5 + (z+19349)^5 - (z+12059)^5 + (z-15309)^5 - (z+15309)^5 = C_6 z + D_6$	Thm 3.4; 6 summands; $31 \nmid C_6$
11. Quadratic (deg 2), Cauchy–Lamé Master	6	6	$(z+x+y)^5 - (z-x-y)^5 - (z+x)^5 + (z-x)^5 - (z+y)^5 + (z-y)^5 = 10xy(x+y)(6z^2 + x^2 + xy + y^2)$	Thm 6.1(1); exact 3-variable factorization
12. Quadratic (deg 2), Minimal Symmetric	6	4	$(z+2)^5 - (z-2)^5 - 2(z+1)^5 + 2(z-1)^5 = 120z^2 + 60$	Thm 6.1(2); $\max\|B_i\| = 2$, proved minimal symmetric $E = 120$
13. Quadratic (deg 2), 4-Term Pure Even	4	4	$(z+16)^5 - (z-16)^5 - (2z+1)^5 + (2z-1)^5 = 81840z^2 + 2097150$	Rem 7.6; 4 summands; $[z^5] = [z^4] = [z^3] = [z^1] = 0$
14. Quadratic (deg 2), 4-Term Parametric	4	4	$(az + kb^4)^5 - (az - kb^4)^5 - (bz + ka^4)^5 + (bz - ka^4)^5 = 20k^3a^2b^2(b^{10} - a^{10})z^2 + 2k^5(b^{20} - a^{20})$	Thm 3.2(1); holds for all $a, b, k \in \mathbb{Z}$
15. Quadratic (deg 2), Weighted 4-Distinct	6	4	$2(z+8)^5 - 2(z-8)^5 - (2z+1)^5 + (2z-1)^5 = 20400z^2 + 131070$	Small-coeff. example; $\max\|B_i\| = 8$, $E = 20400$
16. Quadratic (deg 2), Single-Digit Weighted	6	5	$(2z-2)^5 - (2z-1)^5 + (z+5)^5 + (z+1)^5 - 2(z-5)^5 = 3480z^2 - 2970z + 9345$	Small-coeff. example; $\max\|B_i\| = 5$, $E = 3480$
17. Cubic (deg 3), Low-Coeff. Pure Odd	4	4	$(z+2)^5 + (z-2)^5 - (z+1)^5 - (z-1)^5 = 60z^3 + 150z$	Small-coeff. example; $\max\|B_i\| = 2$; $[z^5] = [z^4] = [z^2] = [z^0] = 0$
18. Cubic (deg 3), Weighted Monomial	4	4	$(2z+1)^5 - (2z)^5 - (z+8)^5 + (z-8)^5 = 80z^3 - 10200z^2 + 10z - 65535$	Small-coeff. example; includes $-(2z)^5$; $[z^5] = [z^4] = 0$
19. Quintic (deg 5), Zero z^4, z^3, z^2, z^0	4	4	$(z+8)^5 + (z-8)^5 - (4z+1)^5 - (4z-1)^5 = -2046z^5 + 40920z$	Uses $c_1 = 2^2, a_2 = 2^3$ to kill z^3

$\{0, \pm 1, \pm 7\} \pmod{25}$ has 2-fold sumset $S_{25} + S_{25} = \{0, \pm 1, \pm 2, \pm 6, \pm 7, \pm 8, \pm 11\} \pmod{25}$ of size $13 < 25$ (omitting $\{\pm 3, \pm 4, \pm 5, \pm 9, \pm 10, \pm 12\}$), whereas Table 3 exhibits an explicit 3-summand decomposition $r \equiv u_1 + u_2 + u_3 \pmod{25}$ for every residue class $r \in \{-12, \ldots, 12\} \pmod{25}$ with a unit fifth power $u_1 = y_1^5 \in \{\pm 1, \pm 7\} \pmod{25}$ and $u_2 = y_2^5, u_3 = y_3^5 \in S_{25}$. For any $N \in \mathbb{Z}/5^e\mathbb{Z}$, choosing $y_2, y_3 \in \{0, \pm 1, \pm 2\}$ according to Table 3 for $r \equiv N \pmod{25}$ gives $N - y_2^5 - y_3^5 \equiv u_1 \in \{\pm 1, \pm 7\} \pmod{25}$, which lies in U_e and hence equals $y_1^5 \pmod{5^e}$ for some $y_1 \in (\mathbb{Z}/5^e\mathbb{Z})^\times$, proving $\Delta(5, 5^e) = 3$ for all $e \geq 2$.

Table 3: Complete modulo-25 three-summand decomposition $r \equiv y_1^5 + y_2^5 + y_3^5 \pmod{25}$ with unit first summand $u_1 = y_1^5 \in \{\pm 1, \pm 7\} \pmod{25}$ ($y_1 \in \{\pm 1, \pm 2\}$) and $y_2, y_3 \in \{0, \pm 1, \pm 2\}$.

r mod 25	$u_1 + u_2 + u_3$	(y_1, y_2, y_3)	r mod 25	$u_1 + u_2 + u_3$	(y_1, y_2, y_3)	r mod 25	$u_1 + u_2 + u_3$	(y_1, y_2, y_3)
-12	$-1+7+7=13$	$(-1,2,2)$	-3	$-1-1-1=-3$	$(-1,-1,-1)$	$+5$	$-1-1+7=5$	$(-1,-1,2)$
-11	$7+0+7=14$	$(2,0,2)$	-2	$-1+0-1=-2$	$(-1,0,-1)$	$+6$	$-1+0+7=6$	$(-1,0,2)$
-10	$1+7+7=15$	$(1,2,2)$	-1	$1-1-1=-1$	$(1,-1,-1)$	$+7$	$1-1+7=7$	$(1,-1,2)$
-9	$-1-1-7=-9$	$(-1,-1,-2)$	0	$1+0-1=0$	$(1,0,-1)$	$+8$	$1+0+7=8$	$(1,0,2)$
-8	$-1+0-7=-8$	$(-1,0,-2)$	$+1$	$1+0+0=1$	$(1,0,0)$	$+9$	$1+1+7=9$	$(1,1,2)$
-7	$1-1-7=-7$	$(1,-1,-2)$	$+2$	$1+0+1=2$	$(1,0,1)$	$+10$	$-1-7-7=-15$	$(-1,-2,-2)$
-6	$1+0-7=-6$	$(1,0,-2)$	$+3$	$1+1+1=3$	$(1,1,1)$	$+11$	$-7+0-7=-14$	$(-2,0,-2)$
-5	$1+1-7=-5$	$(1,1,-2)$	$+4$	$-7-7-7=-21$	$(-2,-2,-2)$	$+12$	$1-7-7=-13$	$(1,-2,-2)$
-4	$7+7+7=21$	$(2,2,2)$						

For $p \equiv 1 \pmod 5$, let $H = \{x^5 : x \in \mathbb{F}_p^\times\}$ be the subgroup of non-zero fifth powers of index 5 and order $(p-1)/5$, and set $H_0 = H \cup \{0\}$. For the eight primes $p \equiv 1 \pmod 5$ with $p < 180$, direct coset evaluation gives:

- $p = 11$: $H = \{\pm 1\}$, $|2H_0| = 5$, $|3H_0| = 7$, $|4H_0| = 9$, $|5H_0| = 11 \implies \Delta(5, 11) = 5$.
- $p = 31$: $H = \{\pm 1, \pm 5, \pm 6\}$, $|2H_0| = 19 < 31$ (missing the two cosets $\pm 3H$), $|3H_0| = 31 \implies \Delta(5, 31) = 3$.
- $p = 41$: $H = \{\pm 1, \pm 3, \pm 9, \pm 14\}$, $|2H_0| = 33 < 41$ (missing the coset $7H$), $|3H_0| = 41 \implies \Delta(5, 41) = 3$.
- $p = 61$: $H = \{\pm 1, \pm 11, \pm 13, \pm 14, \pm 21, \pm 29\}$, $|2H_0| = 49 < 61$ (missing the coset $4H$), $|3H_0| = 61 \implies \Delta(5, 61) = 3$.
- $p \in \{71, 101, 131, 151\}$: $|H| \in \{14, 20, 26, 30\}$ and $2H_0 = \mathbb{F}_p$ in each case $\implies \Delta(5, p) = 2$.

For all primes $p \equiv 1 \pmod 5$ with $p \geq 181$, let χ be a multiplicative character of $\mathbb{F}_p^\times$ of order 5. Every $a \in H \cup \{0\}$ is already a sum of two fifth powers ($x^5 + 0^5$ or $1^5 + (-1)^5$). For $a \in \mathbb{F}_p^\times \setminus H$, $\sum_{i=1}^4 \chi^i(a) = -1$ and $\chi(-1) = \chi((-1)^5) = 1$. For any $u \in \mathbb{F}_p^\times$, the number of fifth roots $x \in \mathbb{F}_p^\times$ satisfying $x^5 = u$ is given by the character indicator $\sum_{i=0}^4 \chi^i(u) \in \{0, 5\}$. Consequently, the number $N^*(a)$ of solutions $(x, y) \in (\mathbb{F}_p^\times)^2$ to $x^5 + y^5 = a$ expands over $u + v = a$ with $u, v \in \mathbb{F}_p^\times$ (i.e., $u \in \mathbb{F}_p \setminus \{0, a\}$ and $v = a - u$) as

$$N^*(a) = \sum_{u \in \mathbb{F}_p \setminus \{0,a\}} \left(\sum_{i=0}^4 \chi^i(u) \right) \left(\sum_{j=0}^4 \chi^j(a-u) \right) = \sum_{i=0}^4 \sum_{j=0}^4 S_{i,j}(a), \qquad S_{i,j}(a) := \sum_{u \in \mathbb{F}_p \setminus \{0,a\}} \chi^i(u)\chi^j(a-u).$$

We evaluate each of the 25 pairs $(i, j) \in \{0, 1, 2, 3, 4\}^2$ explicitly:

1. *Trivial pair* $(i, j) = (0, 0)$*:* Since $\chi^0(u) = \chi^0(a - u) = 1$ for all $u \in \mathbb{F}_p \setminus \{0, a\}$, we have $S_{0,0}(a) = |\mathbb{F}_p \setminus \{0, a\}| = p - 2$.
2. *Boundary terms* $j = 0, 1 \leq i \leq 4$ *and* $i = 0, 1 \leq j \leq 4$*:* Since $\sum_{u \in \mathbb{F}_p^\times} \chi^i(u) = 0$ for $1 \leq i \leq 4$, omitting $u = a$ gives $S_{i,0}(a) = \sum_{u \in \mathbb{F}_p \setminus \{0,a\}} \chi^i(u) = -\chi^i(a)$, and by symmetry $S_{0,j}(a) = -\chi^j(a)$. Summing over $1 \leq i \leq 4$ and $1 \leq j \leq 4$ contributes

$$\sum_{i=1}^4 S_{i,0}(a) + \sum_{j=1}^4 S_{0,j}(a) = -2 \sum_{i=1}^4 \chi^i(a) = -2(-1) = +2.$$

3. *Inverse-character diagonal pairs* $1 \le i, j \le 4$ *with* $i + j \equiv 0 \pmod 5$ *(i.e.,* $j = 5 - i$*):* Setting $w = u(a-u)^{-1}$, as u runs over $\mathbb{F}_p \setminus \{0, a\}$, $w = \frac{a}{a-u} - 1$ runs bijectively over $\mathbb{F}_p \setminus \{0, -1\}$. Thus $\chi^i(u)\chi^{5-i}(a-u) = \chi^i(w)$, so

$$S_{i,5-i}(a) = \sum_{w \in \mathbb{F}_p \setminus \{0,-1\}} \chi^i(w) = -\chi^i(-1) = -1, \qquad \sum_{i=1}^{4} S_{i,5-i}(a) = -4\chi(-1) = -4.$$

4. *Non-conjugate Jacobi sum pairs* $1 \le i, j \le 4$ *with* $i + j \not\equiv 0 \pmod 5$: Substituting $u = a\tilde{u}$ with $\tilde{u} \in \mathbb{F}_p \setminus \{0, 1\}$ factors out $\chi^{i+j}(a)$ and yields $S_{i,j}(a) = \chi^{i+j}(a)\, J(\chi^i, \chi^j)$, where $J(\chi^i, \chi^j) = \sum_{\tilde{u}+\tilde{v}=1} \chi^i(\tilde{u})\chi^j(\tilde{v})$ is the classical Jacobi sum satisfying $|J(\chi^i, \chi^j)| = \sqrt{p}$ for each of the $4 \times 3 = 12$ pairs.

Combining all four contributions yields the exact identity

$$N^*(a) = (p-2) - 2\sum_{i=1}^{4} \chi^i(a) - 4\chi(-1) + \sum_{\substack{1 \le i,j \le 4 \\ i+j \not\equiv 0 \pmod 5}} \chi^{i+j}(a) J(\chi^i, \chi^j) = p - 4 + \sum_{\substack{1 \le i,j \le 4 \\ i+j \not\equiv 0 \pmod 5}} \chi^{i+j}(a) J(\chi^i, \chi^j).$$

By the triangle inequality over the 12 Jacobi sums of modulus $\sqrt{p}$, we obtain

$$N^*(a) \ge (p-2) + 2 - 4 - 12\sqrt{p} = p - 4 - 12\sqrt{p} = (\sqrt{p} - 6)^2 - 40 > 0 \qquad \forall p \ge 152,$$

proving $\Delta(5, p) = 2$ for all $p \equiv 1 \pmod 5$ with $p \ge 71$.

Finally, to show $\Delta(5, p^e) = \Delta(5, p)$ for every prime $p \equiv 1 \pmod 5$ and $e \ge 2$ (where $r = \Delta(5, p) \ge 2$), let $A \in \mathbb{Z}/p^e\mathbb{Z}$ have reduction $a \equiv A \pmod p$. If $a \not\equiv 0 \pmod p$, any representation $a \equiv \sum_{j=1}^{r} y_j^5 \pmod p$ necessarily contains at least one unit summand $y_1 \not\equiv 0 \pmod p$. If $a \equiv 0 \pmod p$, since $r = \Delta(5, p) \ge 2$, we choose the representation $0 \equiv 1^5 + (-1)^5 + \sum_{j=3}^{r} 0^5 \pmod p$, which likewise has unit first summand $y_1 = 1 \not\equiv 0 \pmod p$ (whereas in the separate case $p \not\equiv 1 \pmod 5$ with $e = 1$ where $\Delta(5, p) = 1$, $a = 0$ is represented directly by the single term $0^5 = 0$ without needing Hensel lifting). Because $\frac{d}{dy}(y^5)\big|_{y=y_1} = 5y_1^4 \not\equiv 0 \pmod p$ for $p \ne 5$ and $p \nmid y_1$, Hensel's lemma lifts $y_1 \pmod p$ uniquely to $\tilde{y}_{1,e} \in \mathbb{Z}/p^e\mathbb{Z}$ satisfying $\tilde{y}_{1,e}^5 + \sum_{j=2}^{r} y_j^5 \equiv A \pmod{p^e}$, proving $\Delta(5, p^e) = \Delta(5, p)$.

Moreover, to pass from modular representability modulo every p^e ($e \ge 1$) to exact representability in the p-adic integers $\mathbb{Z}_p$, we provide both an explicit compatible-tower construction and a self-contained topological compactness proof in $\mathbb{Z}_p^5$:

- *Explicit Compatible Lifting Tower in* $\mathbb{Z}_p$*:* For $p \ne 5$, fixing the integer summands $y_2, \ldots, y_r \in \mathbb{Z}$ from the modulo-p decomposition of N mod p with unit first summand $y_1 \not\equiv 0 \pmod p$, the unique Hensel lifts $\tilde{y}_{1,e} \in (\mathbb{Z}/p^e\mathbb{Z})^\times$ satisfy $\tilde{y}_{1,e+1} \equiv \tilde{y}_{1,e} \pmod{p^e}$ for all $e \ge 1$ and define $y_1 = (\tilde{y}_{1,e})_{e \ge 1} \in \varprojlim_e (\mathbb{Z}/p^e\mathbb{Z}) = \mathbb{Z}_p$ satisfying $y_1^5 + \sum_{j=2}^{r} y_j^5 = N$ in $\mathbb{Z}_p$. For $p = 5$, fixing $y_2, y_3 \in \{0, \pm 1, \pm 2\}$ from Table 3 for N mod 25 and starting from $\tilde{y}_{1,2} \in \{\pm 1, \pm 2\}$ satisfying $\tilde{y}_{1,2}^5 + y_2^5 + y_3^5 \equiv N \pmod{25}$, for each $e \ge 2$ write $\tilde{y}_{1,e}^5 + y_2^5 + y_3^5 - N = 5^e m_e$ in $\mathbb{Z}$ and define $\tilde{y}_{1,e+1} := \tilde{y}_{1,e} + 5^{e-1} t_e$ where $t_e \in \{0, 1, 2, 3, 4\}$ satisfies $t_e \equiv -m_e(\tilde{y}_{1,e}^4)^{-1} \equiv -m_e \pmod 5$; then $\tilde{y}_{1,e+1}^5 + y_2^5 + y_3^5 \equiv N \pmod{5^{e+1}}$ and $\tilde{y}_{1,e+1} \equiv \tilde{y}_{1,e} \pmod{5^{e-1}}$, so $(\tilde{y}_{1,e})_{e \ge 2}$ is a 5-adic Cauchy sequence converging to an element $y_1 \in \mathbb{Z}_5^\times$ with $y_1^5 + y_2^5 + y_3^5 = N$ in $\mathbb{Z}_5$.
- *Nested Closed Cylinder Preimages in the Compact Space* $\mathbb{Z}_p^5$*:* Equivalently, for every prime p and integer $e \ge 1$, let $\pi_e : \mathbb{Z}_p^5 \to (\mathbb{Z}/p^e\mathbb{Z})^5$ be the continuous canonical reduction homomorphism and define the closed preimage

$$\mathcal{Z}_e(N) := \pi_e^{-1}\big(X_e(N)\big) = \left\{ \mathbf{y} = (y_1, \ldots, y_5) \in \mathbb{Z}_p^5 : \sum_{j=1}^{5} y_j^5 \equiv N \pmod{p^e} \right\} \subset \mathbb{Z}_p^5,$$

  where $X_e(N) = \{\bar{\mathbf{y}} \in (\mathbb{Z}/p^e\mathbb{Z})^5 : \sum_{j=1}^{5} \bar{y}_j^5 \equiv N \pmod{p^e}\}$ is non-empty since $\Delta(5, p^e) \le 5$. Because π_e is continuous and surjective and $(\mathbb{Z}/p^e\mathbb{Z})^5$ is discrete, each $\mathcal{Z}_e(N)$ is a non-empty closed subset of the compact Hausdorff space $\mathbb{Z}_p^5$, and $\mathcal{Z}_1(N) \supset \mathcal{Z}_2(N) \supset \cdots \supset \mathcal{Z}_e(N) \supset \ldots$ forms a decreasing nested chain of non-empty closed subsets. Hence every finite subcollection $\{\mathcal{Z}_{e_1}(N), \ldots, \mathcal{Z}_{e_k}(N)\}$ has non-empty intersection $\bigcap_{i=1}^{k} \mathcal{Z}_{e_i}(N) = \mathcal{Z}_{\max_i e_i}(N) \ne \emptyset$. By the finite-intersection property of the compact space $\mathbb{Z}_p^5$, the global intersection $\bigcap_{e=1}^{\infty} \mathcal{Z}_e(N) = \{\mathbf{y} \in \mathbb{Z}_p^5 : \sum_{j=1}^{5} y_j^5 = N \text{ in } \mathbb{Z}_p\}$ is non-empty.

Since $y \mapsto y^5$ is surjective on $\mathbb{R}$, every $N \in \mathbb{Z}$ is locally representable as a sum of 5 fifth powers in $\mathbb{R}$ and in $\mathbb{Z}_p$ for every prime p. Conversely, in $\mathbb{Z}_{11}$, fifth powers reduce modulo 11 to $\{-1, 0, 1\}$, so any sum of 4 fifth powers in $\mathbb{Z}_{11}$ lies in $\{-4, \ldots, 4\} \pmod{11}$ and omits $\pm 5 \pmod{11}$, proving $v(5) \geq 5$. □

7.2 The 11^5-Rigidity Sieve for $s = 5$ vs. $s = 6$

Although $\Delta(5, 11^e) = 5$ prevents a local obstruction at $s = 5$, the critical exponent $s = k = 5$ exhibits severe 11-adic rigidity:

Proposition 7.2 (The 11^5-Rigidity Sieve for 5-Term Fifth Powers)**.** *Let $N \in \mathbb{Z}$ have a 5-term representation $N = \sum_{i=1}^{5} x_i^5$ with $x_i \in \mathbb{Z}$.*

1. *If $N \equiv \pm 5 \pmod{11}$ (e.g., $N = 5, 6$), then $x_i^5 \equiv \pm 1 \pmod{11}$ for all $i = 1, \ldots, 5$. In particular, for $N = 6 \equiv -5 \pmod{11}$, a 5-term solution is*

$$6 = 17^5 + 13^5 + 7^5 - 16^5 - 15^5, \tag{49}$$

   which immediately yields 6-term representations for $N = 5$ and $N = 7$:

$$5 = 17^5 + 13^5 + 7^5 - 16^5 - 15^5 - 1^5, \qquad 7 = 17^5 + 13^5 + 7^5 - 16^5 - 15^5 + 1^5. \tag{50}$$

2. *If $N \equiv \pm 4 \pmod{11}$ (such as the open case $N = 7 \equiv -4 \pmod{11}$ [24]), then up to permutation of indices, $x_1^5 \equiv x_2^5 \equiv x_3^5 \equiv x_4^5 \equiv \pm 1 \pmod{11}$ and $11 \mid x_5$. Writing $x_5 = 11w$, we have $11^5 = 161,051 \mid x_5^5$, so the remaining four variables must satisfy the 4-term congruence modulo 161,051:*

$$x_1^5 + x_2^5 + x_3^5 + x_4^5 \equiv N \pmod{161,051}, \qquad x_1^5 \equiv x_2^5 \equiv x_3^5 \equiv x_4^5 \equiv \pm 1 \pmod{11}. \tag{51}$$

Proof. Since $x_i^5 \bmod 11 \in \{-1, 0, 1\}$, the only partition of $+5 \bmod 11$ (or $-5 \bmod 11$) into 5 elements of $\{-1, 0, 1\}$ is $(1, 1, 1, 1, 1)$ (respectively $(-1, -1, -1, -1, -1)$), and the only partition of $+4 \bmod 11$ (or $-4 \bmod 11$) into 5 elements of $\{-1, 0, 1\}$ is $(1, 1, 1, 1, 0)$ (respectively $(-1, -1, -1, -1, 0)$), as $4 - 11 = -7 < -5$. Thus for $N \equiv \pm 4 \pmod{11}$, $x_5^5 \equiv 0 \pmod{11} \implies 11 \mid x_5 \implies 11^5 = 161,051 \mid x_5^5$, yielding (51). Direct arithmetic confirms $17^5 + 13^5 + 7^5 - 16^5 - 15^5 = 1419857 + 371293 + 16807 - 1048576 - 759375 = 6$. □

Remark 7.3 (Finite-Box Computational Search for $s = 5, 6$). Separately from the theorem-level bounds above, an explicit finite-box meet-in-the-middle computation restricted to the box $|x_i| \leq 30$ represents 28 of the first 30 positive integers as a sum of 6 fifth powers (all $1 \leq N \leq 30$ except $N \in \{8, 23\}$ within $|x_i| \leq 30$; the complete list of all 28 verified 6-term decompositions in $|x_i| \leq 30$ is recorded in Table 6 in Appendix A, including $1 = 1^5 + 0^5 + 0^5 + 0^5 + 0^5 + 0^5$, $9 = 16^5 + 13^5 - 17^5 - 1^5 - 1^5 - 1^5$, $10 = 16^5 + 13^5 - 17^5 - 1^5 - 1^5 + 0^5 = 21^5 - 19^5 - 14^5 - 14^5 - 13^5 - 11^5$, and $14 = 15^5 - 13^5 - 13^5 - 7^5 + 2^5 + 0^5 = 27^5 + 20^5 + 8^5 - 28^5 - 13^5 + 0^5$). Enlarging the search box to $|x_i| \leq 60$ represents $N = 8$ as a sum of 6 fifth powers via

$$8 = (-45)^5 + (-33)^5 + (-25)^5 + 31^5 + 40^5 + 40^5 = -184528125 - 39135393 - 9765625 + 28629151 + 102400000 + 102400000,$$

leaving only $N = 23$ unrepresented among $1 \leq N \leq 30$ for $|x_i| \leq 60$, and similarly represents 360 in 5 fifth powers via $360 = (-8)^5 + 7^5 + 7^5 + (-3)^5 + (-3)^5$.

7.3 Three-Tier Multi-Identity Sieve for Compact 9-Term Representations

Rather than using the modulus $C_{\mathrm{I}} = 3,844,230$ for every integer $N \in \mathbb{Z}$, combining Identities 7, 5, and 1 of Table 2 yields a three-tier sieve that dramatically reduces both the maximum modular residue bound $\max |y| \leq C/2$ and the linear parameter scale $|z| = |N - D - y^5|/C$:

Proposition 7.4 (Three-Tier Multi-Identity Sieve for $v(5) \le 9$)**.** *Every integer $N \in \mathbb{Z}$ is represented as a sum of at most 9 signed fifth powers via the following priority hierarchy:*

1. ***Tier 1 (***$C = 720, \max|B_i| = 4$***, covers*** $\frac{7}{16} = \frac{63}{144} = 43.75\%$ ***of*** $\mathbb{Z}$ ***with*** $|y| \le 360$***):*** *Whenever $N + 360 \not\equiv \pm 3 \pmod 9$ (i.e. $N \not\equiv 3, 6 \pmod 9$, since $360 \equiv 0 \pmod 9$) and $N + 360 \in \{1, 3, 5, 7, 9, 11, 13, 15, 0\} \pmod{16}$ (all odd integers and $N \equiv 8 \pmod{16}$ with $N \not\equiv 3, 6 \pmod 9$), there exists $y \in (-360, 360]$ with $y^5 \equiv N + 360 \pmod{720}$. Setting $z = (N + 360 - y^5)/720 \in \mathbb{Z}$ gives*

$$N = (z-4)^5 + (z-1)^5 + z^5 + (z+3)^5 - 2(z-3)^5 - 2(z+2)^5 + y^5,$$

reducing the maximum residue $|y|$ (and the $|y^5|/C$ contribution to the z-parameter scale) by a slope ratio of $3844230/720 = 5,339.2$ compared to Tier 3.

2. ***Tier 2 (***$C = 159,930 = 2 \cdot 3^2 \cdot 5 \cdot 1777, \max|B_i| = 15$***, standalone coverage*** $\frac{7}{9} = \frac{112}{144} \approx 77.78\%$ ***of*** $\mathbb{Z}$ ***and incremental coverage*** $\frac{58}{144} = \frac{29}{72} \approx 40.28\%$ ***of*** $\mathbb{Z}$ ***with*** $|y| \le 79,965$***):*** *For any integer N satisfying $N + 327255 \not\equiv \pm 3 \pmod 9$ (i.e. $N \not\equiv 0, 6 \pmod 9$, since $327255 \equiv 6 \equiv -3 \pmod 9$; this includes all integers $N \equiv 3 \pmod 9$ as well as all even integers with $N \not\equiv 8 \pmod{16}$ and $N \in \{1, 2, 4, 5, 7, 8\} \pmod 9$ missed by Tier 1), there exists $y \in (-79965, 79965]$ with $y^5 \equiv N + 327255 \pmod{159930}$, and $z = (N + 327255 - y^5)/159930 \in \mathbb{Z}$ gives*

$$N = 2(2z-13)^5 - (2z-15)^5 - (2z-12)^5 + (z-15)^5 - (z+2)^5 + (z+14)^5 - (z+13)^5 + y^5,$$

with a modulus slope ratio of $3844230/159930 = 24.037$ compared to Tier 3. Together, Tiers 1 and 2 cover $\frac{63+58}{144} = \frac{121}{144} \approx 84.0278\%$ of all integers in $\mathbb{Z}$.

3. ***Tier 3 (***$C_{\mathrm{I}} = 3,844,230, \max|B_i| = 39$***, covers*** 100% ***of*** $\mathbb{Z}$ ***with*** $|y| \le 1,922,115$***):*** *For the remaining $\frac{23}{144} \approx 15.9722\%$ of integers missed by both Tier 1 and Tier 2—namely, all integers $N \equiv 6 \pmod 9$ ($\frac{1}{9} = \frac{16}{144}$ of $\mathbb{Z}$) together with those integers $N \equiv 0 \pmod 9$ satisfying $N \in \{0, 2, 4, 6, 10, 12, 14\} \pmod{16}$ ($\frac{1}{9} \times \frac{7}{16} = \frac{7}{144}$ of $\mathbb{Z}$)—Theorem 5.3(3) provides the universal representation (41).*

Proof. For Tier 1, $720 = 16 \cdot 9 \cdot 5$. By Lemma 5.2, $y \mapsto y^5 \pmod 5$ is a bijection on $\mathbb{Z}/5\mathbb{Z}$, while $\{y^5 \bmod 9\} = \{0, \pm 1, \pm 2, \pm 4\}$ (7 of 9 residue classes, omitting $\pm 3 \equiv 3, 6 \pmod 9$) and $\{y^5 \bmod 16\} = \{0\} \cup (\mathbb{Z}/16\mathbb{Z})^\times$ (9 of 16 residue classes: all 8 odd residues plus $0 \pmod{16}$). Since $360 \equiv 0 \pmod 9$ and $360 \equiv 8 \pmod{16}$, by the Chinese Remainder Theorem $N + 360 \equiv y^5 \pmod{720}$ is solvable if and only if $N \equiv N + 360 \not\equiv 3, 6 \pmod 9$ and $N + 360 \equiv 0$ or odd $\pmod{16}$ (i.e. $N \equiv 8 \pmod{16}$ or N odd), covering $\frac{7}{9} \cdot \frac{9}{16} = \frac{7}{16} = \frac{63}{144} = 43.75\%$ of $\mathbb{Z}$, and Identity 7 of Table 2 gives N.

For Tier 2, $159,930 = 2 \cdot 9 \cdot 5 \cdot 1777$, where $2, 5, 1777 \not\equiv 1 \pmod 5$ are prime factors of exponent 1; hence $N + 327255 \equiv y^5 \pmod{159930}$ is solvable whenever $N + 327255 \not\equiv 3, 6 \pmod 9$. Since $327255 = 9 \times 36361 + 6 \equiv 6 \pmod 9$, this condition is $N + 6 \not\equiv 3, 6 \pmod 9 \iff N \not\equiv 6, 0 \pmod 9$, which has standalone density $\frac{7}{9} = \frac{112}{144} \approx 77.78\%$ of $\mathbb{Z}$ via Identity 5. Combining Tier 1 and Tier 2 modulo $\mathrm{lcm}(9, 16) = 144$, an integer N is missed by both Tier 1 and Tier 2 if and only if $N \equiv 0$ or $6 \pmod 9$ (missed by Tier 2) and either $N \equiv 3, 6 \pmod 9$ or $N \in \{0, 2, 4, 6, 10, 12, 14\} \pmod{16}$ (missed by Tier 1). This intersection consists of:

- all $N \equiv 6 \pmod 9$ (16 residue classes mod 144, density $\frac{1}{9} = \frac{16}{144}$), and
- all $N \equiv 0 \pmod 9$ with $N \in \{0, 2, 4, 6, 10, 12, 14\} \pmod{16}$ (7 residue classes mod 144, density $\frac{1}{9} \cdot \frac{7}{16} = \frac{7}{144}$),

giving an exact leftover density of $\frac{16+7}{144} = \frac{23}{144} \approx 15.9722\%$ (and combined Tier 1 $\cup$ Tier 2 density $1 - \frac{23}{144} = \frac{121}{144} \approx 84.0278\%$). Finally, Tier 3 covers all remaining integers via Theorem 5.3(3). Every sample decomposition in Table 4 strictly follows the priority rule Tier 1 $\to$ Tier 2 $\to$ Tier 3 and has been verified by exact integer arithmetic. $\square$

Table 4: Explicit compact 9-term fifth-power representations generated by the Three-Tier Sieve under the strict priority rule Tier 1 → Tier 2 → Tier 3 (all rows independently verified by exact integer arithmetic; for $N = 6 \equiv 6 \pmod 9$, which is missed by Tiers 1 and 2, Tier 3 gives $y = -404,049$ and $z = 2,801,307,716,959,828,698,333$).

N	Tier	Base y	Parameter z	Explicit 9-Term Signed Fifth-Power Decomposition
1	Tier 1	-359	8,282,086,203	$8282086199^5 + 8282086202^5 + 8282086203^5 + 8282086206^5 - 2(8282086200)^5 - 2(8282086205)^5 - 359^5$
2	Tier 2	$-24,553$	55,794,691,661,757,925	$2(2z-13)^5 - (2z-15)^5 - (2z-12)^5 + (z-15)^5 - (z+2)^5 + (z+14)^5 - (z+13)^5 - 24553^5$
3	Tier 2	$-29,682$	144,057,497,616,267,233	$2(2z-13)^5 - (2z-15)^5 - (2z-12)^5 + (z-15)^5 - (z+2)^5 + (z+14)^5 - (z+13)^5 - 29682^5$
5	Tier 1	-115	27,935,517	$27935513^5 + 27935516^5 + 27935517^5 + 27935520^5 - 2(27935514)^5 - 2(27935519)^5 - 115^5$
7	Tier 1	-113	25,589,378	$25589374^5 + 25589377^5 + 25589378^5 + 25589381^5 - 2(25589375)^5 - 2(25589380)^5 - 113^5$
8	Tier 1	-352	7,505,520,595	$7505520591^5 + 7505520594^5 + 7505520595^5 + 7505520598^5 - 2(7505520592)^5 - 2(7505520597)^5 - 352^5$
11	Tier 1	131	$-53,582,624$	$(-53582628)^5 + (-53582625)^5 + (-53582624)^5 + (-53582621)^5 - 2(-53582627)^5 - 2(-53582622)^5 + 131^5$
13	Tier 1	133	$-57,799,716$	$(-57799720)^5 + (-57799717)^5 + (-57799716)^5 + (-57799713)^5 - 2(-57799719)^5 - 2(-57799714)^5 + 133^5$

7.4 Congruence-Class Reductions to $s = 8, 7, 6$ and Mixed Waring–Kamke Strategy

We conclude by separating theorem-level congruence-class representations for $s \in \{6, 7, 8\}$ (Corollary 7.5, where $W(5, N)$ denotes the least number of signed fifth powers needed to represent an individual integer $N \in \mathbb{Z}$) from heuristic mixed-degree Waring–Kamke reductions (Remark 7.6).

Corollary 7.5 (Congruence Families with $W(5, N) \le 8, 7, 6$)**.** *1.* 6-Term and 7-Term Arithmetic Progressions: *Every integer N in the arithmetic progressions $N = C_2 z + D_2$ (23), $N = 16C_2 z + D_2'$ (24), or $N = C_6 z + D_6$ (25) (as well as $N = 4,761,285$, evaluating (39) at $z = 0$ where $+2z^5$ vanishes) is a sum of at most 6 signed fifth powers, and every integer of the form $N = C_2 z + D_2 + y^5$ or $N = C_6 z + D_6 + y^5$ ($y \in \mathbb{Z}$) is a sum of at most 7 signed fifth powers.*

2. 8-Term Representations on Positive-Density Congruence Sets: *Every integer in the union of arithmetic progressions $\bigcup_j \{C_j z + D_j : z \in \mathbb{Z}\}$ across all 8-term identities $P_8^{(j)}(z) = C_j z + D_j$ is a sum of at most 8 signed fifth powers. Moreover, adding two fifth powers $y_1^5 + y_2^5$ to the 6-term identity (23) covers a positive proportion of all residue classes modulo $C_2 = 2 \cdot 3^6 \cdot 5^4 \cdot 11 \cdot 13^3 \cdot 31 \cdot 97 \cdot 211$. Specifically, by the Chinese Remainder Theorem and Proposition 7.1, the 2-fold fifth-power sumset $2S_5(p^e) = \{y_1^5 + y_2^5 \bmod p^e\}$ modulo each prime-power factor $p^e \parallel C_2$ has exact cardinality:*

- *For $p^e \in \{2, 3^6 = 729, 13^3 = 2197, 97, 211\}$: since $\Delta(5, p^e) \le 2$, we have $|2S_5(p^e)| = p^e$, giving local coverage fraction 1.*
- *For $p^e = 5^4 = 625$: as shown in Proposition 7.1, a residue mod 5^e is a sum of two fifth powers if and only if its reduction mod 25 lies in the 13-element set $S_{25} + S_{25} = \{0, \pm1, \pm2, \pm6, \pm7, \pm8, \pm11\} \pmod{25}$, giving $|2S_5(625)| = 13 \times 25 = 325$ and local coverage fraction $\frac{325}{625} = \frac{13}{25}$.*
- *For $p^e = 11$: $|2S_5(11)| = |\{0, \pm1, \pm2\}| = 5$, giving local coverage fraction $\frac{5}{11}$.*
- *For $p^e = 31$: $|2S_5(31)| = 19$ (omitting the two 6-element cosets $\pm 3H$), giving local coverage fraction $\frac{19}{31}$.*

Multiplying across all eight prime-power factors of C_2 yields the exact density

$$\frac{2}{2} \cdot \frac{729}{729} \cdot \frac{325}{625} \cdot \frac{5}{11} \cdot \frac{2197}{2197} \cdot \frac{19}{31} \cdot \frac{97}{97} \cdot \frac{211}{211} = 1 \cdot 1 \cdot \frac{13}{25} \cdot \frac{5}{11} \cdot 1 \cdot \frac{19}{31} \cdot 1 \cdot 1 = \frac{247}{1705} \approx 14.4868\%$$

of all residue classes modulo C_2 (and hence of all integers in $\mathbb{Z}$) using $6 + 2 = 8$ signed fifth powers.

Remark 7.6 (Heuristic Mixed-Degree $(2,5,5,5)$ and $(2,5,5)$ Waring–Kamke Strategy). As a conditional/heuristic reduction strategy toward $s \in \{7,8\}$ (distinct from the theorem-level results proved above), substituting the 4-term pure quadratic identity $(z+16)^5-(z-16)^5-(2z+1)^5+(2z-1)^5 = 81840z^2+2097150$ reduces the 7-term fifth-power representation problem to the mixed degree-$(2,5,5,5)$ Diophantine equation

$$81840\,z^2+y_1^5+y_2^5+y_3^5 = N-2097150, \tag{52}$$

whose heuristic reciprocal degree sum is $\frac{1}{2}+\frac{1}{5}+\frac{1}{5}+\frac{1}{5}=\frac{11}{10}>1$. Since $81840 = 11\times 7440 \equiv 0 \pmod{11}$ and $2097150 = 11\times 190650 \equiv 0 \pmod{11}$ (in accordance with the degree-2 $\mathbb{F}_{11}$-Bastien obstruction for $m=4\le 5$), reducing (52) modulo 11 gives $N \equiv y_1^5+y_2^5+y_3^5 \pmod{11}$, so (52) is locally unobstructed on the 7 residue classes

$$N \equiv \{-3,-2,-1,0,1,2,3\} \pmod{11}.$$

At $m=6$, substituting the minimal symmetric quadratic identity (43) ($120z^2+60$, where $120 \equiv -1 \not\equiv 0 \pmod{11}$ and $60 \equiv 5 \pmod{11}$) with two fifth powers $y_1^5+y_2^5$ ($s=8$) gives $120z^2+60+y_1^5+y_2^5 \equiv -z^2+5+y_1^5+y_2^5 \pmod{11}$, which covers all 11 residue classes in $\mathbb{Z}/11\mathbb{Z}$ (since $\{-z^2+5 \bmod 11\} = \{0,1,2,4,5,7\}$ has sumset $\mathbb{Z}/11\mathbb{Z}$ with $\{-2,-1,0,1,2\}$). Note that because (43) is quadratic in z, it lies outside the direct linear hypotheses of Theorem 4.1; by contrast, for any 6-term linear identity $\sum_{i=1}^6 s_i(A_iz+B_i)^5 = Cz+D$, Theorem 4.1 forces $11 \mid C$, requiring $s \ge m+5 = 11$ for any single 6-term linear identity and $s \ge \max(6, 2k+5) \in \{6,7,9,11\}$ across a family where $2k = |\{i : 11 \nmid A_i\}|$ (requiring $s \ge 11$ whenever all six leading coefficients A_i are 11-adic units, i.e., $k=3$). Retaining the quadratic term $120z^2$ thus restores full modulo-11 coverage at $s=8$ with unit leading weights. Establishing global solvability of these mixed-degree ternary and quaternary equations for all admissible integers remains an open analytic problem.

A Appendix: Reproducible Coefficient Verification for the 20-Identity Catalogue and Finite-Box Decompositions

To provide a self-contained, machine-checkable verification of all 20 listed polynomial identities (16 independent up to affine equivalence or specialization) in Table 2, Table 5 records the exact binomial coefficient sums $[z^r] = \binom{5}{r}\sum_i s_i A_i^r B_i^{5-r}$ ($r=5,4,3,2,1,0$) for each univariate identity in $\mathbb{Z}[z]$ (Entries 1–10, including 9b, and 12–19), while Entry 11 is verified symbolically in Theorem 6.1(1) and Entry 14 is verified symbolically by $(az+kb^4)^5-(az-kb^4)^5 = 10a^4kb^4z^4+20a^2k^3b^{12}z^2+2k^5b^{20}$, where subtracting the $(a \leftrightarrow b)$ pair cancels $10a^4b^4kz^4$ identically and leaves $20k^3a^2b^2(b^{10}-a^{10})z^2+2k^5(b^{20}-a^{20})$. In addition, Table 6 records the complete list of 28 verified 6-term fifth-power representations for $1 \le N \le 30$ ($N \notin \{8,23\}$) with $|x_i| \le 30$ referenced in Remark 7.3, and Subsection A.1 archives the complete self-contained Python verification script (`audit_verify.py`, 70 automated checks) and its execution output.

A.1 Reproducible Python Verification Script and Execution Certificate

All exact integer, rational, and symbolic computations in Sections 2–7 and Tables 2–6 are verified by the complete self-contained script `audit_verify.py` below using only the Python standard library and `sympy`. Running the script executes all 70 automated checks—reproducing every coefficient check in Table 5, the hand-expansion brackets in Theorem 3.4, the bijection test of $y \mapsto y^5$ over all 3,844,230 residues modulo C_{I}, 12,001 end-to-end 9-term representations (41) up to $|N| \sim 10^{40}$, the exact Three-Tier Sieve densities and all 9 rows of Table 4 (including the caption row $N=6$) under the priority rule Tier 1 → Tier 2 → Tier 3, the 28 six-term decompositions in Table 6 (plus $N=8$ for $|x_i| \le 60$), the 1,101 degree-3 size-4 PTE pairs in $[-20,20]$ for Proposition 2.2, the elliptic curve arithmetic and Nagell–Lutz coordinates on $E_2(\mathbb{Q})$ and $E_6(\mathbb{Q})$ in Theorem 3.4, and the symbolic

Table 5: Exact degree-by-degree coefficient verification $[z^r]\,P(z) = \binom{5}{r}\sum_i s_i A_i^r B_i^{5-r}$ for Entries 1–19 (including 9b) of Table 2.

Entry	**Weighted Pairs / Tuples** (s_i, A_i, B_i)	$[z^5]$	$[z^4]$	$[z^3]$	$[z^2]$	**Non-Zero Target Coefficients** ($[z^5], [z^3], [z^2], [z^1], [z^0]$)
1	$(+,2,-38),(-,2,-37),(+,1,39),(-,1,35),$ $(-,1,-22),(-,1,22),(+2,1,6)$	0	0	0	0	$[z^1] = 3,844,230, [z^0] = 27,826,665$
2	$(+,2,-50),(-,2,-49),(+,1,33),(-,1,29),$ $(-,1,-28),(-,1,16),(+2,1,0)$	0	0	0	0	$[z^1] = 3,844,230, [z^0] = 4,761,285$
3	$(+,2,-4167),(-,2,-4174),(+,1,9),$ $(-,1,5),(+,1,-2141),(-,1,-2029),$ $(+,1,-2),(-,1,2)$	0	0	0	0	$[z^1] = 7,327,885,530$, $[z^0] = -10,178,299,260,675$
4	$(+,2,-7),(-,2,-8),(+,1,-9),(-,1,5),$ $(+,1,-6),(-,1,-2),(+,1,9),(-,1,7)$	0	0	0	0	$[z^1] = 39,930, [z^0] = -11,715$
5	$(+2,2,-13),(-,2,-15),(-,2,-12),$ $(+,1,-15),(-,1,2),(+,1,14),(-,1,13)$	0	0	0	0	$[z^1] = 159,930, [z^0] = -327,255$
6	$(+,2,-14),(-,2,-15),(+,1,-15),$ $(-,1,-2),(+,1,2),(-,1,7),(+,1,15),$ $(-,1,13)$	0	0	0	0	$[z^1] = 229,350, [z^0] = -166,485$
7	$(+,1,-4),(+,1,-1),(+,1,0),(+,1,3),$ $(-2,1,-3),(-2,1,2)$	0	0	0	0	$[z^1] = 720, [z^0] = -360$
8	$(+,1,5),(+,1,-5),(+2,1,0),(-,1,4),$ $(-,1,-4),(-,1,3),(-,1,-3)$	0	0	0	0	$[z^1] = 2,880, [z^0] = 0$
9	$(+,2,-3636),(-,2,-4221),(+,1,7763),$ $(-,1,4847),(+,1,-6138),(-,1,6138)$	0	0	0	0	$[z^1] = 13,972,565,906,763,750,$ $[z^0] = 8,797,920,299,940,110,625$
9b	$(+,2,-4690),(-,2,-5860),(+,1,16817),$ $(-,1,13567),(+,1,-10985),(-,1,10985)$	0	0	0	0	$[z^1] = 223,561,054,508,220,000 = 16C_2$, $[z^0] = 570,150,770,968,195,560,000$
10	$(+,6,-29436),(-,6,-29454),(+,1,19349),$ $(-,1,12059),(+,1,-15309),(-,1,15309)$	0	0	0	0	$[z^1] = 539,939,269,033,164,000,$ $[z^0] = 842,901,695,917,396,315,200$
11	3-variable Cauchy–Lamé (Theorem 6.1(1))	0	0	0	$E_{x,y}$	$[z^2] = 60xy(x+y)$, $[z^0] = 10xy(x+y)(x^2+xy+y^2)$
12	$(+,1,2),(-,1,-2),(-2,1,1),(+2,1,-1)$	0	0	0	120	$[z^2] = 120, [z^1] = 0, [z^0] = 60$
13	$(+,1,16),(-,1,-16),(-,2,1),(+,2,-1)$	0	0	0	81840	$[z^2] = 81,840, [z^1] = 0, [z^0] = 2,097,150$
14	$(+,a,kb^4),(-,a,-kb^4),(-,b,ka^4),$ $(+,b,-ka^4)$	0	0	0	$E_{a,b}$	$[z^2] = 20k^3a^2b^2(b^{10}-a^{10})$, $[z^0] = 2k^5(b^{20}-a^{20})$
15	$(+2,1,8),(-2,1,-8),(-,2,1),(+,2,-1)$	0	0	0	20400	$[z^2] = 20,400, [z^1] = 0, [z^0] = 131,070$
16	$(+,2,-2),(-,2,-1),(+,1,5),(+,1,1),$ $(-2,1,-5)$	0	0	0	3480	$[z^2] = 3,480, [z^1] = -2,970, [z^0] = 9,345$
17	$(+,1,2),(+,1,-2),(-,1,1),(-,1,-1)$	0	0	60	0	$[z^3] = 60, [z^2] = 0, [z^1] = 150, [z^0] = 0$
18	$(+,2,1),(-,2,0),(-,1,8),(+,1,-8)$	0	0	80	−10200	$[z^3] = 80, [z^2] = -10,200, [z^1] = 10, [z^0] = -65,535$
19	$(+,1,8),(+,1,-8),(-,4,1),(-,4,-1)$	−2046	0	0	0	$[z^5] = -2,046, [z^1] = 40,920, [z^0] = 0$

Table 6: Complete finite-box certificate of 6-term signed fifth-power representations $N = \sum_{i=1}^{6} x_i^5$ ($|x_i| \le 30$) for all 28 integers $1 \le N \le 30$ with $N \notin \{8, 23\}$ (Remark 7.3; for $|x_i| \le 60$, $8 = (-45)^5 + (-33)^5 + (-25)^5 + 31^5 + 40^5 + 40^5$).

N	**Verified 6-Term Decomposition** $\sum_{i=1}^{6} x_i^5$	N	**Verified 6-Term Decomposition** $\sum_{i=1}^{6} x_i^5$
1	$1^5+0^5+0^5+0^5+0^5+0^5$	16	$6^5-5^5-5^5-4^5-3^5-3^5$
2	$1^5+1^5+0^5+0^5+0^5+0^5$	17	$(-15)^5+13^5+13^5+7^5-1^5+0^5$
3	$1^5+1^5+1^5+0^5+0^5+0^5$	18	$(-15)^5+13^5+13^5+7^5+0^5+0^5$
4	$1^5+1^5+1^5+1^5+0^5+0^5$	19	$(-15)^5+13^5+13^5+7^5+1^5+0^5$
5	$17^5-16^5-15^5+13^5+7^5-1^5$	20	$17^5-16^5-13^5+2^5+0^5+0^5$
6	$17^5-16^5-15^5+13^5+7^5+0^5$	21	$5^5-4^5-4^5-4^5-2^5+0^5$
7	$17^5-16^5-15^5+13^5+7^5+1^5$	22	$5^5-4^5-4^5-4^5-2^5+1^5$
9	$(-17)^5+16^5+13^5-1^5-1^5-1^5$	24	$(-10)^5+9^5+9^5-7^5-4^5-3^5$
10	$21^5-19^5-14^5-14^5-13^5-11^5$	25	$(-25)^5+23^5+20^5+11^5-8^5+4^5$
11	$(-17)^5+16^5+13^5-1^5+0^5+0^5$	26	$(-17)^5+16^5+15^5-13^5-7^5+2^5$
12	$(-17)^5+16^5+13^5+0^5+0^5+0^5$	27	$2^5-1^5-1^5-1^5-1^5-1^5$
13	$(-17)^5+16^5+13^5+1^5+0^5+0^5$	28	$2^5-1^5-1^5-1^5-1^5+0^5$
14	$27^5+20^5+8^5-28^5-13^5+0^5$	29	$2^5-1^5-1^5-1^5+0^5+0^5$
15	$15^5-13^5-13^5-7^5+2^5+1^5$	30	$2^5-1^5-1^5+0^5+0^5+0^5$

identities of Theorems 3.2, 4.3, 6.1, 6.2, Lemma 3.3, and Proposition 5.1—and prints `70 of 70 checks passed`.

```
#!/usr/bin/env python3
"""Exact-arithmetic audit checks for the v(5) <= 9 paper. Standard library + sympy only.
Run: python3 audit_verify.py (about a minute; the bijection test over Z/C_I is the slow part)
"""
import itertools, math, random
from fractions import Fraction as F
from math import comb
import sympy as sp
from sympy import factorint, isprime

results = []
def check(name, ok, detail=""):
    results.append((name, bool(ok)))
    print(f"[{'PASS' if ok else 'FAIL'}] {name}" + (f" -- {detail}" if detail else ""))

def coeffs(terms):
    """terms = [(sign, A, B)] meaning sign*(A z + B)^5; returns [z^0 .. z^5]."""
    c = [0]*6
    for s, A, B in terms:
        for r in range(6):
            c[r] += s*comb(5, r)*A**r*B**(5-r)
    return c

# ---------------------------------------------------------------- 1. polynomial identities
IDENT = {
    "(38) C_I": ([(1,2,-38),(-1,2,-37),(1,1,39),(-1,1,35),(-1,1,-22),(-1,1,22),(2,1,6)], (3844230, 27826665)),
    "(39)": ([(1,2,-50),(-1,2,-49),(1,1,33),(-1,1,29),(-1,1,-28),(-1,1,16),(2,1,0)], (3844230, 4761285)),
    "(40) C_II": ([(1,2,-4167),(-1,2,-4174),(1,1,9),(-1,1,5),(1,1,-2141),(-1,1,-2029),(1,1,-2),(-1,1,2)],
                  (7327885530, -10178299260675)),
    "T2-4": ([(1,2,-7),(-1,2,-8),(1,1,-9),(-1,1,5),(1,1,-6),(-1,1,-2),(1,1,9),(-1,1,7)], (39930, -11715)),
    "T2-5": ([(2,2,-13),(-1,2,-15),(-1,2,-12),(1,1,-15),(-1,1,2),(1,1,14),(-1,1,13)], (159930, -327255)),
    "T2-6": ([(1,2,-14),(-1,2,-15),(1,1,-15),(-1,1,-2),(1,1,2),(-1,1,7),(1,1,15),(-1,1,13)], (229350, -166485))
    ,
    "(47)": ([(1,1,-4),(1,1,-1),(1,1,0),(1,1,3),(-2,1,-3),(-2,1,2)], (720, -360)),
    "(45)": ([(1,1,5),(1,1,-5),(2,1,0),(-1,1,4),(-1,1,-4),(-1,1,3),(-1,1,-3)], (2880, 0)),
    "(23) E2": ([(1,2,-3636),(-1,2,-4221),(1,1,7763),(-1,1,4847),(1,1,-6138),(-1,1,6138)], (13972565906763750,
     8797920299940110625)),
    "(24) E2": ([(1,2,-4690),(-1,2,-5860),(1,1,16817),(-1,1,13567),(1,1,-10985),(-1,1,10985)],
     (223561054508220000, 570150770968195560000)),
    "(25) E6": ([(1,6,-29436),(-1,6,-29454),(1,1,19349),(-1,1,12059),(1,1,-15309),(-1,1,15309)],
     (539939269033164000, 842901695917396315200)),
    "4.3 p=3": ([(1,3,-2),(-1,3,-4),(1,1,-179),(-1,1,-163),(1,1,-234),(-1,1,-248),(1,1,-85),(-1,1,75)],
     (-2216246400, 161024169680)),
    "4.3 p=5": ([(1,5,-4),(-1,5,-6),(1,1,2415),(-1,1,2399),(1,1,-2046),(-1,1,-2066),(1,1,-646),(-1,1,640)],
     (1017973008000, 4252444454692752)),
}
for name, (terms, (C, D)) in IDENT.items():
    c = coeffs(terms)
    check(f"linear identity {name}", c[2:] == [0,0,0,0] and (c[1], c[0]) == (C, D), f"C={c[1]}, D={c[0]}")

QUAD = {
    "T2-12": ([(1,1,2),(-1,1,-2),(-2,1,1),(2,1,-1)], [60,0,120,0,0,0]),
    "T2-13": ([(1,1,16),(-1,1,-16),(-1,2,1),(1,2,-1)], [2097150,0,81840,0,0,0]),
    "T2-15": ([(2,1,8),(-2,1,-8),(-1,2,1),(1,2,-1)], [131070,0,20400,0,0,0]),
    "T2-16": ([(1,2,-2),(-1,2,-1),(1,1,5),(1,1,1),(-2,1,-5)], [9345,-2970,3480,0,0,0]),
    "T2-17": ([(1,1,2),(1,1,-2),(-1,1,1),(-1,1,-1)], [0,150,0,60,0,0]),
    "T2-18": ([(1,2,1),(-1,2,0),(-1,1,8),(1,1,-8)], [-65535,10,-10200,80,0,0]),
    "T2-19": ([(1,1,8),(1,1,-8),(-1,4,1),(-1,4,-1)], [0,40920,0,0,0,-2046]),
}
for name, (terms, want) in QUAD.items():
    check(f"identity {name}", coeffs(terms) == want)

z, x, y, a, b, k = sp.symbols('z x y a b k')
check("Thm 6.1 (42)", sp.expand((z+x+y)**5-(z-x-y)**5-(z+x)**5+(z-x)**5-(z+y)**5+(z-y)**5 - 10*x*y*(x+y)*(6*z
    **2+x**2+x*y+y**2)) == 0)
check("Table 2 entry 14 (symbolic)", sp.expand((a*z+k*b**4)**5-(a*z-k*b**4)**5-(b*z+k*a**4)**5+(b*z-k*a**4)**5
    - (20*k**3*a**2*b**2*(b**10-a**10)*z**2 + 2*k**5*(b**20-a**20))) == 0)
c23 = coeffs(IDENT["(23) E2"][0])
check("D2 odd and = 1 mod 8; v2(C2)=1", c23[0] % 8 == 1 and c23[1] % 4 == 2)
P23 = sum(s*(A*z+B)**5 for s,A,B in IDENT["(23) E2"][0]); P24 = sum(s*(A*z+B)**5 for s,A,B in IDENT["(24) E2"
    ][0])
check("(24) is a re-pairing of (23): P24(2z-1291) = 32 P23(z)", sp.expand(P24.subs(z, 2*z-1291) - 32*P23) == 0)
```

```
check("Thm 3.4 hand-expansion brackets for (23), (24), (25)",
      (8*((-3636)**2-(-4221)**2), 7763**2-4847**2) == (-36770760, 36770760) and (4*((-3636)**3-(-4221)**3),
     7763**3-4847**3, -2*6138**3) == (108540649620, 353958190524, -462498840144)
      and (8*((-4690)**2-(-5860)**2), 16817**2-13567**2, 216*((-29436)**2-(-29454)**2)) == (-98748000,
     98748000, -228964320))

# ---------------------------------------------------------------- 2. headline theorem
CI, DI = 3844230, 27826665
fac = factorint(CI)
check("C_I = 2*3*5*13*9857, squarefree, no prime = 1 mod 5", fac == {2:1,3:1,5:1,13:1,9857:1} and all(p % 5 !=
     1 for p in fac))
check("y -> y^5 is a bijection on Z/C_I", len({pow(yv, 5, CI) for yv in range(CI)}) == CI)

def nine_term(N):
    r = (N - DI) % CI
    roots = []
    for p in fac:
        d = pow(5, -1, p-1) if p > 2 else 1
        roots.append((pow(r % p, d, p) if p > 2 else r % 2, p))
    yv = int(sp.ntheory.modular.crt([p for _, p in roots], [v for v, _ in roots])[0])
    zz, rem = divmod(N - DI - yv**5, CI)
    assert rem == 0
    return zz, yv

def nine_sum(N, zz, yv):
    f = lambda t: t**5
    return (f(2*zz-38) - f(2*zz-37) + f(zz+39) - f(zz+35) - f(zz-22) - f(zz+22) + f(zz+6) + f(zz+6) + f(yv))

random.seed(5)
tests = list(range(-3000, 3001)) + [random.randrange(-10**40, 10**40) for _ in range(3000)] + [random.randrange
     (-10**9, 10**9) for _ in range(3000)]
check(f"end-to-end 9-term representation (41), {len(tests)} integers incl. |N|~1e40", all(nine_sum(N, *
     nine_term(N)) == N for N in tests))

# ---------------------------------------------------------------- 3. sieve (Prop 7.4) and Table 4
y5 = lambda m: {pow(t, 5, m) for t in range(m)}
S720, S159930 = y5(720), y5(159930)
L = math.lcm(720, 159930)
n1 = n2 = nn = 0
for N in range(L):
    A_ = ((N+360) % 720) in S720; B_ = ((N+327255) % 159930) in S159930
    n1 += A_; n2 += B_; nn += (not A_) and (not B_)
check("Tier-1 density 7/16, Tier-2 density 7/9, leftover 23/144 (exact over lcm period)", (n1*16, n2*9, nn*144)
      == (7*L, 7*L, 23*L))
T1 = IDENT["(47)"][0]; T2 = IDENT["T2-5"][0]; T3 = IDENT["(38) C_I"][0]
rows = [(1,1,-359,8282086203),(2,2,-24553,55794691661757925),(3,2,-29682,144057497616267233)
     ,(5,1,-115,27935517),
        (6,3,-404049,2801307716959828698333),(7,1,-113,25589378),(8,1,-352,7505520595),(11,1,131,-53582624)
     ,(13,1,133,-57799716)]
for N, tier, yv, zv in rows:
    T = {1:T1, 2:T2, 3:T3}[tier]
    ok = sum(s*(A*zv+B)**5 for s,A,B in T) + yv**5 == N
    t1 = ((N+360) % 9 not in (3,6)) and (((N+360) % 16 == 0) or N % 2 == 1)
    t2 = (N+327255) % 9 not in (3,6)
    chosen = 1 if t1 else (2 if t2 else 3)
    check(f"Table 4 row N={N} arithmetic", ok and chosen == tier, f"Tier {tier}")

# ---------------------------------------------------------------- 4. local analysis (Prop 7.1, Table 3, Cor
     7.5)
def delta(m, cap=12):
    S = y5(m); cur = {0}; r = 0
    while len(cur) < m and r < cap:
        cur = {(u+v) % m for u in cur for v in S}; r += 1
    return r
from sympy import primerange
check("Delta(5,p)=1 for p != 1 mod 5 (p<400); 5,3,3,3 for p=11,31,41,61; 2 for p=1 mod 5, 71<=p<6000",
      all(delta(p) == 1 for p in primerange(2, 400) if p % 5 != 1) and [delta(p) for p in (11,31,41,61)] ==
     [5,3,3,3]
      and all(delta(p) == 2 for p in primerange(71, 6000) if p % 5 == 1))
check("Delta(5,p^e): 4,8,9,27 -> 2; 25,125,625 -> 3; 121 -> 5; 961,1681 -> 3",
      [delta(m) for m in (4,8,9,27,25,125,625,121,961,1681)] == [2,2,2,2,3,3,3,5,3,3])
S25 = y5(25)
tab3 = {-12:(-1,2,2),-11:(2,0,2),-10:(1,2,2),-9:(-1,-1,-2),-8:(-1,0,-2),-7:(1,-1,-2),-6:(1,0,-2),-5:(1,1,-2)
     ,-4:(2,2,2),
        -3:(-1,-1,-1),-2:(-1,0,-1),-1:(1,-1,-1),0:(1,0,-1),1:(1,0,0),2:(1,0,1),3:(1,1,1),4:(-2,-2,-2),
```

```
        5:(-1,-1,2),6:(-1,0,2),7:(1,-1,2),8:(1,0,2),9:(1,1,2),10:(-1,-2,-2),11:(-2,0,-2),12:(1,-2,-2)}
check("Table 3 (25 rows)", all((sum(t**5 for t in tr) - r) % 25 == 0 and abs(tr[0]) in (1,2) for r, tr in tab3.
    items()))
S2_25 = {(u+v) % 25 for u in S25 for v in S25}
check("|S25+S25| = 13 and |2S5(625)| = 325", len(S2_25) == 13 and len({(u+v) % 625 for u in y5(625) for v in y5
    (625)}) == 325)
dens = F(1)
for p_, e_ in factorint(13972565906763750).items():
    m = p_**e_; S = y5(m); dens *= F(len({(u+v) % m for u in S for v in S}), m)
check("Cor 7.5(2) density 247/1705", dens == F(247, 1705), str(float(dens)))

# ------------------------------------------------------------------ 5. small representations (Remark 7.3, Table
    6)
T6 = {1:[1,0,0,0,0,0],2:[1,1,0,0,0,0],3:[1,1,1,0,0,0],4:[1,1,1,1,0,0],
      5:[17,-16,-15,13,7,-1],6:[17,-16,-15,13,7,0],7:[17,-16,-15,13,7,1],
      9:[-17,16,13,-1,-1,-1],10:[21,-19,-14,-14,-13,-11],11:[-17,16,13,-1,0,0],
      12:[-17,16,13,0,0,0],13:[-17,16,13,1,0,0],14:[27,20,8,-28,-13,0],
      15:[15,-13,-13,-7,2,1],16:[6,-5,-5,-4,-3,-3],17:[-15,13,13,7,-1,0],
      18:[-15,13,13,7,0,0],19:[-15,13,13,7,1,0],20:[17,-16,-13,2,0,0],
      21:[5,-4,-4,-4,-2,0],22:[5,-4,-4,-4,-2,1],24:[-10,9,9,-7,-4,-3],
      25:[-25,23,20,11,-8,4],26:[-17,16,15,-13,-7,2],27:[2,-1,-1,-1,-1,-1],
      28:[2,-1,-1,-1,-1,0],29:[2,-1,-1,-1,0,0],30:[2,-1,-1,0,0,0]}
check("Table 6: all 28 rows", all(sum(t**5 for t in v) == N for N, v in T6.items()))
def six_term_reps(box):
    pw = {t: t**5 for t in range(-box, box+1)}
    half = {}
    for a_, b_, c_ in itertools.combinations_with_replacement(range(-box, box+1), 3):
        half.setdefault(pw[a_]+pw[b_]+pw[c_], (a_, b_, c_))
    out = {}
    for N in range(1, 31):
        for s, tr in half.items():
            if N - s in half: out[N] = tr + half[N-s]; break
    return out
r30 = six_term_reps(30)
check("Remark 7.3: exactly N in {8,23} missing among 1..30 for |x|<=30", sorted(set(range(1,31)) - set(r30)) ==
     [8, 23])
r60 = six_term_reps(60)
check("23 still missing for |x|<=60; 8 represented via (-45,-33,-25,31,40,40)", 23 not in r60 and 8 in r60)

# ------------------------------------------------------------------ 6. Prop 2.2 (PTE parity gap and divisibility)
R = range(-20, 21)
bucket = {}
for ms in itertools.combinations_with_replacement(R, 4):
    bucket.setdefault((sum(ms), sum(t*t for t in ms), sum(t**3 for t in ms)), []).append(ms)
npair = bad = mixed = 0
for Lst in bucket.values():
    for u, v in itertools.combinations(Lst, 2):
        d4 = sum(t**4 for t in u) - sum(t**4 for t in v)
        if d4 == 0: continue
        d5 = sum(t**5 for t in u) - sum(t**5 for t in v); C, D = 5*d4, d5
        npair += 1
        if C % 720 or D % 360 or ((C//720 - D//360) % 2) or 4*D != sum(u)*C: bad += 1
        if sum(t & 1 for t in u) != sum(t & 1 for t in v): mixed += 1
check(f"Prop 2.2 on {npair} PTE pairs in [-20,20]: 720|C, 360|D, parity, D = e1*C/4", bad == 0)
check("no PTE pair with different numbers of odd entries", mixed == 0)
ms3 = list(itertools.combinations_with_replacement(range(3), 4))
def e123(ms): return tuple(sum(math.prod(c) for c in itertools.combinations(ms, r)) % 3 for r in (1,2,3))
check("15 four-element multisets of F3 have 15 distinct (e1,e2,e3)", len({e123(m) for m in ms3}) == 15 == len(
    ms3))

# ------------------------------------------------------------------ 7. elliptic curves (Thm 3.4)
def ec_add(P, Q, cv):
    if P is None: return Q
    if Q is None: return P
    x1, y1 = P; x2, y2 = Q
    if x1 == x2 and y1 == -y2: return None
    lam = (3*x1*x1 + 2*cv[0]*x1 + cv[1])/(2*y1) if P == Q else (y2-y1)/(x2-x1)
    x3 = lam*lam - cv[0] - x1 - x2
    return (x3, lam*(x1-x3) - y1)
def ec_mul(n, P, cv):
    R_ = None
    for _ in range(n): R_ = ec_add(R_, P, cv)
    return R_
A2 = 2**10; M2 = (A2-1)//3; cv2 = (F(M2+A2), F(2*A2*M2), F(A2*M2*M2))
toW = lambda p, yy: (M2*p, M2*yy); fromW = lambda P_: (P_[0]/M2, P_[1]/M2)
```

```
on = lambda P_, cv: P_[1]**2 == P_[0]**3 + cv[0]*P_[0]**2 + cv[1]*P_[0] + cv[2]
P1 = toW(F(144,25), F(43888,125)); P0 = toW(F(0), F(32))
check("P1 on E2; P0 has order 4 with 2P0 = (-1,0)", on(P1, cv2) and ec_mul(4, P0, cv2) is None and fromW(ec_mul
    (2, P0, cv2)) == (-1, 0))
check("P1 - P0 = (260/81, -132308/729), 2P1 as printed", fromW(ec_add(P1, (P0[0], -P0[1]), cv2)) == (F(260,81),
     F(-132308,729))
     and fromW(ec_add(P1, P1, cv2)) == (F(71784764,188101225), F(116043399920052,2579808300875)))
A6 = 6**10; N6 = 3*(A6-1); cv6 = (F(N6+9*A6), F(18*A6*N6), F(9*A6*N6*N6)); Q1 = (N6*F(16,5), N6*F(659568,5))
dbl = ec_add(Q1, Q1, cv6)
check("Q1 on E6; x_W(2Q1) = -1151448732500/151^2", on(Q1, cv6) and dbl[0] == F(-1151448732500, 151**2))
def no_torsion(P, cv):
    R_ = None
    for m in range(1, 13):
        R_ = ec_add(R_, P, cv)
        if R_ is None: return False
    return True
check("m*P1 != O and m*Q1 != O for m <= 12 (Mazur bound)", no_torsion(P1, cv2) and no_torsion(Q1, cv6))
def short_model_ok(a_, p_, Y_):
    AA = a_**10; Na = 3*(AA-1); beta = 4*AA - 1
    XS, YS = Na*p_ + beta, Na*Y_
    return YS**2 == XS**3 + 3*(2*AA*AA-10*AA-1)*XS - (AA+2)*(7*AA*AA-26*AA+1)
check("points map onto the printed short Weierstrass models (a=2: P1; a=6: Q1)",
     short_model_ok(2, F(144,25), 3*F(43888,125)) and short_model_ok(6, F(16,5), F(659568,5)))

# ------------------------------------------------------------------ 8. symbolic checks (sympy)
from sympy import Rational, symbols
t_ = symbols('t'); A_, B_, C_ = symbols('A B C')
Ft = B_**2*(C_**2-A_**2)*t_**3 - 3*A_**2*B_**2*t_**2 - 3*A_**2*B_**2*t_ + A_**2*(C_**2-B_**2)
printed_disc = -27*A_**4*B_**4*C_**4*(A_+B_+C_)*(A_+B_-C_)*(A_-B_+C_)*(A_-B_-C_)
check("Thm 3.2(2): disc F(t,1) = -27 a1^20 a2^20 a3^20 * prod(Heron factors)", sp.expand(sp.discriminant(Ft, t_
    ) - printed_disc) == 0)
p_, q_, kap = symbols('p q kappa'); w1_, w2_, w3_ = symbols('w1 w2 w3', positive=True)
dd = (kap*p_/w1_, kap*q_/w2_, -kap*(p_+q_)/w3_); XX = (kap*p_, kap*q_, -kap*(p_+q_)); uu = (-q_, p_, 0)
Cg = 40*sum(Xi*ui*(ui**2+di**2) for Xi, ui, di in zip(XX, uu, dd))
check("Thm 3.2(2): C on gauge (-q,p,0) equals printed 40*kappa*p*q*[...]",
      sp.simplify(Cg - 40*kap*p_*q_*((p_**2-q_**2) + kap**2*(q_**2/w2_**2 - p_**2/w1_**2))) == 0)
pp, aa = symbols('p a'); Aq = aa**10; Qa = (Aq-1)*pp**2 + 3*Aq*pp + 3*Aq
check("Lemma 3.3(1): C(p)/(40 kappa p) = (a^10-1)p^2 (p+1)(p+2)/Q_a",
      sp.simplify((pp**2-1) + 3*Aq*(pp+1)/Qa*(1-pp**2/Aq) - (Aq-1)*pp**2*(pp+1)*(pp+2)/Qa) == 0)
check("Lemma 3.3(3)(b): d'_3 = (beta_5 - beta_4)/2 = -p(1+kappa)/2", sp.simplify((-kap*(pp+1) - (pp-kap))/2 +
    pp*(1+kap)/2) == 0)
X1, X2, u2 = symbols('X1 X2 u2'); b1, b2, b3, b4 = symbols('b1 b2 b3 b4')
u1 = (X1**4*(b1-b3) + X1*X2**3*(b2-b4) - 3*X2*(X2-X1)*u2**2)/(6*X1*X2*u2); u3 = u1 + X2*u2/X1
Xs = (X1, X2, -X1, -X2); us = (u1, u2, u3, 0); bs = (b1, b2, b3, b4)
Ck4 = sp.simplify(40*sum(Xi*ui*(ui**2+Xi**2*bi) for Xi, ui, bi in zip(Xs, us, bs)))
printed4 = (10*X2*(X1**2-X2**2)/X1**2*u2**3 + 20*X2*(X2**2*(b2+b4)-X1**2*(b1+b3))*u2
           + 10*(X1**6*(b1-b3)**2-X2**6*(b2-b4)**2)/(3*X2)/u2)
check("Thm 3.2(4): slice satisfies (12) and C equals the printed Laurent polynomial in u2",
      sp.simplify(sum(Xi*ui for Xi, ui in zip(Xs, us))) == 0 and sp.simplify(3*sum(Xi*ui**2 for Xi, ui in zip(
    Xs, us)) + sum(Xi**3*bi for Xi, bi in zip(Xs, bs))) == 0
      and sp.simplify(Ck4 - printed4) == 0)
lam, W1, W2 = symbols('lambda W1 W2', positive=True)
Rl = (W1**2*lam**4 + W2**2 + 4*W1*W2*(lam+lam**3) - 6*W1*W2*lam**2)/(W1*lam+W2)**4
check("Remark 4.5: R'(lambda) = 12 lambda w1 w2 (w1+w2)(lambda-1)/(lambda w1 + w2)^5",
      sp.simplify(sp.diff(Rl, lam) - 12*lam*W1*W2*(W1+W2)*(lam-1)/(lam*W1+W2)**5) == 0)
dl = symbols('d1:5'); sg = symbols('s1:5'); n_, Rr = symbols('n R')
Bs = []
for i in range(4): Bs += [(sg[i]+dl[i])/2, (sg[i]-dl[i])/2]
Pg = sp.expand((2*z+Bs[0])**5 - (2*z+Bs[1])**5 + sum((z+Bs[2*i])**5-(z+Bs[2*i+1])**5 for i in (1,2,3)))
e34 = 12*dl[0]*sg[0]**2 + 3*sum(dl[i]*sg[i]**2 for i in (1,2,3)) + 4*dl[0]**3 + sum(dl[i]**3 for i in (1,2,3))
check("Prop 5.1: [z^4]/5 = (32), [z^3]/10 = (33), [z^2] = (5/2)*((34) lhs+rhs)",
      sp.expand(Pg.coeff(z,4)/5 - (16*dl[0]+dl[1]+dl[2]+dl[3])) == 0 and sp.expand(Pg.coeff(z,3)/10 - (8*dl[0]*
    sg[0]+sum(dl[i]*sg[i] for i in (1,2,3)))) == 0
      and sp.expand(Pg.coeff(z,2) - Rational(5,2)*sp.expand(e34)) == 0)
sig3 = -(8*dl[0]*sg[0] + dl[1]*sg[1] + dl[3]*sg[3])/dl[2]
lhs35 = sp.expand((4*dl[0]*sg[0]**2 + dl[1]*sg[1]**2 + dl[2]*sig3**2 + dl[3]*sg[3]**2 - Rr)*dl[2])
Tt = 8*dl[0]*sg[0] + dl[3]*sg[3]
rhs35 = dl[1]*(dl[1]+dl[2])*sg[1]**2 + 2*dl[1]*Tt*sg[1] + 4*dl[0]*dl[2]*sg[0]**2 + Tt**2 + dl[2]*dl[3]*sg[3]**2
      - dl[2]*Rr
check("Prop 5.1: (35) follows from (34) with sigma3 eliminated", sp.simplify(lhs35 - rhs35) == 0)
u40 = ((-1023*28**2 + 3*16*27)//24, 112, (-1023*28**2 + 3*16*29)//24, 0)
check("Thm 5.3(2): (36) at (2,28,1,4) gives u = (-33364,112,-33360,0) and (40)", u40 == (-33364, 112, -33360,
    0)
     and [ai*(ui+s*di)//16 for ai, ui, di in zip((2,1,1,1), u40, (28,32,-896,-32)) for s in (1,-1)] ==
    [-4167,-4174,9,5,-2141,-2029,-2,2])
```

```
bj = [-2*n_**2+12*n_-682, -2*n_**2+4*n_-682, 16*n_**2+8*n_, -16*n_**2+8*n_, n_**2-60*n_-341, n_**2+68*n_-341,
     -16*n_**2, 16*n_**2]
Bn = [t/8 for t in bj]
Pn = sp.expand((2*z+Bn[0])**5 - (2*z+Bn[1])**5 + sum((z+Bn[2*i])**5-(z+Bn[2*i+1])**5 for i in (1,2,3)))
check("(36)-(37): P8(z;n) is linear in z with C(n) = 30 n^3 (85n^4 - 678n^2 + 116281)/8",
      all(sp.simplify(Pn.coeff(z, r)) == 0 for r in (2,3,4,5)) and sp.simplify(Pn.coeff(z,1) - 30*n_**3*(85*n_
     **4-678*n_**2+116281)/8) == 0)
check("(37): v2(C(n)) = 1 and all B_j(n) integral for odd n = 1..99", all(
      (30*nn**3*(85*nn**4-678*nn**2+116281)//8) % 4 == 2 and all(int(b.subs(n_, nn)) == b.subs(n_, nn) for b in
      Bn) for nn in range(1, 100, 2)))
u_, v_, w_ = symbols('u v w')
l46 = (z+(u_-1)/2)**5+(z-(u_+1)/2)**5+(z+(v_-1)/2)**5+(z-(v_+1)/2)**5-2*(z+(w_-1)/2)**5-2*(z-(w_+1)/2)**5
d46 = sp.expand(l46 - Rational(5,16)*(u_**4+v_**4-2*w_**4)*(2*z-1))
check("Thm 6.2(2) (46) holds modulo u^2+v^2=2w^2", sp.rem(sp.Poly(d46, v_), sp.Poly(v_**2-(2*w_**2-u_**2), v_))
     .is_zero)
check("Delta^4 z^5 = 120 z + 240", sp.expand(sum((-1)**(4-j)*comb(4,j)*(z+j)**5 for j in range(5))) == 120*z
     +240)
check("Cauchy-Lame (x+y)^5-x^5-y^5 = 5xy(x+y)(x^2+xy+y^2)", sp.expand((x+y)**5-x**5-y**5-5*x*y*(x+y)*(x**2+x*y+
     y**2)) == 0)
ok43 = True
for name, (terms, (C, D)) in IDENT.items():
    ok43 &= C % 30 == 0 and all((D - sum(s*B for s, A, B in terms if A % pr == 0)) % pr == 0 for pr in (2,3,5))
check("Thm 4.3: 30 | C and D = sum_{p|A} s*B (mod p), p=2,3,5, for all 13 linear identities", ok43)
def clear(a_, p0, k0):
    sh = []
    for ai, ui, di in zip((a_,1,1), (F(-1), p0, F(0)), (k0*p0/a_**5, k0, -k0*(p0+1))): sh += [ai*(ui+di), ai*(
     ui-di)]
    Lc = math.lcm(*[t.denominator for t in sh]); g = math.gcd(*[int(t*Lc) for t in sh])
    return [int(t*Lc)//g for t in sh]
check("Lemma 3.3(2): printed B_j of (23),(24),(25) are reproduced from (a,p,kappa)",
      clear(2, F(260,81), F(72,97)) == [-3636,-4221,7763,4847,-6138,6138] and clear(2, F(144,25), F(130,211))
     == [-4690,-5860,16817,13567,-10985,10985]
      and clear(6, F(16,5), 3*6**5*(F(16,5)+1)/F(659568,5)) == [-29436,-29454,19349,12059,-15309,15309])

print()
print(f"{sum(ok for _, ok in results)} of {len(results)} checks passed")
```

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